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A Comprehensive Mapping of European-African Actors and Activities

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Christina Giannopapa

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http://www.espi.or.at
Tel. +43 1 7181118-0; Fax -99

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Executive Summary

The Context

Sustainable development in Africa requires access to data, information, knowledge where space technologies through the use of satellites can be of tremendous benefit. The transversal nature of space touches various policies (e.g. environment, agriculture, health, security, education, disaster management) and can facilitate growth and sustainable development. This year several high level meetings are taking place where discussions will be focused on identifying steps forward. Only five years are left to achieve the Millennium Development Goals and in September the United Nations summit will be held in New York with the aim of evaluating progress and identifying the further efforts required in order to achieve the goals. In the EU-Africa Summit in November, the European and African partners will reaffirm their commitment to each other for a strong partnership and discuss the implementation of the next Action Plan 2011-2013 for the implementation of the Africa-EU strategic partnership that is expected to maintain the space component of Area 8. Presently, the Belgian EU Council Presidency is making "Space and Africa" a priority issue, highlighted with the conference "Space for the African Citizen" to be held in September and is aiming at demonstrating the benefits space can bring to the African citizen.

Space activities in Africa are utilized by numerous collaborations at various levels between various actors from African countries, European and international partners, where there has been very little overview. Therefore, the present study has been conducted under the auspices of the Belgian EU Council Presidency to tackle three main questions: 1) "who is who"; 2) "who is doing what"; and 3) "who is cooperating with whom". This study provides a comprehensive overview of the main African actors and their activities in the continent as well as gathering information regarding activities run by the main European actors and the United Nations organizations. This mapping was conducted in consultation with European and African stake holders and the presented material reflects mainly a European perspective. This study is meant to serve as a catalogue providing general information and contact points about main actors and activities in Africa where satellite applications are or could potentially be used. Furthermore, the study provides recommendations to policy makers regarding the way forward. Figure 1 shows the policy relevant actors in Africa involved in its economic development. This figure was developed during consultations with African representatives and proved to be useful in understanding the links between different actors in Africa that also need to be addressed for coordinating African activities for development through the use of space. The methodology used for the study was: performing a desktop research, organising a round table with African and European representatives, organising a workshop with European representatives, participating in the UNSPIDER regional workshop in Africa, and gathering information via questionnaires from European member states and African countries.

Recommendations

Social factor

Space applications can assist in providing solutions to the African population regarding their basic needs such as providing food and water security, healthcare, education, early warning, disaster management and emergency response, etc. Nevertheless, the benefits of space applications are not sufficiently communicated to decision makers as well as the wider population and there is not enough basic education at various levels to perform, manage and operate space-based assets. Various space projects in different areas have been developed for Africa but very few are sustainable beyond the pilot phase. This is due to the fact that often the local community of the end users is not involved from the beginning and does not have the feeling of ownership. Often, appropriate bodies have not been identified within the government structure to take up the responsibility for running the project and its maintenance. The projects developed in Africa are often conceived by developed countries without being

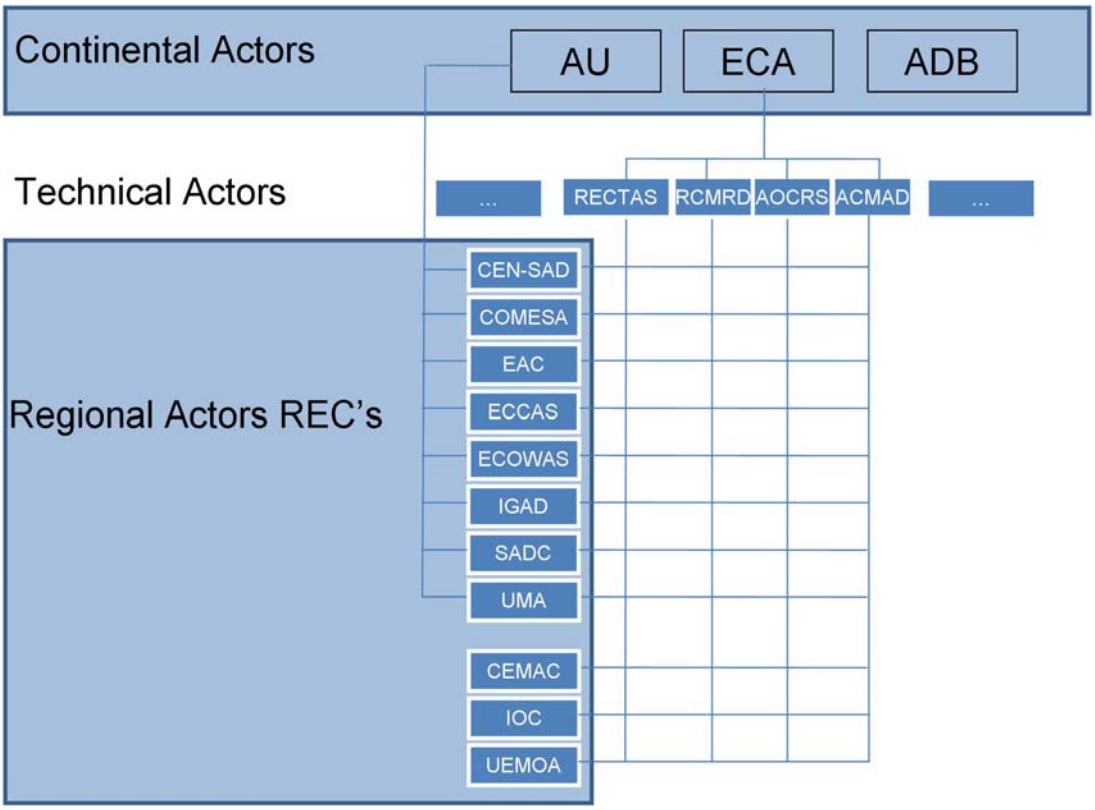


Figure 1: Policy relevant actors in Africa.

able to capture properly the societal need and infrastructure restrictions of the underdeveloped countries. The recommendations are:

- *Enhance communication about the benefits of space.* Appropriate information mechanisms need to be set up to communicate the benefits of various space applications and projects at local community level. Such communication mechanisms could be local information centres, radio and TV info notes.
- *Enhance education regarding space and its applications.* Education regarding space should be enhanced at school and university level. Adequate learning material on the benefits of space and its contribution to skills development should be provided. Capacity building at all levels of education (primary, secondary, tertiary and continuous learning) should be emphasised. Educational programs are a long-term measure with longer social benefits.
- *Enhance the involvement of the community in ensuring continuation and sustainability after the pilot phase is completed.* The African citizen should be involved in the ownership of each project. This would enhance continuation beyond the completion of the project. Involvement at the community level is expected to be the most effective. A coherent mechanism of knowledge transfer has to be ensured from the very beginning of pilot-projects.

Legal factor

In Africa the areas of telecommunications and satellite imaging are fragmented into small national markets and the licence requirements and regulations vary from country to country. Current regulatory aspects regarding satellite applications may hinder private initiatives by African and foreign investors and delay sustainable development. Furthermore, there is very little coordination between various policy and regulatory frameworks in various African countries regarding space and the space component of other sectors. Often different African government or different ministries within a government purchase the same asset from satellites. This duplication increases administrative and economic overheads. Current taxation systems on goods import affects the development of satellite applications equipment since equipment, devices and services need to be imported. It

appears that taxation is duplicate for the same goods and services at various levels resulting in accumulative excessive burden for the importer. The recommendations are:

- *Implement appropriate regulations.* It is necessary to foster initiatives to build institutional capacity in making treaties, setting standards, formulating policy, drafting regional integration protocols and harmonising regulations at a pan-Africa level.
- *Coordinate policies and regulations.* African governments should coordinate their policies and regulations regarding commercial and non-commercial space applications (e.g. data policy). They should foster policy and regulatory harmonisation to create larger common markets based on regional economic communities while increasing private sector participation. This would facilitate exchange of assets and reduced duplication in purchasing the same asset.
- *Harmonise licensing procedures.* Establish common licensing and authorisation procedures instead of having them country by country. Creation of an internet based licence application and dissemination of regulatory information would significantly assist potential investors.
- *Facilitate cross border exchange and harmonise information infrastructure.* With regard to the purchase of information from satellites, initiatives should be promoted for investment in access and cross-border information infrastructure through effective partnerships between public, private and not-for-profit sectors, in order to achieve universal access and full inter-country exchange of information and data.

Economic factor

Even though during the past 10 years there have been significant efforts by the REC's in integrating the various markets, many of them are still fragmented. At the same time the low income levels in Africa and limited economies of scale make it unattractive for Foreign Direct Investment (FDI). Licence fees for satellite systems are generally very high and the cost of satellites is also high, which makes it difficult for companies to survive. There is a need to have low cost systems to facilitate improved access. The African market is mostly dominated by government activities in the field of space, satellite applications and related sectors. It is important that the market is opened up. The recommendations are:

- *Enhance institutional coordination.* Common economic targets, complementarities and how they can be achieved need to be identified. The current institutional framework in Africa is appropriate but concrete links of cooperation need to be established between the main African actors like the AU, UNECA and the REC's. Common strategic partnerships between them are essential for sustainable development. Adequate resources such as funding, personnel and technical expertise are essential for the success of the partnership.
- *Align funding mechanisms.* The various funds provided from Europe, United Nations, European member states, private bodies, etc need to be aligned in terms of funding, time scale and expected targets in order to better utilise the available resources, avoid duplication and create complementarities. The funding mechanisms should also be aligned such that they cover the entire cycle of a satellite applications project, from pilot project to hand over, operation, capacity building and maintenance.
- *Refocus allocation of funding.* A cost-benefit analysis should be performed before projects are started. The criteria should not necessarily be based on direct monetary factors but should be based factors such as benefits to the citizen, safety, job creation, etc.
- *Improve economic regulatory frameworks for FDI,* making them far more open, permitting profit repatriation and providing economic measures and other incentives to attract investment.
- *Enhance and regulate the pre-competition environment.* Promote liberalization and privatization in order to open access to satellite operators and service providers through partnerships with local stakeholders in terms of public entities, private firms and not-for-profit sectors. The pre-competition framework has to facilitate the involvement of African SMEs.
- *Improve taxation.* African countries are also encouraged to accelerate the conclusion of double taxation treaties (DTTs). DTTs can make it more attractive for foreign investors to invest in a country by helping them to avoid paying taxes twice on the same transaction[1].

[1] The majority of African countries have signed multilateral agreements dealing with the protection of FDI, such as the Convention establishing the Multilateral Investment Guarantee Agency (MIGA) and the Convention on the Settle-

Taxation is a leverage to be considered also in order to support demand for satellite services in terms of tax-emption or negative taxation – financial support - in order to improve citizens' rights as members of society (freedom of self-expression, right to be informed, freedom of press, etc).

Political factor

The political commitment of African leaders to work together and with other international partners to solve problems for the African citizen and achieve societal, economic and political integration is being materialised step by step through the African Union and the various Regional Economic Communities. Nevertheless, political commitment for space and satellite applications is lacking. This is mainly due to the fact that there is insufficient understanding of the benefits and communication of the success stories. The context of space policy is different in Europe and in African and also varies from country to country in Africa. The main objectives of a space policy for Africa would be to focus on implementing those policy aspects that would allow successful use of space based information and applications for sustainable development. Space policy is uncoordinated at the continental level as well as the regional level. Furthermore, within each African country there are insufficient links between the space component and other policy areas (e.g. agriculture, environment, health, transport, security). The link between the priorities of the various African countries and how space can assist in achieving them is missing. Even though at the continental level the appropriate institutional mechanisms are present for harvesting the benefits of space, there is insufficient coordination between the various institutions.

- *Capture adequately Member States needs and translate them into concrete areas where space can assist in meeting them.* The national needs of African countries where space competence is present need to be translated and reflected in other national policies (e.g. agriculture, environment, health, transport, security) in a coherent manner. Coordination meetings, workshops and conferences can be used as tools for the appropriate capturing of the need.
- *Identify responsible actors.* At the beginning of every project the appropriate potential governmental actors needed for the sustainability of projects need to be identified. Communication and establishment of appropriate links with the respective authorities should be part of the tasks before the completion of the pilot project.
- *Establish appropriate governance links.* The AU, ECA and REC's are important actors in Africa. The coordination and interaction between them regarding space based information and applications should be strengthened and formalised.
- *Establish harmonized international relations between African countries.* This would increase political stability and facilitate FDI's attractiveness which would further technological and industrial development.
- *Increase the involvement and commitment of political leaders.* African leaders and politicians need to be more aware of and committed to the benefits of space and use of satellite applications for the sustainable development of Africa.
- *Coordinate space policy and other policy areas.* Space policy and the space component in other policies (e.g. agriculture, environment, health, transport, and security) need to be identified in a systematic way. Coordination should be achieved at the three levels: national, regional and continental.
- *Develop the basis for a pan-African space policy.* The common aspects of the space component should be treated at regional and continental levels. After establishing national needs, African countries should work together on the regional and continental levels to tackle their common goals. As a second step, a pan-African Space Policy that can reflect common interests such as disaster management and emergency response, climate change (including desertification, deforestation, drought etc.), telecommunications and GMES would better facilitate the coordination of efforts. The area that is in vital need of such a pan-African approach is disaster management and emergency response. Thus, a common policy in disaster management and emergency response could set the first area of the pan-African space policy.
- *Formalise institutional relations for space based information and applications.* As a first step, existing institutions are sufficient enough to work together in building sustainable development for Africa but there is lack of cooperation and coordination between institutions and agencies

ment of Investment Disputes between States and Nationals of Other States.

involved in space based information and applications. The first point of coordination that is needed is in the field of access, use and sharing of space based information. In particular, maps and GIS would be the first step to formalise interactions and institutionalise use of these assets on the regional level. A particular topic where this coordination is essential is early warning and emergency response in disaster management. An effort should be made to foster horizontal agreements to establish institutional groups. A second step to be targeted in the long run would be to set the basis for institutionalisation at the continental level. Particular fields of coordination could be environmental monitoring and disaster management. Discussions about the establishment of a continental institution which would work in cooperation with the AU date back to 1991. The AU project for the establishment of an African Space Agency was finally launched last month. Such an institution would be beneficial for centralise purchase of space data like maps, satellite images etc. and would be beneficial for avoiding duplications. It could also be involved as a central point to the outsider with respect to disaster management and emergency response. Nevertheless, this should be a long term plan starting first at national and regional levels since different countries have different needs and different levels of development.

- *Enhance communication of the benefits of space.* The success stories of the use of satellite applications for the benefit of the African citizen and for sustainable development for Africa need to be communicated in way that political figures are able to use as arguments for supporting their case. Relevant information material needs to be prepared by consortiums including representatives from technical experts, end users, entrepreneurs, administrators, societal and political groups, etc.
- *Promote bilateral and/or multilateral investment treaties (BITs).* This should be conducted in conjunction with other countries who aim at protecting and promoting FDI to clarify the terms under which FDI can take place between the partner countries.

Technological factor

The different African countries focus their space activities and run projects under three main categories: a) Infrastructure development; b) use of satellite applications; and c) capacity building. From these categories the majority of the projects focus on space applications and there is a need for more projects on capacity building. A pan-African University is currently being established where space will be covered by the South Africa unit.

- *Increase projects on capacity building.* Projects with a focus on capacity building should be enhanced. These projects should cover all levels including schools, universities, government employees, politicians, and the public at large.
- **I***dentify relevant space technology in relation to providing solutions to needs of African population.* Developed countries often initiate projects without being able to sufficiently capture the needs of the African population. African-led initiatives should be given priority, which capture African needs. These projects should be chosen at a regional, national and continental level. Appropriate workshops should be taking place that involve the local community and the technical experts.
- *Implement technological innovation scoreboard.* This should be implemented in order to map and to monitor the degree of technology. This tool will help to understand the current degree of technology and whether this technology is suitable for the needs of the end user.
- *Increase cooperation between those that have technology know-how and users.* The end user often has very limited understanding or receives insufficient capacity building from those that have the technology know-how. After training they often move out to more senior managerial (not hand-on) levels. Increased intensity of education at the various user levels could be enhanced. Community to community training may reduce the cost of implementation.
- *Enhance transfer-of-knowledge mechanisms through initiatives of mutual learning platforms.* Academic cooperation with all other actors in order to facilitate know-how development and exchange should be enhanced. Existing academic networks could provide the initial necessary links needed. The pan-African University, currently under establishment, is expected to play an important role in the development of sustainable development of Africa.

Furthermore, specific recommendations were made with respect to the main European and African Actors.

1. Introduction

1.1 The setting

The African continent, with over one billion people, has the fastest growing population in the world. The majority of the population lives in poverty and the 53 states on the continent are still classified by the United Nations and International Monetary Fund as developing countries. African countries are facing problems in covering the basic needs of their people such as food, water, housing, healthcare and education. Space and satellite applications can be used as fundamental tools to assist in solving basic problems and contribute to Africa's sustainable development.

The international community, the European Union and the African countries themselves have set goals and strategies for the sustainable development of the African continent. The eight Millennium Development Goals (MDG), which also target Africa, include reducing extreme poverty, reducing child mortality rates, fighting disease epidemics such as AIDS, and developing global partnership for development. They show the commitment of 192 nations and at least 23 international organisations, to achieve them by 2015. Various organisations within the United Nations are involved in activities with Africa and, in particular, UNECA (United Nations Economic Commission for Africa), as the UN regional commission serving Africa, plays a central role in the coordination of international efforts. Even though not all UN organisations have dedicated programmes for space applications, many of them use the assets provides by satellites in their work including disaster management and emergency response, resources management, healthcare, energy, climate change, security.

African countries themselves have a long history of attempting to work together for political and socioeconomic integration and for solving their common problems. In particular, regarding economic cooperation and integration, Africa has the highest number of Regional Economic Communities (RECs), with overlapping memberships, mandates and programmes. Even though the REC's do not explicitly target space systems and its applications, these technologies can assist in achieving their goals. Currently, eight of the REC's have become pillars of the African Union which was established by 53 member states with very ambitious targets to create a united and integrated Africa with political, economic and social integration. This vision to approach Africa in a holistic manner rather than in an incremental approach makes the African Union project unique. The African Union also seeks to establish international partnerships to achieve its goals.

Africa and Europe have strong ties as geographic neighbours and share a long standing history, traditions, cultural, social and trade patterns. Over the last decade, the link between Africa and Europe has been deepened by political commitment from their respective leaders. The Africa-European Union Strategic Partnership verifies this commitment and the 2008-2010 First Action Plan outlines the eight areas of this partnership in which Area 8 explicitly sets space as a target. In this it already identifies the importance of the involvement of the European Space Agency, EUMETSAT and the private sector. It identifies activities in the area of telecommunications; navigation; climate change, resources security (including water and food) through earth observation; humanitarian aid and security through integrated applications; and knowledge based society. It also foresees the involvement of European initiatives such as Galileo, EGNOS and GMES. Currently the evaluation of the First Action Plan is taking place and preparations are being made for the Second Action Plan of 2011-2013. Furthermore, individual European Member States are undertaking their own activities directly in partnership with African Countries.

Thus, there are numerous actors and activities engaged in the African continent but there is a lack of a comprehensive overview that is vital for assessing present developments and identifying the way forward. There is a need to systematically catalogue actors and activities in Africa by African, European and International actors.

1.2 Approach of the Study

Even though there is much activity in the African continent regarding the use of satellite applications, there is a lack of a compre-

hensive mapping of actors and activities from African, European and international stakeholders. Under the auspices of the Belgian EU Council Presidency, ESPI focused on providing a comprehensive mapping of activities and actors in Africa. This project aimed at tackling three main questions regarding the use of satellite applications: 1) "who is who"; 2) "who is doing what"; and 3) "who is cooperating with whom".

Chapter 2 of the study provides the setting for this study. Chapter 3 gives an overview of international and European goals and strategies. Chapter 4 provides an overview of the main African, European and international actors whereas the follow up chapters provide in-depth information about them. Chapter 5 highlights the main African actors at the continental, regional and national levels. Chapter 6 describes the European actors and their roles. It also describes the role of the African-European partnership from a European perspective. Chapter 7 highlights the United Nations efforts in Africa and the main agencies that are active in Africa. Chapter 8 lists space activities in Africa by African countries and European Member States that have direct involvement in Africa. Chapter 9 provides information about the main projects in Africa and private initiatives.

The methodology used to conduct this study was based around three tools: an extensive desktop study, interviews and workshops/roundtables. For this purpose two questionnaires were prepared: one for the African actors and one for the European actors. From the answers collected a microenvironment analysis of social, legal, economic, political and technological factors was conducted to draw policy recommendations.

In the framework of this study a round table and a workshop were organised by the European Space Policy Institute (ESPI). On 10 February 2010, utilising the opportunity of the session of the Scientific and Technical Subcommittee of the United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS-STSC), ESPI organised a roundtable assessing the status of European-African cooperation in space applications and discussing further developments and initiatives. Representatives from Africa and Europe were brought together to publicly highlight benefits of ongoing cooperation and to look into perspectives for the future. On 24 June 2010 a workshop was organised assessing the views of the European stakeholder's workshop on European-African partnership. Through a systematic approach with targeted questions on the social, legal, economic, political and technological factors, the workshop aimed at identifying weak and strong points of the current approach and drawing policy perspectives from the European perspective. Furthermore, during this project the author of this study was invited to participate on 9 July 2010 at the UN-SPIDER workshops in Addis Ababa. The workshop was organised by the UN-SPIDER Programme, in cooperation with the United Nations Economic Commission for Africa (UNECA), and with support from the government of Austria and the Secure World Foundation, to promote access to and use of space-based technologies and solutions for disaster management and emergency response within relevant communities in Africa. The author presented a paper on "Policy Perspectives in European-African Partnership through the Use of Satellite Applications", and acted as a moderator for the Panel Session on "Ways to Institutionalise Space-Based Information for Disaster Risk Reduction and Emergency Response" where panellists from civil protection and disaster management agencies engaged in discussion on exchanging best practices and institutional and regulatory aspects. Furthermore, she actively participated in the Discussion Sessions of the Working Group on "Policies and Strategies". This group met for four sessions and discussed topics on networking and long term sustainability, capacity building efforts in Africa and ways to institutionalise space based information and disaster risk reduction and emergency response. The main recommendations of this group were that there is a need to identify "who is who" and "who is doing what" in Africa and the need for a pan-African space policy in disaster management and emergency response which, in the short term, can be supported by the existing actors of the African Union, Economic Commission for Africa, its technical centres and the Regional Economic Communities.

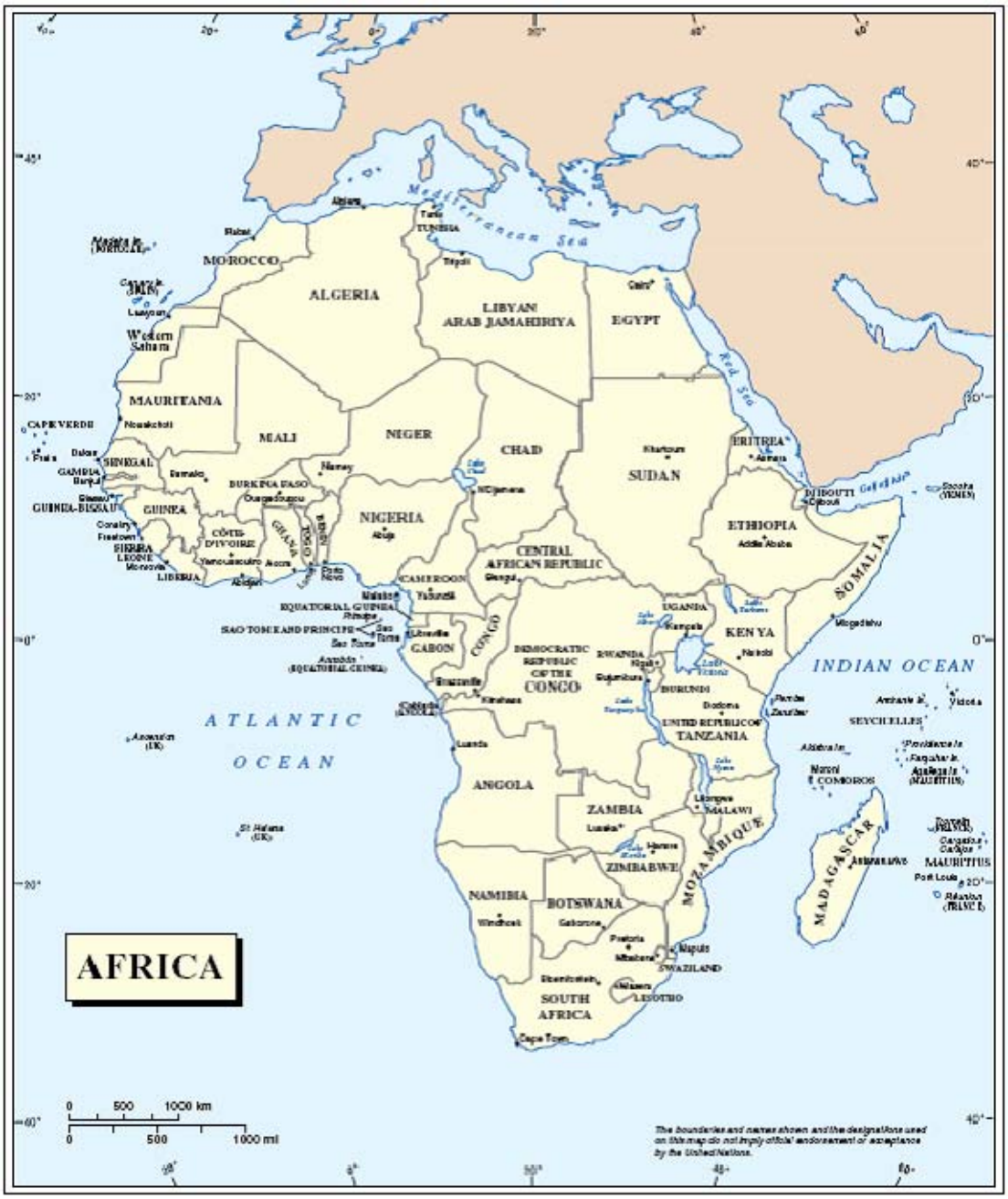


Figure 2: Map of Africa, United Nations 2004[2]

[2] Map No. 4045 rev.4, January 2004, United Nations, Department of Peacekeeping Operations, Cartography Section.

2. International and European Goals and Strategies

2.1 Millennium Development Goals

The Millennium Development Goals (MDGs) are eight international development goals that 192 United Nations member states and at least 23 international organizations have agreed to achieve by the year 2015. They include reducing extreme poverty, reducing child mortality rates, fighting disease epidemics such as AIDS, and developing a global partnership for development. These goals are:

Goal 1: Eradicate extreme poverty and hunger

Goal 2: Achieve universal primary education

Goal 3: Promote gender equality and empower women

Goal 4: Reduce child mortality

Goal 5: Improve maternal health

Goal 6: Combat HIV/AIDS, malaria, and other diseases

Goal 7: Ensure environmental sustainability

Goal 8: Develop a global partnership for development

2.2 Joint Africa-EU Strategy

Overview

As geographic neighbours, Africa and Europe have a long-standing history and traditions of cultural and social exchange. The cooperation at various levels between European and Africa countries goes way back but over the last decade has been significantly strengthened and formally sealed by the political commitment of leaders. In 2000 the first historic Africa-EU summit in Cairo set out the basis for constructive dialogue and joint actions. Since then, with the establishment of the African Union in 2002, Europe has found an institutional partner for dialogue. Over the last ten years the European Union has been enlarged and, since December 2009, when the EU High Representative for Foreign Affairs and Security Policy took office and the new Lisbon Treaty came into force, Africa-EU relations are expected to become even closer by sharing common values: the respect for human rights, freedom, equality, solidarity, justice and rule of law and democracy.

Over the last decade Africa and the EU have developed political strategies, policy documents and action plans as a basis for their cooperation. These include the 2004-2007 Strategic Plan of the AU[3], the 2005 "EU and Africa: Towards a Strategic Partnership" and the 2008 "The Africa-European Union Strategic Partnership"[4]. Subsequently, the First Action Plan (2008-2010) for the implementation of the Africa-EU strategic partnership was adopted in December 2007.This Action Plan is currently tasked with assessing the status of progress regarding the Joint Africa-EU partnership, derive lessons learned and define follow up actions. In particular, with regards to the GMES and Africa Action Plan, there is an ongoing consultation process[5]. Currently, the Second Action Plan (2011-2013) is under preparation and is scheduled for adoption in the upcoming 3rd Africa-EU Summit that will take place in Libya on 29-30 November 2010.

Shared vision and common principles

The joint Africa-EU strategic partnership defines long-term policy orientations, based on a shared vision and common principles. The vision for the partnership between Africa and the European Union should be based on the common values of human rights, freedom, equality, solidarity, justice, the rule of law and democracy. It should intend to bridge the efforts of the Africans and the European Union and its Member States for development in the African continent through coordination of economic cooperation and promotion of sustainable development.

[3] African Union, "Strategic Plan of the Commission of the African Union, Volume 3: 2004-2007 Plan of Action. Programmes to speed up integration of the continent", Addis Ababa, Ethiopia, May 2004.
[4] Council of the European Union, General Secretariat, "The Africa-European Union Strategic Partnership", June 2008, Brussels, ISBN 978-92-824-23752.
[5] The document currently under consultation can be downloaded from http://aquaknow.net/gmesafrica-action-plan

The policy orientations based on the shared vision and common principles are:

- African unity;
- Interdependence between Africa and the EU;
- strong political dialogue;
- co-management and co-responsibility;
- burden-sharing and mutual accountability;
- solidarity and mutual confidence;
- equality and justice;
- common human security;
- gender equality and non-discrimination;
- right to development;
- coherence and effectiveness of existing agreements, policies, and instruments;
- long term approach for sustainable development.

Objectives

The four main objectives of the partnership are[3]:

- To reinforce and elevate the Africa-EU political partnership to address issues of common concern.
- To strengthen and promote peace, security, democratic governance and human rights, fundamental freedoms, gender equality, sustainable economic development, including industrialisation, regional and continental integration in Africa, and to ensure that all the Millennium Development Goals (MDG) are meet in all African countries by the year 2015.
- To jointly promote and sustain a system of effective multilateralism, with strong, representative and legitimate institutions, and the reform of the United Nations (UN) system and of other key international institutions, and to address global challenges and common concerns, such as human rights, including children's rights and gender equality, fair trade, migration, HIV/AIDS, malaria, tuberculosis and other pandemics, climate change, energy security and sustainability, terrorism, the proliferation of Weapons of Mass Destruction and the illicit trafficking of Small Arms and Light Weapons, and knowledge-based society issues as ICT, science, technology and innovation.
- To facilitate and promote a broad-based and wide-ranging people-centred partnership, Africa and the EU will empower non-state actors and create conditions to enable them to play an active role in development, democracy building, conflict prevention and the post-conflict reconstruction process.

Strategy Implementation

It is foreseen that the joint strategy will provide the basis for a long term approach and will be implemented via short term (three year) Action Plans with concrete and measurable outcomes. In order to finance the Action Plans financial instruments are used such as[3]: the European Development Fund (EDF), the Development Cooperation Instrument (DCI), the European Neighbourhood Policy Instrument (ENPI), the Instrument for Stability, as well as the Thematic Programmes and through EU financial institutions, such as the European Investment Bank (EIB). It is also foreseen that individual European member states will also provide support when needed. From the African side, African financial institutions involved are the African Development Bank.

First Action Plan 2008-2010

The Strategy's First Action Plan adopted in December 2007 outlines eight areas for strategic partnership with priority action for the period 2008-2010:

Area 1: Peace and Security
- Enhance dialogue on challenges to peace and security;
- Full operationalization of the African Peace and Security Architecture;
- Predictable Funding for African-led Peace Support Operations.

Area 2: Democratic Governance and Human Rights
- Enhance dialogue at global level and in international fora;
- Promote the African Peer Review Mechanism and support the African Charter on
- Democracy, Elections and Governance;
- Strengthen cooperation in the area of cultural goods.

Area 3: Trade, Regional Integration and Infrastructure
- Support the African integration agenda;
- Strengthen African capacities in the area of rules, standards, and quality control;
- Implement the EU-Africa Infrastructure Partnership.

Area 4: Millennium Developments Goals
- Ensure the finance and policy base for achieving the MDGs;
- Accelerate the achievement of the Food Security Targets of the MDGs;
- Accelerate the achievement of the Health Targets of the MDGs;

- Accelerate the achievement of the Education Targets of the MDGs.

Area 5: Energy
- Implement the Energy Partnership to intensify cooperation on energy security and energy access.

Area 6: Climate Change
- Build a common agenda on climate change policies and cooperation;
- Cooperate to address land degradation and increasing aridity, including the "Green Wall for the Sahara Initiative".

Area 7: Migration, Mobility and Employment
- Implement the Declaration of the Tripoli Conference on Migration and Development;
- Implement the EU-Africa Plan of Action on Trafficking of Human Beings;
- Implement and follow up the 2004 Ouagadougou Declaration and Action Plan on Employment and Poverty Alleviation in Africa.

Area 8: Science, Information Society and Space
- Support the development of an inclusive information society in Africa;
- Support S&T Capacity Building in Africa and Implement Africa's Science and Technology Consolidated Plan of Action;
- Enhance cooperation on space applications and technology.

The eighth area explicitly mentions space but other areas also make use and benefit from satellite applications. Area 8 as presented in the First Action Plan 2008-2010 will be described in greater detail. . Furthermore, Area 6 climate change and Area 3 infrastructures are also areas where space based information and applications are used and various projects are developed (see projects in Section 8.1).

Area 8: Science, Information Society and Space

The importance and role of science and technology and ICT as already identified in the Plan of Action have been developed in the Millennium Development Goals. The Africa Science and Technology Consolidated Plan of Action and the component of the NEPAD programme related to science and technology, capacity building, knowledge production and technological innovation have been consolidated. It is recognised that space applications provide an essential platform for Africa's regional and continental sustainable development and can assist in tackling the problems Africa is facing including monitoring climate change

Regarding Priority Action 3, the objective is to enhance cooperation in the use of space applications and technology to support Africa's sustainable development objectives. The expected outcome is to fully integrate space related issues in specific dialogues and cooperation initiatives in areas such as environment and resource management, climate change, peace and security. It expected to create specific projects based on space technologies to achieve regional and global development goals. The activities foreseen in the Action Plan are:

- Jointly analyse the potential to use space applications to better manage natural resources, improve living conditions of populations and promote sustainable development, in particular in the areas of:
  - Telecommunications and bridging the digital divide;
  - Monitoring of climate change, desertification or fires, and water and food resources through Earth Observation, such as Global Monitoring for Environment and Security (GMES) for Africa;
  - Navigation applications such as GALILEO, the European Geostationary Navigation Overlay Service (EGNOS);
  - Facilitating humanitarian aid operations and improving security of populations through integrated space applications,
  - Space technologies and scientific applications as contributions to the knowledge based society.
- Ensure effective follow-up to the event “Space for Development - the Case of GMES and Africa”;
- Develop concrete joint cooperation initiatives in selected areas.

The actors foreseen in these activities were:

- AU Commission/NEPAD, African States;
- European Commission, EU Member States;
- European agencies such as the European Space Agency (ESA) and EUMETSAT;
- Private Sector.

Area 6: Climate Change

This area of the First Acton Plan focuses on climate change and space systems can contribute mainly in:

- promoting climate observation, in particular for the African continent, and enhance links to global climate observatory systems; and

- strengthening climate-monitoring and forecasting capacities.

The implementation of African Monitoring of Environment and Sustainable Development (AMESD) (€20 million – 9th European Development Fund, implemented by AUC) is recognised as one of the achievement of the Partnership 6.

The ClimDev Africa programme, which also recognises the usefulness of space system, is one of the major programmes implemented within Partnership 6. (See details of projects in Section 8.1).

Second Action Plan 2011-2013

During the 14th Africa-EU Ministerial Meeting that took place in Luxembourg on 26 April 2010 within the framework of the Africa-EU Dialogue, the general options for the Action Plan 2011-2013 were discussed[6]. This plan will be endorsed at the 3rd Africa-EU Summit that will take place on 29-30 November 2010 in Libya.

It is expected that the current areas of cooperation of the eight thematic partnerships will remain unchanged and the Second Action Plan will be along the same lines as the First Action Plan. Nevertheless, the Second Action Plan should be more focused, realistic, manageable, and effective. The primary focus should be on activities that[6]:

- Have a clear regional, continental, or global dimension and adequate steps should be taken to ensure coordination and coherence among activities at the continental, regional and national levels.
- Have a clear added value, are focused and streamlined, and reinforce complementarities and coherence with existing initiatives and fora, and align with African strategic priorities, their structures and their mechanisms (for example Ministerial Conferences, the PIDA program in infrastructure, etc.).
- Have a proven buy-in of a critical mass of competent actors on both sides, including the necessary political, human and financial resources.

The Ministerial Meeting foresees that the implementation should be improved through the following:

- enhanced political and policy dialogue;
- enhanced involvement of Joint Africa-EU Strategy Actors;
- Joint Expert Groups (JEG);
- resources;
- communication.

It was agreed that the involvement of Africa's regional and sub-regional organisations (REC's) is essential for the success of the Joint Strategy. In the next phase the REC's will be invited to come forward with proposals for tracking the implementation of the Action Plan. The prioritisation of regional and continental activities and effective communication is important. The AUC is expected to take concrete steps to engage the involvement of the REC's and create the conditions for the involvement of all stakeholders and identify and ensure regular participation of core experts in meetings to guarantee a critical mass of competent actors from Africa and Europe. The JEG's are expected to be reformed in order to make links with national and regional levels, as well as existing structures in thematic areas and to better link implementation to existing structures/bodies that could be of use in each partnership. Regarding resources including funding and human and technical resources, it is recognised that this is essential for the implementation of the technical cooperation. In the Joint Strategy, Africa and Europe have agreed to make efforts to build together synergies also via the progressive establishment of a Pan-African financial support programme. Better utilisation of existing instruments is needed and the need to explore innovative financial agreements including pool funding when necessary is recognised. Furthermore, it was recognised that it is important to increase the visibility of the Joint Africa-EU partnership in Africa, the EU and internationally.

[6] Council of the European Union, "14th Africa- EU Ministerial Meeting", Luxembourg, 26th April, 9041/10 (Press 92).

3. African, European and International Actors

The main actors operating in the African continent can be categorised as African actors, European actors, United Nations and private initiatives. In Figure 3 the various actors operating in Africa can be seen. The African Actors can be categorised into those that operate in Africa on a continental level, regional organisations, technical organisations and African Space Agencies. In Europe the European Commission directly cooperates with the African Union on various levels. At the same time many of the European Member States have direct cooperation agreements and memorandums of understanding with African actors and states. Other organisations residing in Europe, such as the European Space Agency and EUMETSAT, are also active in Africa in their respective fields of expertise. The United Nations has numerous agencies involved in activities in Africa and many of them use satellite applications as a means to achieve their goals. For many years the private sector also has shifted its attention to Africa as it offers various business opportunities. The various actors will be described in the following subsections.

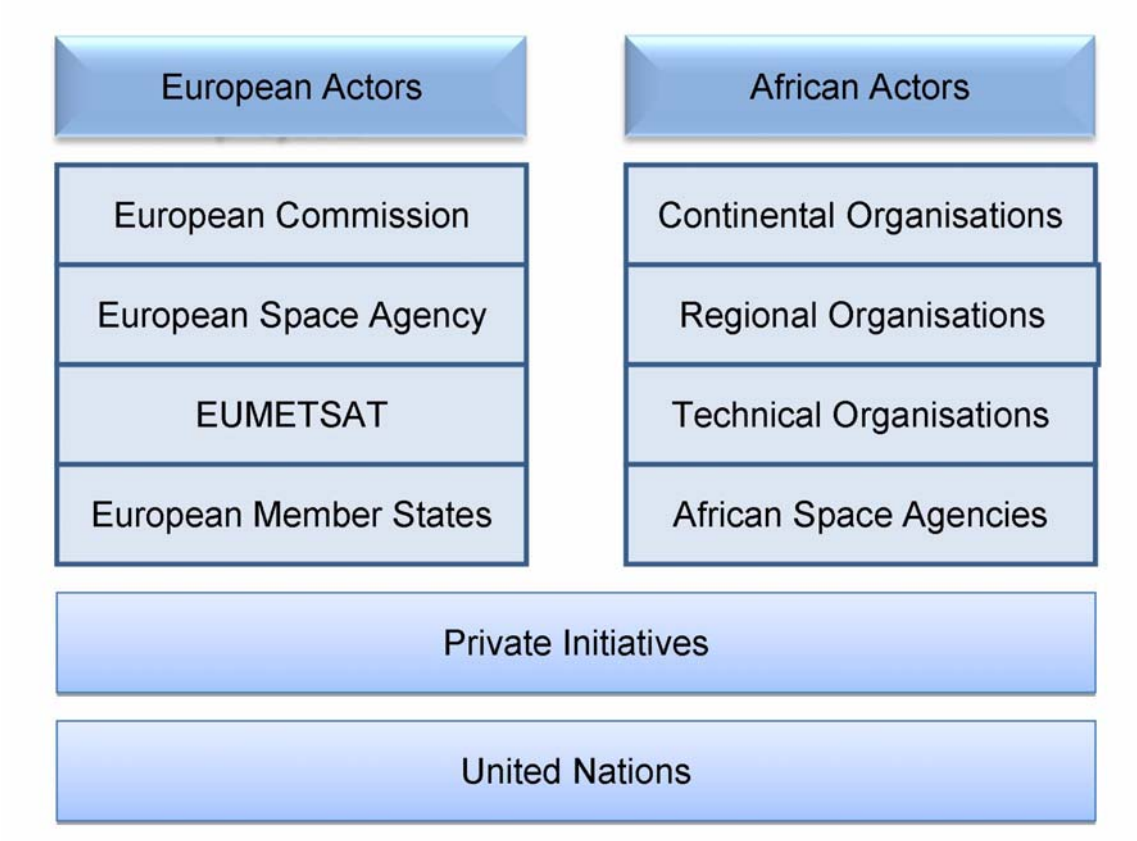


Figure 3: African, European and International Actors involved in African satellite application projects.

4. African Actors

The African actors include organisations that operate at the continental level such as the African Union, Regional Organisations typically the Regional Economic Communities (RECs), technical organisations such as specialised agencies and institutes in different areas and the African space agencies. Figure 4 shows the various policy relevant actors in Africa and the matrix relationship for cooperation between them.

At the continental level the African Union (AU), the United Nations Economic Commission for Africa (UNECA) and the African Development Bank (ADB) have a joint secretariat. These three institutions have distinct roles: the AU has the political mandate, the ADB has the financial mandate and the ECA is responsible for economic issues. The three institutions largely operate independently with their own initiatives. The role of the secretariat is to assist in coordinating their actions and see how best to advocate for common policy positions.

Various technical institutions that have been created often act as the technical executive for various projects under the auspices of UNECA. They cooperate in a matrix like relationship with UNECA and the AU and the RECS. These institutions are the Centre for Training in Aerospace Surveys (RECTAS), the Regional Center for Mapping of Resources for Development (RCMRD), the African Organization of Cartography and Remote Sensing (AOCRS) and the Centre of Meteorological Application for Development (ACMAD).

There are numerous regional blocks in Africa, also known as Regional Economic Communities, which have been established by various African countries to facilitate mutual economic development amongst various African states. Some of the REC's have become pillars of the African Union and others have not. The CEN-SAD, COMESA, EAC, ECCAS, ECOWAS, IGAD, SADC and UMA are pillars of

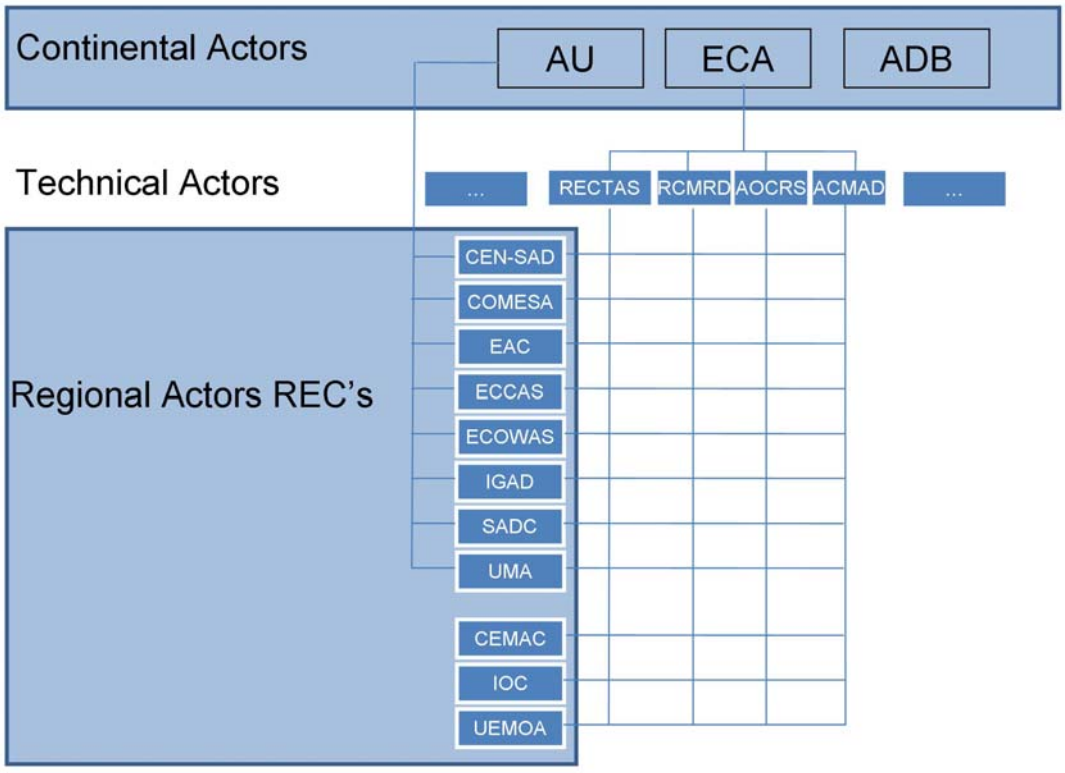


Figure 4: Policy relevant actors in Africa.

the African Economic Community and organisation of the African Union. In Figure 5 the REC pillars of the African Economic Community (AEC) and their respective African member states can be seen. The other RECs namely CEMAC, IOC and UEMOA have not become part of the AU umbrella but are fully functional and active in Africa. There are some overlaps in many of the RECs in membership from the African states as well as in activities.

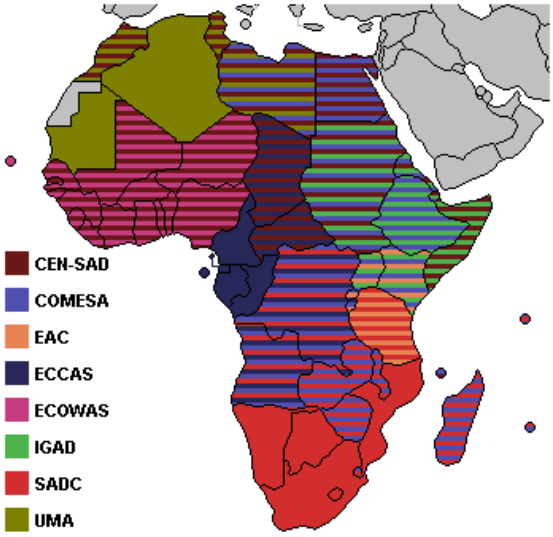


Figure 5: Regional Economic Communities (RECs) which are part of the African Economic Community (AEC) and their African member states.

4.1 African Union (AU)

Overview

The African Union is an intergovernmental organisation with 53 member states and six official languages. The only African state that is not a member is Morocco. The AU was established on the 9th of July 2002 as the successor of the Organisation of African Unity (OAU). Among the objectives of the AU are: to accelerate the political and socioeconomic integration of the continent; to promote and defend African common positions on issues of interest to the continent and its people; to achieve peace and security in Africa; and to promote democratic institutions, good governance and human rights. The AU aims at establishing an African Economic Community with a single currency, the "Afro", by 2023. The AU consists of various bodies and the most important decisions are made by the Assembly of the AU during a semi-annual meeting of the Heads of States of its member states. The AU Secretariat and the African Union Commission (AUC) are located in Addis Ababa in Ethiopia.

Historical Background

The Organisation of African Unity (OAU) was the predecessor to the AU. It was established on 25 May 1963 in Addis Ababa by 32 governments that signed the OAU Charter. The main objectives of the OAU were, inter alia, to rid the continent of the remaining vestiges of colonization and apartheid; to promote unity and solidarity among African states; to coordinate and intensify cooperation for development; to safeguard the sovereignty and territorial integrity of Member States and to promote international cooperation within the frame work of the United Nations. Over the years 21 states joined gradually and in 1994 South Africa become the 53rd member.

Four Summits led to the establishment of the AU on 9 July 2002. The first Summit held in September 1999, adopted the Sirte Declaration with a view, inter alia, to accelerating the process of integration in the continent to enable it to play its rightful role in the global economy while addressing multifaceted social, economic and political problems, com-

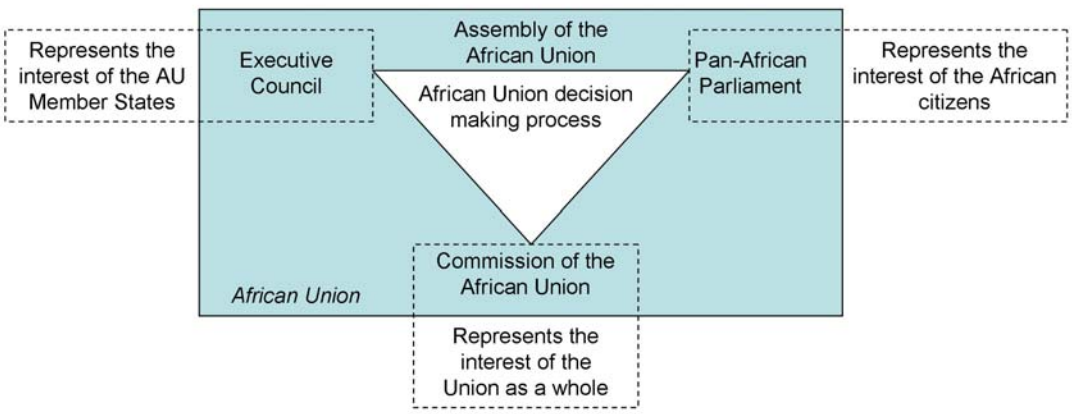


Figure 6: African Union decision making process

pounded as they are by certain negative aspects of globalisation. It was decided to establish an African Union in conformity with the objectives of the Charter of the continental organisation and the provisions of the Treaty establishing the African Economic Community. In particular, it decided to shorten the implementation periods of the Abuja Treaty; ensure the speedy establishment of all the institutions provided for in the Abuja Treaty; such as the African Central Bank, the African Monetary Union, the African Court of Justice and especially, the Pan-African Parliament; strengthen and consolidate the RECs as the pillars for achieving the objectives of the African Economic Community and realising the envisaged Union; and convene an African Ministerial Conference on Security, Stability, Development and Cooperation in the Continent, as soon as possible. In 2000, the Lome Summit adopted the Constitutive Act of the Union; the 2001 Lusaka Summit drew the road map for the implementation of the AU and, finally, in 2002 the Durban Summit launched the AU and convened the 1st Assembly of the Heads of State of the AU.

The Vision of the AU

The African Union has set for itself the ambition of building by the year 2030[7]: "A united and integrated Africa; an Africa imbued with the ideals of justice and peace; an interdependent and robust Africa determined to map for itself an ambitious strategy; an Africa underpinned by political, economic, social and cultural integration which would restore to Pan-Africanism its full meaning; an Africa able to make the best of its human and material resources, and keen to ensure the progress and prosperity of its citizens by taking advantage of the opportunities offered by a globalized world; an Africa capable of promoting its values in a world rich in its disparities."

This ambition is reflected in the following Vision: "... build an integrated Africa, a prosperous and peaceful Africa, driven by its own citizens and representing a dynamic force in the international arena".

The Objectives of the AU

The objectives of the African Union as contained in the Constitutive Act, Article 3, are to:

- Achieve greater unity and solidarity between the African countries and the peoples of Africa;
- Defend the sovereignty, territorial integrity and independence of its Member States;
- Accelerate the political and socio-economic integration of the continent;
- Promote and defend African common positions on issues of interest to the continent and its peoples;
- Encourage international cooperation, taking due account of the Charter of the United Nations and the Universal Declaration of Human Rights;
- Promote peace, security, and stability on the continent;
- Promote democratic principles and institutions, popular participation and good governance;
- Promote and protect human peoples' rights in accordance with the African Charter on Human and Peoples' Rights and other relevant human rights instruments;
- Establish the necessary conditions which enable the continent to play its rightful role in the global economy and in international negotiations;
- Promote sustainable development at the economic, social and cultural levels as well as the integration of African economies;
- Promote cooperation in all fields of human activity to raise the living standards of African peoples;
- Coordinate and harmonise the policies between the existing and future Regional Economic Communities for the gradual attainment of the objectives of the Union;
- Advance the development of the continent by promoting research in all fields, in particular in science and technology; and
- Work with relevant international partners in the eradication of preventable diseases and the promotion of good health on the continent.

Member States of the AU

The AU has 53 member countries, each holding the annually rotating presidency in turn. The member states are:

[7] "Strategic Plan of the African Union Commission", Volume 1: Vision and Mission of the African Union, May 2004, African Union.

Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Comoros, Côte d'Ivoire, Democratic Republic of the Congo, Republic of the Congo, Djibouti, Egypt, Equatorial Guinea, Ethiopia, Gabon, Gambia, Ghana, Guinea-Bissau, Kenya, Lesotho, Liberia, Libya, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Nigeria,Rwanda, São Tomé and Príncipe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, Sudan, Swaziland, Tanzania, Togo, Tunisia, Uganda, Sahrawi Arab Democratic Republic, Zambia, Zimbabwe.

A former member of the African Union is Morocco which left the AU's predecessor OAU in 1984 after the majority of member states granted membership in the OAU to the Saharan Arab Democratic Republic set up by the independence movement in the disputed territory of Western Sahara.

Languages of the AU

According to the Constitutive Act of the African Union, its working languages are Arabic, English, French, and Portuguese, as well as Afrikaans. In 2003 a protocol amending the Constitutive Act added Spanish, Swahili and "any other African language" and termed all six "official" languages of the African Union.

Organisation of the AU

According to the Constitutive Act the organs of the African Union are as follow:

- The *Assembly of the Union*, the supreme organ of the Union, is composed of Heads of State and Government or their duly accredited representatives. This organ meets at least once in extraordinary session.
- The *Executive Council*, composed of the Foreign Ministers or such other Ministers or representatives as are designated by the governments of Member States, is responsible for coordinating and taking decisions on policies in areas of common interest to Member States.
- The *Permanent Representatives' Committee*, composed of the Permanent Representatives or other Plenipotentiaries of Member States, is responsible for preparing the work of the Executive Council and acts on the latter's instructions.
- The *Pan-African Parliament* will be unicameral with representatives from all Parliaments of the countries of Africa. It will adopt legislation by two-thirds majority of its members. It will be composed of five members per country with at least one woman per country. It will subsequently evolve into a parliament elected by universal suffrage.
- The *African Court of Justice* will adjudicate in civil cases and be responsible for human rights protection and monitoring human rights violations. It will also constitute itself into a real criminal court in the long term.
- The *Commission of the African Union* is the Secretariat of the Union. It is composed of the Chairperson, his or her Deputy and Commissioners. It represents the Union and defends its interests under the direction of the Assembly and the Executive Council. It can initiate proposals for submission to the other organs of the Union and executes decisions taken by them. It assists Member States in executing the policies and programmes of the Union, particularly NEPAD[8] and CSSDCA[9]. It formulates common positions of the Union and coordinates the work of Member States during international negotiations.
- The *Economic, Social and Cultural Council (ECOSOCC)* is an advisory organ composed of different social and professional groups from Member States of the Union, particularly youth and women's associations. The ECOSOCC is assisted by CIDO (African Citizens Directorate, the AU Commission department in charge of relations with civil society and the diaspora).
- The *African Court of Human and People's Rights* is composed of 11 members elected by the Assembly of Heads of State and Government for a 6 year mandate. It has jurisdiction over all disputes and requests submitted to it in respect of interpretation and implementation of the African Charter on Human and People's Rights
- Three *Financial Institutions*: a) The African Central Bank; b) The African Monetary Fund; and c) The African Investment Bank. This integration process has the constant support of the African Development Bank and the United Nations Economic Commission for Africa.

[8] NEPAD: New Partnership for Africa's Development

[9] CSSDCA: Conference on Security, Stability, Development and Cooperation in Africa
Subsequent to the adoption of the Solemn Declaration on the CSSDCA, the NEPAD initiative has come into being as Africa's principal agenda for development, providing a holistic, comprehensive integrated strategic framework for the socio-economic development of the continent, within the institutional framework of the African Union.

- Specialized *Technical Committees* are composed of Ministers or senior officials responsible for the sectors falling within their respective areas of competence. Seven Technical Committees, the numbers and composition of which are not limited, are provided for in the Constitutive Act of the Union; namely:
  - The Committee on Rural Economy and Agriculture;
  - The Committee on Monetary and Financial Matters;
  - The Committee on Trade, Customs and Immigration Matters;
  - The Committee on Industry, Science and Technology, Energy, Natural Resources and the Environment;
  - The Committee on Transport, Communications and Tourism;
  - The Committee on Health, Labour and Social Affairs; and
  - The Committee on Education, Culture and Human Resources.
- The *Peace and Security Council* composed of 15 Member States is responsible for the promotion of peace, security and stability in Africa, preventive diplomacy and the restoration of peace. It is also responsible for disaster management and humanitarian activities. It replaces the Central Organ of the Conflict Prevention, Management and Resolution Mechanism established in 1993 by Heads of State at the Tunis Summit.

Contact Information

H.E. Dr. Jean PING
Tel: (251) 11 5514554 / (251) 11 551 77 00 ext 120
Fax:(251) 11 5513036
E-mail: chairperson@africa-union.org

African Union Headquarters
P.O. Box 3243
Roosevelt Street
(Old Airport Area)
W21K19
Addis Ababa
Ethiopia

Tel: (251) 11 551 77 00
Fax: (251) 11 551 78 44

Website: www.africa-union.org and www.au.int

4.2 The African Union Commission (AUC)

The African Union Commission is the Secretariat of the Union with executive functions. It represents the Union and protects its interests under the auspices of the Assembly of Heads of State and Government as well as the Executive Committee. The headquarters of the Commission are located in Addis Ababa, Ethiopia. The role of the AUC is similar to its counterpart in Europe - the European Commission of the European Union.

As the secretariat of the Union the AUC consists of a chairperson, his deputy and number of commissioners dealing with different policy areas. The leadership of the African Union Commission is:

- Chairperson: H.E Dr Jean Ping (Gabon)
- Deputy Chairperson: H.E. Erastus Jarnalese Onkundi Mwencha (Kenya)
- Commissioner for Political Affairs: Mrs. Julia Dolly Joiner (Gambia)
- Commissioner for Trade and Industry: H.E Mrs Elisabeth Tankeu (Cameroons)
- Commissioner for Peace and Security: H.E Mr. Ramtane Lamambra (Algeria)
- Commissioner for Human Resources, Science and Technology: H.E. Dr Jean Pierre Onvehoun Ezin(Benin)
- Commissioner for Rural Economy and Agriculture: H.E Mrs Rhoda Peace Tumusiime (Uganda)
- Commissioner for Infrastructure and Energy: H.E Dr. Elham Mahmoud Ahmed Ibrahim (Egypt)
- Commissioner for Economic Affairs: H.E. Dr. Maxwell M. Mkwezalamba (Malawi)
- Commissioner for Social Affairs: Advocate. Bience Philomina Gawanas (Namibia)

The functions of the Commission are, among others, to:

- Represent the Union and defends its interests under the guidance of and as mandated by the Assembly and the Executive Council;
- Initiate proposals to be submitted to the Organs as well as implement the decisions that are taken by the other Organs;
- Act as a custodian of the Constitutive Act, its protocols, the treaties, legal instruments, decisions adopted by the Union and those inherited from the OAU;
- Organise and manage the meetings of the organs;

- Assist Member States in implementing the Union programmes and policies, including CSSDCA and NEPAD;
- Coordinate and monitor the implementation of the decisions of other organs of the Union in close collaboration with the PRC and report regularly to the Executive Council;
- Work out draft common positions of the Union and coordinate the actions of Member States in international negotiations;
- Prepare the Union's Programme and Budget for approval by the policy organs as well as strategic plans and studies for the consideration of the Executive Council;
- Elaborate, promote, coordinate and harmonize the programmes and policies of the Union with those of the RECs;
- Ensure the mainstreaming of gender in all programmes and activities of the Union;
- Manage the assets and liabilities of the Union according to laid down regulations and procedures;
- Prepare strategic plans and studies for the consideration of the Executive Council;
- Take action in the domains of responsibility as may be delegated by the Assembly and the Executive Council. The domains include:
  - Control of pandemics;
  - Disaster management;
  - International crime and terrorism;
  - Environmental management;
  - Negotiations relating to external trade;
  - Negotiations relating to external debts;
  - Population, migration, refugee and displaced persons;
  - Food security;
  - Socio-economic integration; and
  - All other areas in which a common position has been established.
- Mobilize resources and devise appropriate strategies for self-financing, income generating activities and investment for the Union;
- Promote integration and socio-economic development;
- Strengthen cooperation and coordination of activities between Member States in fields of common interest;
- Prepare and submit an annual report on the activities of the Union to the Assembly, the Executive Council and the Parliament;
- Prepare the Staff Rules and Regulations for approval by the Assembly;
- Implement the decisions of the Assembly regarding the opening and closing down of sections and administrative or technical offices;
- Follow-up and ensure the application of the Rules of Procedure and Statutes of the organs of the Union;
- Negotiate, in consultation with the PRC, with the host countries, the Host Agreements of the Union and those of its administrative or technical offices;
- Build capacity for scientific research and development for enhancing socio-economic development in the Member States;
- Strive to promote and popularise the objectives of the Union;
- Coordinate the actions of the Members States during international negotiations as well as collect and disseminate information on the Union;
- Assist the Member States in ensuring the implementation of the various programs of the Union as well as the policies;
- Provide operational support to the PSC and ensure the promotion of peace, democracy, security and stability.

Seven missions defined for the Commission of the African Union translate the Vision of the African Union into concrete action:

- Mission 1: Building the capacity of integration players.
- Mission 2: Ensuring Overall Coherence of the programmes aimed at speeding up the integration process through concrete actions.
- Mission 3: Organizing Brainstorming and Instituting a Strategic Watch Tower on Key Issues Affecting the Future of the Continent
- Mission 4: Assuming a Dynamic Information and Advocacy Role for Africa vis-à-vis the World.
- Mission 5: Playing a Leadership Role for Promotion of Peace, Human Security and Good Governance in the Continent
- Mission 6: Stimulating Economic, Social and Cultural Development in the Continent.

- Mission 7: Establishing a Standing Follow-up/Evaluation Mechanism.

Contact Information

Ms. Esther Azaa Tankou
Senior Editorial Officer
Division of Communication and Information,
African Union Commission
Tel.Mob. +251 911 3611 85
Email: yamboue@africa-union.org

4.3 Special Programmes of the AU

Two special programmes of the African Union are the NEPAD and CIDO.

4.3.1 African Citizens' Directorate Civil Society and Diaspora (CIDO)

The African Citizens' Directorate Civil Society and Diaspora (CIDO) was established in 2005 to liaise with civil society on the continent, reach out to the African Diaspora, and function as an interim secretariat for ECOSOCC. CIDO is mandated to facilitate civil society contributions to the decision-making processes of the AU, including the Summits. It has also been instrumental in the establishment of the ECOSOCC.

4.3.2 New Partnership for Africa's Development (NEPAD)

NEPAD was created in July 2001, as a programme of the Organization of African Unity the predecessor of the African Union. NEPAD was given the task of providing a framework that would eradicate poverty; place African countries, both individually and collectively, on a path of sustainable growth and development; halt the marginalization of Africa in the globalization process and enhance its full and beneficial integration into the global economy; and accelerate the empowerment of women.

Some of the key areas for NEPAD to focus on were: policy reforms and increased investment in agriculture; human development with a focus on health, education, science and technology and skills development; building and improving infrastructure, including Information and Communication Technology (ICT), Energy, Transport, Water and Sanitation; promoting diversification of production and exports, particularly with respect to agro-industries, manufacturing, mining, mineral beneficiation and tourism; accelerating intra-African trade and improving access to markets of developed countries.

NEPAD is a very active organization. There are no space projects or programs in NEPAD, but space technologies could be useful in the sectors highlighted above.

4.4 Regional Economic Communities (RECs)

From the 1960's until the 1990's Africa witnessed the creation of various regional economic communities targeting economic cooperation and integration. Many were initially created and functioned for some years and then after decades were revised and re-established. Africa has the highest number of such cooperation and integration agreements. This process has been characterised by a variety of African state conglomerates with overlapping memberships mandates and programmes. There is no country in Africa that does not belong to at least one grouping; most belong to two or more of these groups.

Africa's regional integration was given a boost in 1991 by the adoption of the Abuja Treaty which established the African Economic Community (AEC) with the objective "to promote economic, social and cultural development and integration of African Economies in order to increase economic self-reliance and promote an endogenous and self-sustained development"[10]. The Abuja Treaty describes the relationship between the AEC and the RECs. In particular, in accordance with Article 88 of the Abuja Treaty, the foundation of the Treaty is that the African Economic Community must be established mainly through the coordination, harmonisation and progressive integration of the activities of the RECs. Nowadays, the AEC is a project of the African Union and thus Africa's regional integration groupings as they have evolved can be grouped into those that have become pillars of the African Union and those that are not. Nevertheless, all regional integration groups are factional and play an important role in the sustainable development of Africa. The REC's that have become pillars of the African Union are: CEN-SAD, COMESA, EAC, ECCAS, ECOWAS, IGAD, SADC, and UMA. The other REC's, often described as sub-regional economic communities (SECs), are: CEMAC, IOC, UEMOA, CEPGL, MRU, SACU, WAMZ.

[10] OAU, Abuja Treaty, Article 4.

Africa Union Regional Economic Communities

4.4.1 Community of Sahel-Saharan States (CEN-SAD)

The Community of Sahel-Saharan States (CEN-SAD) was established in 1998 during the Summit Meeting in Tripoli. The Treaty on the establishment of the Community was signed by the Leader of Great El-Fateh Revolution and the Heads of State of Burkina Faso, Mali, Niger, Chad and Sudan. The Central African Republic and Eritrea joined the Community during the first Summit of the organization held in Syrte in April 1999. Senegal, Djibouti and Gambia joined during the N'djamena Summit in February 2000. Others countries joined later, and still more are in the process of joining the Organization. On July 2000 it became a Regional Economic Community during the Conference of Heads of State and Government of the Organisation of African Unity held in Lomé in the Republic of Togo.

The objectives of CEN-SAD are:

- the establishment of a global Economic Union based on the implementation of a community development plan that complements the local development plans of member states and which comprises the various fields of a sustained socio-economic development: agriculture, industry, energy, social, culture, health…
- the removal of all restrictions hampering the integration of the member countries through the adoption of necessary measures to ensure:
  - Free movement of persons, capitals and interests of nationals of member States;
  - Right of establishment, ownership and exercise of economic activity;
  - Free trade and movement of goods, commodities and services from member states;
- the promotion of external trade through an investment policy in member states
- the increase of means of land, air and maritime transport and communications among member States and the execution of common projects
- the same right, advantages and obligations granted to their own citizens to nationals of the signatory countries in conformity with the provisions of their respective constitutions
- the harmonization of educational, pedagogical, scientific and cultural systems of the various cycles of education.

Programmes of CEN-SAD are the special food security programme, the great green wall programme and the visa exemption programme between the CEN-SAD Member States.

The list of CEN-SAD countries is: Republic of Benin, Burkina Faso, Central African Republic, Union of the Comoros, Republic of Côte d'Ivoire, Republic of Djibouti, Arab Republic of Egypt, State of Eritrea, Republic of The Gambia, Republic of Guinea, Republic of Guinea-Bissau, Republic of Ghana, Great Socialist People's Libyan Arab Jamahiriya, Republic of Liberia, Republic of Kenya, Republic of Mali, Islamic Republic of Mauritania, Kingdom of Morocco, Republic of Niger, Federal Republic of Nigeria, Republic of Senegal, Republic of Sierra Leone, Somali Republic, Democratic Republic of Sao Tome and Principe, Republic of Sudan, Republic of Chad, Togolese Republic, Tunisian Republic.

Contact Information

CEN-SAD Secrétariat
P.O.Box 4041
Place d'Algeria

Tel: (00218) 21 3614832, (00218) 21 3614832
Fax: (00218) 21 3343670, (00218) 21 333216
E-mail: censadsg@yahoo.com, info@cen-sad.org

http://www.cen-sad.org/

4.4.2 Common Market for Eastern and Southern Africa (COMESA)

The Common Market for Eastern and Southern Africa (COMESA) was established in 1993 as a successor of the Preferential Trade Area for Eastern and Southern Africa (PTA). COMESA formally succeeded the PTA in 1994. The idea of regional economic co-operation traces back to the mid 1960's. In 1981 the PTA was established by a treaty which came into force in 1982. The PTA was established to take advantage of a larger market size, to share the region's common heritage and destiny and to allow greater social and economic co-operation, with the ultimate objective being to create an economic community. The PTA Treaty envisaged its transformation into a Common Market and, in conformity with this, the Treaty establishing the Common Market for Eastern and Southern Africa, COMESA came into force.

COMESA's Vision is to "be a fully integrated, internationally competitive regional economic community with high standards of living for all its people ready to merge into an African Economic Community". Its mission is to "Endeavour to achieve sustainable economic and social progress in all Member States through increased co-operation and integration in all fields of development particularly in trade, customs and monetary affairs, transport, communication and information, technology, industry and energy, gender, agriculture, environment and natural resources". The Secretariat of COMESA has a specific Mission Statement as follows: "To provide excellent technical services to COMESA in order to facilitate the region's sustained development through economic integration".

COMESA's programmes have the following focus: achievement of zero tariffs for all tradable goods among COMESA members (free trade area status) by the year 2000 with continued development of COMESA as a common investment area, establishment of a common external tariff, CET, or customs union by the year 2004 (common market status), the implementation of programmes to enable free movement of services, labour (including right of establishment) and capital (leading to monetary union) between 2000 and 2025.

The core of COMESA's development integration agenda is:

- Trade Development with focus on formation of a free trade area 2000, formation of a customs union with a common external tariff, 2004, formation of a monetary union.
- Investment Development with focus on industry, agriculture, fisheries, livestock development, irrigation development, tourism, industrial and private sector support, and gender mainstreaming
- Infrastructure Development with focus on transport, communication and information, environment and energy
- Science and Technology (S&T) Development with focus on promotion of a supportive policy environment for increased investment in R&D by both the public and private sector, adaptation and transfer of new and emerging technologies, promotion of collaborative research and development, including commercialisation of R&D results, promotion of indigenous technology, promotion of capacity building, including literacy in S&T, promotion and strengthening of inter-firm linkages, in particular collaboration between local firms and scientific institutions and foreign multinational corporations.

The Member States of COMESA are: Burundi, Comoros, D.R. Congo, Djibouti, Egypt, Eritrea, Ethiopia, Kenya, Libya, Madagascar, Malawi, Mauritius, Rwanda, Seychelles,

Sudan, Swaziland, Uganda, Zambia, Zimbabwe.

Contact Information

Mr. Masiko Kachamba, Secretary General
Ben Bella Road
Lusaka
Zambia

Tel (+260-211) 229 725;
(+260-211) 229 725
Email: webmaster@comesa.int

http://www.comesa.int

4.4.3 East African Community (EAC)

The Permanent Tripartite Commission for East African Co-operation was first formed in 1967 as the East African Community. It collapsed in 1977 due to political differences. The Treaty for the Establishment of the East African Community was signed in Arusha on 30 November 1999. The Treaty entered into force on 7 July 2000 following the conclusion of the process of its ratification and deposit of the Instruments of Ratification with the Secretary General by all the three Partner States. The EAC was inaugurated in January 2001. The treaty was amended on 14 December 2006 and August 2007. The Treaty calls for a customs union (the framework of which was to be negotiated over the next four years), common market and monetary union and sets the ultimate objective as the birth of a political federation of east African states. Among the key institutions are an East African parliament, a regional stock exchange and a joint court of justice.

The objectives of the community according to the Treaty shall be to develop policies and programmes aimed at widening and deepening co-operation among the Partner States in political, economic, social and cultural fields, research and technology, defence, security and legal and judicial affairs, for their mutual benefit.

The Community shall ensure:

- the attainment of sustainable growth and development of the Partner States by the promotion of a more balanced and harmonious development of the Partner States;

- the strengthening and consolidation of cooperation in agreed fields that would lead to equitable economic development within the Partner States and which would in turn, raise the standard of living and improve the quality of life of their populations;
- the promotion of sustainable utilization of the natural resources of the Partner States and the taking of measures that would effectively protect the natural environment of the Partner States;
- the strengthening and consolidation of the long standing political, economic, social, cultural and traditional ties and associations between the peoples of the Partner States so as to promote a people centred mutual development of these ties and associations;
- the mainstreaming of gender in all its endeavours and the enhancement of the role of women in cultural, social, political, economic and technological development;
- the promotion of peace, security, and stability within, and good neighbourliness among, the Partner States;
- the enhancement and strengthening of partner ships with the private sector and civil society in order to achieve sustainable socioeconomic and political development; and
- the undertaking of such other activities calculated to further the objectives of the Community, as the Partner States may from time to time decide to undertake in common.

Over the years, EAC has laid the foundation for strategic partnership with the rest of the world and benefited from technical and financial assistance from a large number of international aid and development agencies, including the World Bank, European Union, European Investment Bank, African Development Bank, Swedish International Development Co-operation Agency (SIDA), United Nations Economic Commission for Africa, German Agency for Technical Co-operation (GTZ), East African Development Bank, United Kingdom Department for International Development (DFID), Norwegian Agency for Development (NORAD), Commonwealth Secretariat and Danish International Development Agency (DANIDA). Other development partners with whom EAC expects to enter into co-operation programmes or who have entered into Memoranda of Understanding include France, India, Finland, Canada, Sweden, Norway, International Labour Organisation, UN Food and Agriculture Organisation, Austrian Development Co-operation and the World Meteorological Organization. Being a new organization, EAC does not have any space project.

The Partner States of the EAC are Kenya, Uganda and Tanzania.

Contact Information

East African Community
Arusha International Conference Centre
5th Floor Kilimanjaro Wing
P.O. Box 1096
Arusha
Tanzania

Tel: +255 27 2504253/8
Fax: +255 27 2504255
Email: eac@eachq.org

http://www.eac.int

4.4.4 Economic Community of Central Africa States (ECCAS)

At a summit meeting in December 1981, the leaders of the Central African Customs and Economic Union (UDEAC) agreed in principle to form a wider economic community of Central African states. CEEAC/ECCAS was established on 18 October 1983 by the UDEAC members and the members of the Economic Community of the Great Lakes States (CEPGL) (Burundi, Rwanda and the then Zaire) as well as Sao Tome and Principe. Angola remained an observer until 1999, when it became a full member. ECCAS began functioning in 1985, but has been inactive since 1992 because of financial difficulties (non-payment of membership fees) and the conflict in the Great Lakes area. The war in the DRC has been particularly divisive, as Rwanda and Angola fought on opposing sides.

CEEAC has been designated a pillar of the AEC, but formal contact between the AEC and CEEAC was only established in October 1999 due to the inactivity of CEEAC since 1992 (CEEAC signed the Protocol on Relations between the AEC and the RECs in October 1999). The AEC again confirmed the importance of CEEAC as the major economic community in Central Africa at the third preparatory meeting of its Economic and Social Council (ECOSOC) in June 1999.

CEEAC aims to achieve collective autonomy, raise the standard of living of its populations and maintain economic stability through harmonious cooperation. Its ultimate goal is to establish a Central African Common Market.

At the Malabo Heads of State and Government Conference in 1999, four priority fields for the organization were identified:

- to develop capacities to maintain peace, security and stability, which are essential prerequisites for economic and social development;
- to develop physical, economic and monetary integration;
- to develop a culture of human integration [sic]; and
- to establish an autonomous financing mechanism for ECCAS.

The member states of ECCEAS are: Angola, Burundi, Cameroon, Central African Republic, Chad Congo (Brazzaville), Democratic Republic of Congo, Equatorial Guinea, Gabon, Rwanda, Sao Tome and Principe

Contact Information

Luís Silva Nguema
Secretary General of the Economic Community of Central African State (ECCAS),
CEEAC, BP 2112
Libreville, Gabon
Tel. +241-73 3547/8

www.ceeac.org (website not functional)

4.4.5 Economic Community of West African States (ECOWAS)

On 28 May 1975, 15 West African Countries signed the Treaty of Lagos for the creation of an Economic Community of Western African States. The protocols launching ECOWAS were signed on 5 November 1976 in Lomé, Togo. The objective of ECOWAS was to provide a forum for economic cooperation and to eliminate mistrust between its members. In July 1993, a revised ECOWAS Treaty designed to accelerate economic integration and to increase political co-operation, was signed.

The revised treaty of 1993, which was to extend economic and political co-operation among member states, designates the achievement of a common market and a single currency as economic objectives, while in the political sphere it provides for a West African parliament, an Economic and Social Council and an ECOWAS Court of Justice to replace the existing Tribunal and enforce Community decisions. The treaty also formally assigned the Community with the responsibility of preventing and settling regional conflicts.

The mission of ECOWAS was to promote economic integration in "all fields of economic activity, particularly industry, transport, telecommunications, energy, agriculture, natural resources, commerce, monetary and financial questions, social and cultural matters."

Over the years it aims to promote co-operation and integration in economic, social and cultural activity, ultimately leading to the establishment of an economic and monetary union through the total integration of the national economies of member states. It also aims to raise the living standards of its peoples, maintain and enhance economic stability, foster relations among member states and contribute to the progress and development of the African Continent. The community is financed by a community tax of 0.5% levied on all goods from third countries.

There exists a health initiative with the European Union in collaboration with ECOWAS called the West African Regional Programme (PRSAO). The objective of the program is to strengthen regional integration in West Africa through the coordination and the harmonization of health policies, the facilitation of free movement of populations, goods, services and technologies and to contribute to the improvement of the performance of health systems in areas of common interest. ECOWAS does not have any space project.

It is composed of 15 Member States- Nigeria, Niger, Republic of Benin, Burkina Faso, Liberia, Cape Verde, Mali, Sierra Leone, Togo, Guinea Bissau, Cote D'Ivoire, The Gambia, Senegal, Ghana, and Guinea.

Contact Information

Tel: (234) (9) 31 47 647-9
Fax: (234) (9) 31 43 005, 31 47 646
Email: info@ecowas.int

Website: www.ecowas.int

4.4.6 Intergovernmental Authority on Development (IGAD)

The recurring and severe droughts and other natural disasters between 1974 and 1984 which caused widespread famine, ecological degradation and economic hardship in the Eastern Africa region, led to the creation of the Intergovernmental Authority on Drought and Development (IGADD) in 1986. IGADD was replaced by the Intergovernmental Authority on Development (IGAD) in 1996.

The IGAD mission is to assist and complement the efforts of the Member States to achieve, through increased cooperation:

- Food security and environmental protection
- Promotion and maintenance of peace and security and humanitarian affairs, and,

- Economic cooperation and integration.

The objectives of IGAD are to:

- Promote joint development strategies and gradually harmonize macro-economic policies and programmes in the social, technological and scientific fields;
- Harmonize policies with regard to trade, customs, transport, communications, agriculture, and natural resources, and promote free movement of goods, services, and people within the region.
- Create an enabling environment for foreign, cross-border and domestic trade and investment;
- Achieve regional food security and encourage and assist efforts of Member States to collectively combat drought and other natural and man-made disasters and their natural consequences;
- Initiate and promote programmes and projects to achieve regional food security and sustainable development of natural resources and environment protection, and encourage and assist efforts of Member States to collectively combat drought and other natural and man-made disasters and their consequences;
- Develop and improve a coordinated and complementary infrastructure, in the areas of transport, telecommunications and energy in the region;
- Promote peace and stability in the region and create mechanisms within the region for the prevention, management and resolution of inter- and intra-state conflicts through dialogue;
- Mobilize resources for the implementation of emergency, short-term, medium-term and long-term programmes within the framework of regional cooperation;
- Promote and realize the objectives of the Common Market for Eastern and Southern Africa (COMESA) and the African Economic Community;
- Facilitate, promote and strengthen cooperation in research development and application in science and technology.

IGAD neither has a space project nor a space program. However space technologies will go a long way in helping IGAD achieve its missions.

IGAD is comprised of 7 East African States Djibouti, Ethiopia, Kenya, Somalia, Sudan, Uganda, and Eritrea.

Contact Information

IGAD Secretariat
Avenue Georges Clemenceau
P.O. Box 2653 Djibouti
Republic of Djibouti

Telephone +253-354050
Fax: +253-356994

Website: http://www.igad.int/

4.4.7 South African Development Community (SADC)

The South African Development Community (SADC) was founded in 1980, with the aim of achieving development and economic growth, alleviating poverty, enhancing the standard and quality of life of the peoples of Southern Africa and support the socially disadvantaged through regional integration, promoting and defending peace and security, promoting self-sustaining development on the basis of collective self-reliance, and the interdependence of Member States; achieving sustainable utilisation of natural resources and effective protection of the environment.

It comprises 15 member states- Angola, Botswana, Democratic Republic of Congo, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Swaziland, Tanzania Zambia, and Zimbabwe. SADC headquarters are located in Gaborone, Botswana. SADC aims to

- Harmonise political and socio-economic policies and plans of Member States;
- Mobilise the peoples of the region and their institutions to take initiatives to develop economic, social and cultural ties across the region, and to participate fully in the implementation of the programmes and projects of SADC;
- Create appropriate institutions and mechanisms for the mobilisation of requisite resources for the implementation of the programmes and operations of SADC and its institutions;
- Develop policies aimed at the progressive elimination of obstacles to free movement of capital and labour, goods and services, and of the peoples of the region generally within Member States;
- Promote the development of human resources;
- Promote the development, transfer and mastery of technology;
- Improve economic management and performance through regional cooperation;

- Promote the coordination and harmonisation of the international relations of Member States;
- Secure international understanding, co-operation and support, mobilise the inflow of public and private resources into the region; and
- Develop such other activities as Member States may decide in furtherance of the objectives of SADC.

The SADC Common Agenda is based on various principles, such as development orientation; subsidiarity; market integration and development; facilitation and promotion of trade and investment and variable geometry. The SADC Common Agenda includes:

- the promotion of sustainable and equitable economic growth and socio-economic development that will ensure poverty alleviation with the ultimate objective of its eradication;
- promotion of common political values, systems and other shared values which are transmitted through institutions which are democratic, legitimate and effective; and
- the consolidation and maintenance of democracy, peace and security.

SADC runs projects on Combating HIV&AIDS, environment and sustainable development, gender equality and development, information & communication technologies, poverty eradication, science and technology, statistics. SADC has no space project however; space technologies could be useful in achieving the aim of sustainable utilisation of natural resources, and the effective protection of the environment.

Contact Information

Telephone +2673951863
Fax+2673972848
Email: registry@sadc.int

Website: www.sadc.int

4.4.8 Union du Maghreb Arabe (UMA)

The Arab Maghreb Union (Union du Maghreb Arabe – UMA) was formed in 1989 in Marrakech by a Treaty that was signed by the Heads of States of five countries (Algeria, Libya, Morocco and Tunisia). The history of its creation dates back to1964 with the first conference of Maghreb Economic Ministers in Tunis, which established the Conseil Permanent Consultatif du Maghreb (CPCM) between Algeria, Libya, Morocco and Tunisia. The aim was to coordinate and harmonise the development plans of the four countries, their international trade and relationships with the European Union.

The main objectives of the AMU Treaty are to strengthen all forms of ties among Member States (in order to ensure regional stability and enhance policy coordination), as well as to introduce gradually free circulation of goods, services, and factors of production among them. Common defence and non-interference in the domestic affairs of the partners are also key aspects of the Treaty. The Treaty highlights the broad economic strategy to be followed, namely, the development of agriculture, industry, commerce, food security, and the setting up of joint projects and general economic cooperation programs. Finally, the agreement provides the possibility for other Arab and African countries to join the Union at a later stage.

The AMU had been in a dormant status for a while but recently significant efforts have been made to revive its activities.

AMU Secretariat
14 Rue Zalagh
Rabat-Agdal
Morocco

Phone: +212-37-671 274/78/80/85
Fax:+212-37-671 253

www.maghrebarabe.org

Other Regional Economic Communities

4.4.9 Communauté Économique et Monétaire de l'Afrique Centrale (CEMAC)

In 1964 in Brazzaville, the states of Central Africa, Congo, Gabon and Chad signed the Treaty establishing the Customs and Economic Union of Central Africa (UDEAC). After the 1980-1990 successive economic crises, it seemed essential to re-launch the economic and social integration by signing in 1994 in N'Djamena, Chad, the Treaty to establish the Economic and Monetary Community of Central Africa (CEMAC). CEMAC replaced UDEAC, which was suspended in 1999. The primary mission of CEMAC is to promote the harmonious development of Member States in the context of establishing a genuine common market.

The objectives of CEMAC and its Member States reaffirm their commitment to the principles of freedom and democracy and their commitment to expand all human and natural resources for the welfare of their peoples in all areas; provide a new and decisive impulse for the integration process in Central Africa through greater harmonisation of policies and

laws of their states; and ensure the convergence of their economic policies.

CEMAS' objectives are to:

- Establish a system of multilateral surveillance of economic policies of member states;
- Ensure stable management of the common currency;
- Secure the environment of economic activities and business in general;
- Harmonize regulations of sectoral policies in key areas required by the organizational instruments (agriculture, livestock, fisheries, industry, trade, transportation and telecommunications, energy and environmental research, education and vocational training.)
- Create a common market based on the free movement of persons, goods, capital and services.

The Member States of CEMAS are Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea and Gabon.

Contact Information

Jean Nkuete Executive Secretary
CEMAC Building,
Avenue des Martyrs
BP 969 Bangui

Tel : +236 61 18 85
+236 61 21 79
+236 61 47 81
+236 61 65 84
+236 61 21 35
Email: secema@hotmail.com

http://www.cemac.int/

4.4.10 Indian Ocean Commission (IOC)

The Indian Ocean Commission was established in 1984 by the General Agreement of Victoria (Seychelles). Its objective is to promote sustainable development of the Western Indian Ocean Islands. Its headquarters are located in Quatre Bornes in Mauritius. It is mainly financed by the European Development Fund (EDF). It executes and implements projects financed by EDF and others such as the World Bank, African Development Bank, French Cooperation, UNESCO and UNAIDS.

The objectives and missions of IOC are primarily to strengthen links between the peoples of its member states and improve their standards of living, promoting cooperation in a number of areas: diplomacy, economy, trade, agriculture, fishing, the conservation of resources and ecosystems, culture, science and education.

Ten projects are currently underway, focused on four strategic priorities adopted at the 2005 Summit that define the mandates of the IOC and are under the following axes:

- Political and diplomatic cooperation to support regional stability and good governance. Participation in political and strategic dialogue between the member states
- Economic and trade cooperation to insert the island economies into the regional and global economy.
- Sustainable regional development focuses in part on the protection of people and improving their quality of life and also the preservation of terrestrial and marine natural resources.
- Strengthening of regional identity and bringing people who are needed, by the intensification of cultural exchanges and networking professionals and organizations on issues related to youth, gender, training and regional mobility

The IOC includes five member states: Comoros, France (Réunion Island), Madagascar, Mauritius and Seychelles.

Contact Information

Secretary General
Callistus D'Offay
Q4, Avenue Sir Guy Forget
Quatre Bornes
Mauritius

Tel: +230 427 3366 or +230 427 6172 or +230 427 6199
Fax: +230 425 2709 / 425 2487
Email: secretariat@coi-ioc.org

http://www.coi-ioc.org

4.4.11 Union Economique et Monétaire Ouest Africaine (UEMOA)

The Union Economique et Monétaire Ouest Africaine (UEMOA) was established by Treaty signed on 10 January 1994 in Dakar by the Heads of State of seven countries in West Africa with the idea of creating a common currency. The Treaty entered into force on 1August 1994, after ratification by the Member States.

The objectives of UEMOA were:

- to enhance the competitiveness of economic and financial activities of the Member States through an open and

competitive environment and a rationalised and harmonised legal market;

- to ensure convergence of economic performance and policies of the Member States by establishing a multilateral surveillance procedure;
- to create among Member States a common market based on the free movement of persons, goods, services, capital and right of establishment of persons, freelance or employed, as well as a common external tariff policy in commerce;
- to coordinate national policies in different sectors by the implementation of joint actions and common policies including human resources, planning, agriculture, energy, industry, mining, transport, infrastructure and telecommunications;
- to harmonise the legislation of Member States and particularly the system of taxation in favour of the proper functioning of the common market.

The UEMOA has close cooperation with the European Union and has developed active cooperation with ECOWAS, the Permanent Interstate Committee Fight against Drought in the Sahel (CILSS), the Economic and Monetary Community of Central Africa (CEMAC), with whom it manages the site www.izf.net through the promotion of investment in the Franc Zone, the Common Organisation of Cooperation Against Major Endemic Diseases (OCCGE), the Organization for the Harmonization of Business Law in Africa (OHADA), the Organization of African Unity (OAU), the African Development Bank (ADB) and the United Nations Economic Commission for Africa (ECA). It has recently strengthened its cooperation with the Organization of the United Nations Food and Agriculture Organization (FAO), the Islamic Development Bank (IDB), the International Fund for Agricultural Development (IFAD), the International Organization of La Francophonie, the World Bank and the International Monetary Fund (IMF). UEMOA has strengthened its cooperation, bilaterally, with the United States of America, through the signing of an Agreement on Trade Development and Investment and with France, with the signing of a financing agreement under the Priority Solidarity Fund (FSP).

The countries that signed the initial agreement were Benin, Burkina Faso, Côte d'Ivoire, Mali, Niger, Senegal and Togo. On 2 May, 1997, Guinea-Bissau became the eighth member of the UEMOA.

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Contact Information

Mr. Soumaila Cisse
President of the UEMA Commission
Avenue du Professeur Joseph Ki-Zerbo,
01 BP 543 Ouagadougou 01
Burkina Faso

Tel.: + 22650 31 88 73 / 76
Fax: + 22650 31 88 72
Email: commission@uemoa.int Ainsi une Convention de financement, d'un montant de quatorze millions quatre

http://www.uemoa.int/

4.4.12 Mano River Union (MRU)

The Mano River Union (MRU) was established in 1973 as a custom union between Liberia and Sierra Leone with the aim to foster economic cooperation among the partner countries, to promote trade and a unified tariff and customs policy. In 1980 Guinea joined the Union. The MRU can be regarded as the first attempt at close regional cooperation and economic integration in the Mano River West Africa. Unfortunately, due to political instabilities in the region the objectives of the Union could not be achieved. As there were conflicts between the two founding members of the union, it became dormant for two decades. However, it was reactivated on May 2004 at a Summit with the leaders of the three member states. Since then, there have been several high level meetings. In 2008 Cote d'Ivoire joined the Union. The current chair of the union is the Liberian MRU Chair President Ellen Johnson Sirleaf.

MRU's development partners include the United Nations Population Fund (UNFPA), the United Nations Development Program (UNDP), the Asia Development Bank/Asia Development Fund (ADB/ADF), the Economic Community of West Africa (ECOWAS), the African Capacity Building Foundation (ACBF), the African Union (AU), the United Nations (UN), the European Union (EU), the United States Agency for International Development (USAID), and the Joint United Nations Program on HIV/AIDS (UNAIDS).

Contact Information

Not available.

4.4.13 Communauté Economique de Pays des Grands Lacs (CEPGL)

The Communauté Economique de Pays des Grands Lacs (CEPGL) -Great lakes River Basin- was established in 1976 with the aim of economic integration and to facilitate the movement of goods and people between the countries of the Great lakes region. The CEPGL project was suspended following major political problems in the region. In 2004Louis Michel, Foreign Minister of Belgium invited foreign ministers of DR Congo, Rwanda and Burundi in the Palais d'Egmont in Brussels to discuss the details of a possible revival of the CEPGL. Since then there have been various attempts related to the revival of CEPGL. The CEPGL is in Gisenyi in Rwanda. The member states are Burundi, DR Congo, and Rwanda. The Community operates through four specialised agencies:

- L'Institut de Recherche Agronomique et Zootechnique (IRAZ). Its primary mission is to conduct research in agriculture and livestock and to participate in the execution of community projects.
- La Banque de Dévelopement des Etats des Grands Lacs (BDEGL). Its main objective is to mobilize financial resources to finance projects whose purpose is to pursue economic integration and development in the region.
- La Société Internationale pour l'Electricité des Grands Lacs (SINELAC). Its mission is the operation of the hydroelectric Community Ruzizi II and the marketing of energy to the three member countries through their national societies of electricity, namely the National Society Electricité (SNEL) for Congo and Burundi and of the REGIDESO ELECTROGAZ for Rwanda.
- L'Organisation de la CEPGL pour l'Energie des Grands Lacs (EGL). Its primary mission is to ensure cooperation between Member States in the field of energy. Its role is that of a planning body, conductor of studies and project implementation.

Contact Information

Not available.

4.4.14 Southern Africa Customs Union (SACU)

The Southern Africa Customs Union is the oldest customs union dating back to the 1889 Customs Union Convention between the British Colony of Cape of Good Hope and the Orange Free State Boer Republic. In 1920 a new agreement was signed, extending SACU to the Union of South Africa and the British High Commission Territories. The primary goal was to promote economic development through regional coordination of trade. The Southern African Customs Union came into existence on 11 December 1969 with the signature of the Customs Union Agreement between South Africa, Botswana, Lesotho, Namibia and Swaziland. It entered into force on1 March 1970, thereby replacing the Customs Union Agreement of 1910. In 2002 the latest SACU agreement was signed. The objectives of SACU as contained in Article 2 of the 2002 SACU Agreement are:

- to facilitate the cross-border movement of goods between the territories of the Member States;
- to create effective, transparent and democratic institutions which will ensure equitable trade benefits to Member States;
- To promote conditions of fair competition in the Common Customs Area;
- to substantially increase investment opportunities in the Common Customs Area;
- to enhance the economic development, diversification, industrialization and competitiveness of Member States;
- to promote the integration of Member States into the global economy through enhanced trade and investment;
- to facilitate the equitable sharing of revenue arising from customs, excise and additional duties levied by Member States; and
- to facilitate the development of common policies and strategies.

The vision of the Union is an economic community with equitable and sustainable development, dedicated to the welfare of its people for a common future. The SACU mission is:

- to serve as an engine for regional integration and development, industrial and economic diversification, the expansion of intra regional trade and investment, and global competitiveness;
- to build economic policy coherence, harmonization and convergence to meet the development needs of the region;
- to promote sustainable economic growth and development for employment creation and poverty reduction;
- to serve as a building block of an ever closer community among the peoples of Southern Africa;

- to develop common policies and strategies for areas such as trade facilitation; effective customs controls; and competition;
- to develop effective, transparent and democratic institutions and processes.

The member states of SACU are South Africa, Botswana, Lesotho, Namibia and Swaziland.

Contact Information

The Executive Secretary
Private Bag 13285
Windhoek
Namibia

Tel: (+264) 61 295 8000
Fax: (+264) 61 245 611
Email: info@sacu.int

Website: www.sacu.int

4.4.15 West African Monetary Zone (WAMZ)

The West African Monetary Zone (WAMZ) came into existence on 15 December, 2000 when the five countries, Gambia, Ghana, Guinea, Nigeria and Sierra Leone (also members of ECOWAS), signed the Articles of Agreement of the zone at the Second Summit of the WAMZ Heads of State and Government held in Bamako, Mali. On 20 April 2002, the Accra Declaration was signed, launching the Second Monetary Zone after ECOWAS, the West African Monetary Zone (WAMZ). The objective of the WAMZ is to establish a monetary union characterized by a common central bank and a single currency to replace the existing five national currencies. The group of five countries are planning to introduce a common currency, the Eco, by the year 2015.

Contact Information

Not available.

4.5 Centres, Institutes, Organisations, Associations and Conferences

4.5.1 Regional Centre for Training in Aerospace Surveys (RECTAS)

The Regional Centre for Training in Aerospace Surveys (RECTAS), which is located at the Obafemi Awolowo University (O.A.U) Campus, Ile-Ife, Osun State, Nigeria, was established in 1972 under the auspices of the UN Economic Commission for Africa (UNECA) with a mandate for Training, Research, Consultancy and Advisory Services in Geoinformatics. The Centre is a joint project of African countries. Current participating countries are: Benin, Burkina, Cameroon, Ghana, Mali, Niger, Nigeria and Senegal.

The objectives of the Centre are:

- To provide theoretical and practical training in the field of geo-informatics, integrating photogrammetry, remote sensing, geographic information systems, cartography and airborne geophysical surveys.
- To conduct seminars, workshops and short term courses with a view to providing an opportunity for disseminating information in the field of geo-informatics.
- To promote studies and research in the field of geo-informatics.
- To provide advisory and consultancy services upon request on geo-informatics matters, including instrument maintenance, to governments of the member states of ECA, and to other survey and mapping organizations in the region.

Contact information

Dr. Olajide Kufoniyi
Director
kufoniyi@skannet.com.ng
kufoniyi@rectas.org
jidekufoniyi@yahoo.com

Mr. Massaër Mbaye
Deputy Director
massaer@gannetcity.com

RECTAS, P.M.B 5545,
Ile-Ife, Osun State,
Nigeria.

Tel: 234-36-230050; -230225
Fax: 234-36-230481
E-mail: RECTAS@oauife.edu.ng

4.5.2 Regional Centre for Mapping of Resources for Development (RCMRD)

The Regional Centre for Mapping of Resources and Development was founded in Nairobi, Kenya, in 1975, under the auspices of the United Nations Economic Commission for Africa and the then Organization of African Unity (now the African Union). The goal of RCMRD is to be a premier Centre of excellence in the provision of Geo-Information and Information Technology applications in Africa and beyond. The promotion of the development and use of Geo-information and Information technology in the sustainable development of Africa is the main mission of the

RCMRD. This mission includes assistance in human resource and institutional capacity building.

The objectives of RCMRD are:

- To provide quality Geo-Information.
- To ally ICT products and services in environmental and resource management for sustainable development in its member countries and beyond.

At the beginning the RCMRD programmes contained service technology frameworks (e.g. remote sensing, geodesy, cartography etc.). The programmes have now varied into problem solving applications in natural resource and environmental management. RCMRD is now providing service on a demand driven basis and in a business-like manner.

The Regional Centre for Mapping of Resources for Development (RCMRD) and the United Nations Office for Outer Space Affairs (UNOOSA) signed a cooperation agreement on the occasion of the UNSPIDER Regional Workshop – "Building Upon Regional Space-based Solutions for Disaster Management and Emergency Response for Africa" in Addis Ababa on 7 July 2010. In its role as UNSPIDER Regional Support Office, RCMRD will support development and strengthening of national capacities in the Eastern African region, contribute to horizontal cooperation and technical assistance to countries and organizations involved in disaster prevention and mitigation, as well as assist in generating and sharing knowledge for the use of valuable space based information in the decision making processes.

The RCMRD is an inter-governmental organisation composed of 15 Member States in the Eastern and Southern Africa Regions, namely; Botswana, Comoros, Ethiopia, Kenya, Lesotho, Malawi, Mauritius, Namibia, Somalia, South Africa, Sudan, Swaziland, Tanzania, Uganda and Zambia.

Contact Information

Dr. Hussein Farah
Director General
farah@rcmrd.org

Dr. Tesfaye Korme
Director of Remote Sensing, Geo-Information Systems and Mapping
korme@rcmrd.org

Kasarani Road, Off Thika Road
P. O. Box 632-00618
Nairobi
Kenya

Tel: +254-020-8560227, -8560265, -8561775, -8560335, -8560336, -8564131, -8561673, -8563767

www.rcmrd.org

4.5.3 African Organisation of Cartography and Remote Sensing (AOCRS)

The African Organisation of Cartography and Remote Sensing (AOCRS) is an intergovernmental organization established in Addis Ababa in 1988 by the merger of the African Association of Cartography (AAC) with the African Remote Sensing Council (ARSC).

The objectives of AOCRS are:

- To promote, encourage, co-ordinate and harmonize member states policies in the fields of surveys, cartography and remote sensing (understood in their widest senses of the terms):
- Promote the development of cartographical and remote sensing activities in member states, encourage the setting up of services for cartographical production and processing remote sensing in order to assist economic planning and improve the development of natural resources, desertification and disaster management, and other activities related to geo-information and space sciences and co-operate with member states' governments for the establishment of efficient coordination of its activities.
- Favour the exploration and development of surveys, cartography, remote sensing and space science activities and ensure the co-ordination of these activities in the territories of its member states for the better exploration, inventory and development of each nation's natural resources and those of common economic interest to more than one of its member states.
- Encourage the establishment of close relationships between its member states in the fields of cartography and remote sensing at the bilateral and multilateral levels, ensure the coordination of activities and harmonization of efforts for the success of cooperation and in this context promote exchange of personnel.
- Be informed of potentialities in Africa or put at its disposal, introduce new techniques and management of techniques of cartographical and remote sensing activities, inform member states and African bodies operating in this field about them through magazines, directories and other publications and fight for the national

and optimal use of existing means according to the principle “depend on oneself first” in accordance with the Lagos Plan of Action.

- Promote the development of surveys, cartography, remote sensing and space sciences through the establishment in territories of member states of reception, processing, training and assistance centres to use and participate closely in the coordination of activities of existing centres of member states.
- Allow member states to have access to all techniques of cartography, remote sensing and space sciences and promote the development of them by creating at the levels of the continent and its regions training and services centres and fostering those that already exist
- Set-up and support, in close collaboration with co-operating bodies, high level training initiatives for member states as well as organizations at the national, regional and continental level, and organise conferences, seminars, exhibitions and other meetings and demonstrations dealing with all aspects of activities in the fields of cartography, remote sensing and space sciences.
- Sustain regional specialized centres and help them in implementing projects related to mapping and geo-information.

AOCRS conducts programs in the following fields: Basic Geodetic Network; Cartography; Natural Resource Mapping; Remote Sensing and Space Sciences; Geographical Information System; Early warning; Training.

Principal national mapping and remote sensing organizations/agencies represent the governments of 24 African countries: Algeria, Burkina Faso, Cameroon, Central Africa, Congo, DR Congo, Cote d’Ivoire, Ethiopia, Gabon, Ghana, Guinea, Liberia, Libya, Madagascar, Mali, Mauritania, Morocco, Niger, Senegal, Sudan, Tanzania, Togo, Tunisia, and Uganda.

Contact Information

http://www.oact.dz/oact/ (not functioning)

4.5.4 African Centre for Meteorological Applications for Development (ACMAD)

The creation of the African Centre for Meteorological Applications for Development (ACMAD) was initiated by the Conference of the Ministers of the United Nations Economic Commission for Africa (UNECA) and the World Meteorological Organization (WTO) in 1987 and has been operational since 1992 in Niamey. All 54 states of Africa are members of ACMAD.

The main goal of ACMAD is to contribute to the sustainable development of African socio-economic sectors by using information related to weather, climate and environment as resources for development(within the context of national strategies for poverty eradication) in the fields of agriculture, water resources, health, public safety and renewable energy.

To fulfil its goals, ACMAD works together with meteorologists from member states and capacity-building professionals from the National Meteorological Services (NMS). The NMS specialise in weather prediction, climate monitoring (extreme events…), transfer of technology (telecommunications, computing and rural communication) and in research. The NMS are encouraged by ACMAD to prepare strategic development plans that integrate new African initiatives (NEPAD, regional integration) and the socio-economic conditions related to the changing global environment (post Rio Conventions, Kyoto Protocol). Since 1996 ACMAD has been cooperating with EUMETSAT and bringing into focus the use of METEOSAT satellite data. EUMETSAT and ACMAD have a long-standing partnership where ACMAD is responsible for the activities related to climate monitoring in Africa and assembles on a periodic basis the meteorological bulletin. This meteorological bulletin is disseminated throughout Africa through EUMETCast.

The objectives of ACMAD are:

- to prepare and disseminate products and services, specialising in early morning advisories for agriculture and water management;
- to ensure development and transfer of tools and technology to the NMS;
- to monitor communications with users notably in rural communities;
- to network with NMS and regional development aid institutions;
- to perform data collection and analysis;
- to monitor climate variability and change, especially relating to economic activities;
- to provide a window to technology partners, under conditions which are typical of the African situation; supplement member states and partners contributions through a Resource Mobilisation policy.

Contact Information

85 Avenue des Ministères, BP 13184
Niamey
Niger

Fax: +227 20723627
Email: dgacmad@acmad.ne

www.acmad.ne

4.5.5 African Association of Remote Sensing of the Environment (AARSE)

The African Association of Remote Sensing of the Environment was founded in 1992 for the development of GIS and remote sensing in Africa as part of the regional csollaboration process. It currently has over 1000 individual and institutional members from within and outside Africa. It has annual conferences that are increasingly popular for bringing together Africa's growing communities of academics, professionals and enthusiasts working in the field of remote sensing and Geographic Information Systems (GIS). It seeks to address the range of Africa's developmental needs, from resource management to governance through the application of space technologies while paying attention to related factors like policy and infrastructure. Being the largest conference of its type on the continent, it also serves as a hub to gather professionals around other space-related initiatives with related themes such as the activities of the Group on Earth Observation System of Systems in Africa, the development of open source GIS platforms and even proprietary geo-informatics software.

Contact Information:

Tsehaie Woldai

Tel: +31 53 487 42 79
Fax: +31 53 487 43 36
E-mail: woldai@itc.nl

Website: http://www.itc.nl/aarse

4.5.6 AGRHYMET Regional Centre (ARC)

The Regional Centre (ARC) of Agrometeorological and Hydrometeorological Operations and their Applications (AGRHYMET) is an interstate public institute and has international status. It was established in 1974 in Niamey (Niger) as a specialised institution of the Permanent Inter-State Committee for Drought Control in the Sahel (CILSS). It is a regional institute specialized in the science and techniques applied to agricultural development, rural development and natural resource management.

The mission of ARC is to promote information and training about food security, desertification control and the management of natural and water resources; focusing on the science and techniques applied to agricultural development, rural development and natural resource management.

The objectives of ARC are to contribute to achieving food security and increased agricultural production in CILSS member states; and to improve natural resource management in the Sahelian region by providing training and information to development stakeholders and partners in agroecology taken as a whole (agroclimatology, hydrology, crop protection…).

ARC uses METEOSAT data for operational and development activities, including rainfall estimates, training, and agro-meteorological and hydrological applications, through the established cooperation of EUMETSAT and AGRHYMET from1996. For Western Africa, AGRHYMET has been the reference Regional centre for the framework of the PUMA Project and for the Regional Implementation Centre in AMESD. ASMESD is responsible for the THEMA "Cropland and Rangeland Management" in the ECOWAS region.

ARC has nine member states: Chad, Niger, Bukina Faso, Mali, Mauritania, Senegal, Gambia, Guinea Bissau, Cape Verde.

Contact Information:

Mr. Mohamed YAHYA Ould MAHMOUD
(Director General)
Tel: + 22 72 03 15 20 2
Email: m.yahya@agrhymet.ne

Mrs. Fatchima YAHAYA
Tel: + 22 72 03 15 20 3
Email: f.yahaya@agrhymet.ne

Mr. Philippe MORANT
Tel: +22 72 03 15 26 5
Email: p.morant@agrhymet.ne

Mrs. Mariama SALIFOU
Tel: + 22 72 03 15 26 6
Email: m.salifou@agrhymet.ne

www.agrhymet.ne

4.5.7 IGAD Climate Prediction and Applications Centre (ICPAC))

The Greater Horn of Africa (GHA) is prone to extreme climate events such as droughts and floods. These extreme events have severe negative impacts on key socio-economic sectors of all the countries in the sub-region. In 1989, twenty four countries in Eastern and Southern Africa established a Drought Monitoring Centre with its headquarters in Nairobi

(the DMCN) and a sub centre in Harare (Drought Monitoring Centre Harare – DMCH) in response to the devastating weather related disasters. In October 2003, the Heads of State and Governments of the Intergovernmental Authority on Development (IGAD) held their 10th Summit in Kampala, Uganda, where DMCN was adopted as a specialized IGAD institution. The name of the institution was at the same time changed to IGAD Climate Prediction and Applications Centre (ICPAC) in order to better reflect all its mandates, mission and objectives within the IGAD system. A Protocol integrating the institution fully into IGAD was however signed on 13 April 2007.

The vision of ICPAC is to become a viable regional centre of excellence in climate prediction and applications for climate risk management, environmental management, and sustainable development. Its mission is to provide a timely climate early warning information and supporting specific sector applications to enable the region cope with various risks associated with extreme climate variability and change for poverty alleviation, environment management and sustainable development of the member countries.

The centre is responsible for seven member countries namely: Djibouti, Eritrea, Ethiopia, Kenya, Somalia, Sudan and Uganda as well as Burundi, Rwanda and Tanzania

Contact Information:

Laban A. Ogallo
Director
IGAD Climate Prediction and Applications Centre (ICPAC).
P.O. Box 10304, 00100 Nairobi,
KENYA.

Tel: 254 20 3514426
Fax: 254 20 3878343 ,
Email: director@icpac.net

www.icpac.net

4.5.8 SADC- Drought Monitoring Centre (SADC-DMC)

The start of the SADC-Drought Monitoring Center (SADC-DMC) dates back to 1983 where during the sub-regional (SADC) Drought management meeting sponsored by WMO, Zimbabwe National Meteorological And Hydrological Services (NMHSs) agreed to cooperate, exchange climate data and establish a regional telecommunication hub in Lusaka. SADC-DMC is provides the sub-region with weather and climate advisories as well as timely early warnings on droughts, floods and other extreme climate related events.

The mission is to avail timely climate information and prediction products, and enhance application of the products to the reduction of climate and weather related risks to food security, water resources and health for sustainable development in Southern Africa.

The main objective of the SADC DMC is to carry out climate monitoring and prediction for early warning and mitigation of adverse impacts of extreme climatic events on agricultural production, food security, water resources, energy, and health among other socio-economic sectors.

The SADC-DMC is currently hosted by the Botswana Department of Meteorological Services.

Contact Information:

Mr B.J. Garanganga
The Coordinator
The Drought Monitoring Centre
P/Bag 0095
Gaborone, Botswana

Tel: +267 395 3413
Fax: +267 397 2848
Email: dmc@sadc.int

http://www.sadc.int/dmc

4.5.9 Agence pour la Sécurité de la Navigation Aérienne en Afrique et à Madagascar (ASECNA)

ASECNA is an international organisation created in 1959 with the mandate of ensuring general air traffic safety for its 18 Member States, managing their air space and providing services for air traffic and aeronautical meteorology.

The Agency is responsible for the design, implementation and transmission of technical messages and traffic, guide traffic, control of air traffic information theft, forecasting and reporting in the meteorological field for route approach and landing at airfield.

ASECNA operates the Ecole Africaine de la Météorologie et de l'Aviation Civile (EAMAC) in Niamey, Niger. ASECNA has close collaboration with EUMETSAT which began in 1998 and aims at coordinating training activities on satellite meteorology for Africa, including the use of EUMETSAT data and products for educational purposes at EAMAC. EUMETSAT and EAMAC are regularly jointly organising training session on Satellite Meteorology for the French-speaking African users.

ASECNA has also participated actively to the EGNOS Africa test bed implementation.

The member states of ASECNA are Benin, Burkina Faso, Cameroon, Central, Comoros, Congo, Cote d'ivore, Gabon, Guinea Bissau, Equatorial Guinea, Madagascar, Mali, Mauritania, Niger, Senegal, Chad, and Togo.

Contact Information:

Mahamat Youssouf
ASECNA Direction Générale
32-38, Avenue Jean Jaurès
BP 3144 Dakar, Senegal

Tel: +221 33 849 66 00
Fax: +221 33 823 46 54
Email: contact@asecna.aero

http://www.asecna.aero/

4.5.10 Sahara and Sahel Observatory (OSS)

Sahara and Sahel Observatory (OSS) is an independent international organisation based in Tunis, Tunisia. It was founded in 1992 to improve early warning and monitoring systems for agriculture, food security and drought in Africa. It has the role of the North-South partnership platform.

The mission of OSS is to provide an adequate framework for North-South-South partnership. Through this partnership, it aims to mobilise and reinforce member countries' capacity to tackle the environmental issues they face with a view to underpinning sustainable development and combating desertification in circum-Saharan Africa. The main focus is on issues related to water and land degradation.

The OSS community includes 22 member countries, 5 countries in Europe and North America. These are: Algeria (AMU), Burkina Faso (CILSS), Canada, Cape-verde (CILSS), Chad (CILSS), Côte d'Ivoire, Djibouti (IGAD), Egypt, Eritrea(IGAD), Ethiopia (IGAD), France, Germany, Guinea Bissau (CILSS), Italy, Kenya (IGAD), Libya (AMU), Mali (CILSS), Mauritania (CILSS / AMU), Morocco (AMU), Niger (CILSS), Senegal (CILSS), Somalia (IGAD), Sudan (IGAD), Switzerland, The Gambia (CILSS), Tunisia (AMU), Uganda (IGAD). It also includes, 4 sub-regional organisations: AMU, IGAD, CILSS, CEN-SAD; regional and international organisations: ACMAD, AOCRS, FAO, UNCCD, UNESCO, UN-GS; and civil society: Environmental Development Action in the Third World (Enda-TM).

Contact Information:

Mr Chedli Fezzani,
Sahara and Sahel Observatory
Boulevard du Leader Yasser Arafat, BP 31
Tunis Carthage, Tunisia, 1080

Tel: +216 71 206 633/634
Fax: +216 71 206 636
Email: boc@oss.org.tn

http://www.oss-online.org/

4.5.11 Commission Internationale du Bassin Congo-Oubangui-Sanga (CICOS)

The agreement establishing a Uniform framework for the river system and creating CICOS was signed on 6 November 1999 by the 4 heads of state of Cameroon, Central Africa, Congo and Democratic Republic of Congo (Angola is now observer). An Addendum to the agreement extending the mandate of CICOS to IWRM was signed on 22 February 2007.

The aim of CICOS is to ensure efficient and effective coordination to enhance water man agent in the Congo river basin.

CICOOS works together with various partners such as the African Water Facility, GTZ, France Coopération, FAO, WMO, African Network of Basin Organizations in Central Africa (RAOB). Furthermore, CICOS is involved in the AMESD project.

The member states of CICOS are Cameroon, Central African, Congo and democratic republic of Congo.

Contact Information:

Benjamin Ndala
General Secretary
Building Kilou, BP 12645, Kinshasha
Democratic Republic of Congo

Tel: +243 81 555 36 76, +243 81 555 36 75
Email: cicos_inst@cicos.info

http://www.cicos.info

4.5.12 African Institute of Space Science

The African Science and Technology Consolidated Plan of Action were developed in 2005 to give direction to the continent's aspirations and efforts to collaborate on common initiatives that would enhance its technological capabilities. It follows a series of consultations and workshops that aimed to consolidate the plans on science and technology for the continent developed by the African Union and the New Partnership for African Development (NEPAD). It "articulates Africa's common objectives and commitment to collective actions to develop and use science and technology for the socio-economic transformation of the continent and its integration

into the world economy"[11]. The establishment of an African Institute of Space Science is one of the programs under this plan and it aims to do the following:

"(a) building public and policy-makers' awareness of the benefits of space science to Africa's economic transformation and sustainable development;

(b) Mapping global trends and identifying specific technological opportunities; and *Africa's Science and Technology Plan of Action* 37;

(c) Identifying specific institutional arrangements for space science in Africa".

Website:
http://www.nepadst.org/platforms/aiss.shtml

4.5.13 Regional African Satellite Communications Organisation (RASCOM)

In February 1991, African countries agreed to co-operate in order to build a telecommunication satellite that will satisfy Africa's telecommunications needs, globally and optimally. This decision led to the establishment of the Regional African Satellite Communications Organisation (RASCOM) in 1992. In December 2001, RASCOM launched its first satellite RASCOMStar-QAF, however, the satellite suffered power failure 8 days after its launch, thus reducing the orbit life by 13 years from its expected 15 year life. The satellite will be replaced by Thales-Alenia, the company that built the first satellite.

4.5.14 African Leadership Conference (ALC)

The idea of the ALC was conceived in 2004 by South Africa, Algeria and Nigeria. The objective of the ALC is promoting the use of space tools towards achieving Africa's sustainable development. The ALC also seeks to sensitize African leaders on the importance of science and space technology for the sustainable socio-economic development of Africa. It also provides a regular forum for information exchange and will promote inter-African cooperation in the development and the application of space technology. The ARMC collaboration is an initiative of the ALC.

Prof. Ade Abiodun
Tel: +1-516-333-5474 or
+1-516-884-0559
Email: aaabiodun@optonline.net

4.5.15 International Academy of Astronautics African Regional Conference

The International Academy of Astronautics holds annual African Regional Conferences. These conferences bring together space professionals on the continent. The membership of the academy is growing on the continent and the regional conferences help to expand interest and awareness in astronautics and other space activities especially as they pertain to Africa's unique needs and level of development.

General Secretary:
Jide AJAYI, National Space Research and Development Agency,

Tel: 234-802-340-1381, 234-807-584-9733 or 234-(9)-2342203
Fax: +234-(9)-4132624
Email: ajayijide2000@yahoo.com, ajayijide@hotmail.com or jajayi@nasrda.net
http://iaaweb.org/content/view/255/392

4.5.16 The African Institute of Corporate Citizenship (AICC)

The African Institute of Corporate Citizenship (AICC) was established in 2001 and promotes sustainable business practice in Africa as an NGO. The aim of AICC is to facilitate the competitiveness of countries, companies, and communities through responsible business activity and to be a centre of excellence in corporate citizenship in Africa, promoting the role of business in building sustainable communities.

The objective of AICC is to facilitate, promote, support, and work in partnership with a range of development stakeholders in order to influence the underlying values of corporate citizenship and sustainable business practice. It aims to develop programmes in the following fields: good governance and eliminating corruption; promoting a sustainable workplace; and influencing the global corporate social responsibility architecture.

AICC takes the role of the key player in a range of international corporate citizenship initiatives by providing African perspectives to emerging global standards of best practice. Additionally AICC offers advisory services to public and private sector businesses on a range of sustainability issues. Through these processes AICC seeks to be an agent of progressive change in advancing the corporate citizenship agenda.

AICC's portfolio contains a wide range of activities and services. Moreover AICC assembles a number of multi-stakeholder project networks spanning many countries in Africa.

[11] AU and NEPAD. 2005. Africa's Science and Technology Consolidated Plan of Action. [Online] Available at: www.nepapdst.org/doclibrary/pdfs/ast_cpa_2007.pdf

Contact Information

Johannesburg office
1st Floor, 09 St Davids Place
Parktown, Johannesburg
P.O. Box 37357
Birnam Park
Johannesburg
South Africa

Tel: +27 11 643 6604
Email: nonhlanhla@aiccafrica.org

Malawi Office
Amina House Ground Floor
Lilongwe 3 Malawi
Private Bag 382
Lilongwe 3
Malawi

Tel. +265 1 772 404
Email: khambile@aiccafrica.org

4.5.17 African Development Forum (ADF)

The African Development Forum (ADF) was founded in 2006 through the encouragement of Britain's African and Caribbean Christian community and implemented as a team in 2007. The African Development Forum (ADF) is formed by African and Caribbean Church leaders and is guided by the motto, 'We believe in Africa'.

The objectives of ADF are:

- *Economic Justice.* ADF will engage with the global movement calling for rich country governments and institutions to implement pro-poor and pro-development trade policies. This includes international trade rules and trade agreements; an end to damaging liberalisation conditions attached to loans and debt relief by working with and through its partners.
- *Mitigation against Climate Change.* ADF works with partners to develop accessible theological resources on 'creation care' to enable learning on global warming and its disproportionate effects on the continent of Africa. This will include raising awareness about the way in which northern governments and businesses react and respond
- *Preventing HIV/AIDS.* ADF works with partners to increase understanding of HIV/AIDS by developing a theological framework in which to respond to stigma and discrimination. In addition support the campaign to make life saving treatments available to African countries.

The ADF targets the mobilisation of the African and Caribbean Christian community and wider diaspora through the confrontation with poverty and the response to poverty by challenging unjust structural systems, which keep millions of Africans poor. Through educational and advocacy activities, the AFD aims for economic and social progress.

In order to reinforce its vision, AFD extends its collaboration with Christian Aid and Dr Robert Beckford, and partners with Tipping Point Film Fund to develop the ADF Progressive Forum.

Contact Information

admin@africandevelopmentforum.org

http://www.africandevelopmentforum.org/

5. European Actors

5.1 European Union (EU)

Overview

The European Union is an economic and political union with 27 member states and 23 official languages. It has established a single market and 16 member states have the Euro as their common currency. The EU maintains exclusive competencies in monetary policy for member states in the Euro, customs, commerce, and conservation of marine biological resources. Other policy area competences such as social, economic, social and territorial cohesion, agriculture, fisheries, environment, consumer protection, transport, energy, freedom, security and justice, and public health safety matters are shared between the EU and its member states. The first seeds for the EU were put in place after the Second World War with the aim to end the wars between the neighbouring countries in Europe. The EU was established in 1993 with the Treaty of Maastricht, succeeding the European Communities. On 1 December 2009, the European Community was replaced by the European Union and takes all its rights and responsibilities. The European Union consists of various institutions and bodies of which the three main institutions involved in the decision making process of the Union are the European Commission, the European Parliament and the Council of the European Union often referred to as the Council. The European Commission (EC) is located in Brussels in Belgium; the Council is located in Brussels; and the European Parliament (EP) is in Brussels and in Strasbourg in France. The Council is one of the key actors in the European decision-making process. The Ministers of the Member State meet within the Council to approve legal regulations and budgets of the EU and other decisions. The Presidency of the Council is held for six months by each Member State on a rotating basis. The European Parliament represents the EU citizens and is responsible together with the Council for the other half of the legislature and budget approval of the Union. The European Commission represents the interests of the Union as a whole and has the role of the executive of the Union, where it is responsible for implementing the decisions of the Parliament and the Council. It is also responsible for initiating legislation. The President of the Commission is Mr. Barroso. In the decision making process the European Parliament, the Council and the Commission shall be assisted by an Economic and Social Committee and a Committee of the Regions acting in an advisory capacity. Under the latest treaty of the Union, the Lisbon Treaty which entered into force on 1 December 2009, the European Council became an institution and has a permanent President. The European Council is responsible for defining the general political directions and priorities of the Union and has no legislative functions. The President of the European Council will ensure the representation of the Union on issues concerning foreign and security policy. The European Council consists of Heads of States or Governments of Member States, together with the President of the Commission and its President Mr. Van Rompuy. The High Representative of the Union for Foreign Affairs and Security Policy takes part in its work. With the entry into force of the Lisbon treaty Ms. Ashton is the High Representative of the Union of Foreign Affairs and Security with the duties to conduct the Union's common foreign and security policy and has authority over the European External Action Service (EEAS), also set up by the Treaty of Lisbon, and over some 130 Union delegations in third countries and international organisations. Furthermore she is the Council's representative for Common Foreign and Security policy, the President of the Foreign Affairs Council and the Vice-President of the Commission. Regarding the space competences, before the Lisbon treaty it was left up to the Member States of the Union. Now, Articles 4 and 189 call for the Union to develop competences to carry out activities and implement programmes. In particular it calls for the Union to draw up a European Space Policy and take up activities that may take a form of a European Space Programme. The treaty also recognises the importance of the European Space Agency and calls for establishing all appropriate relations at the Union level.

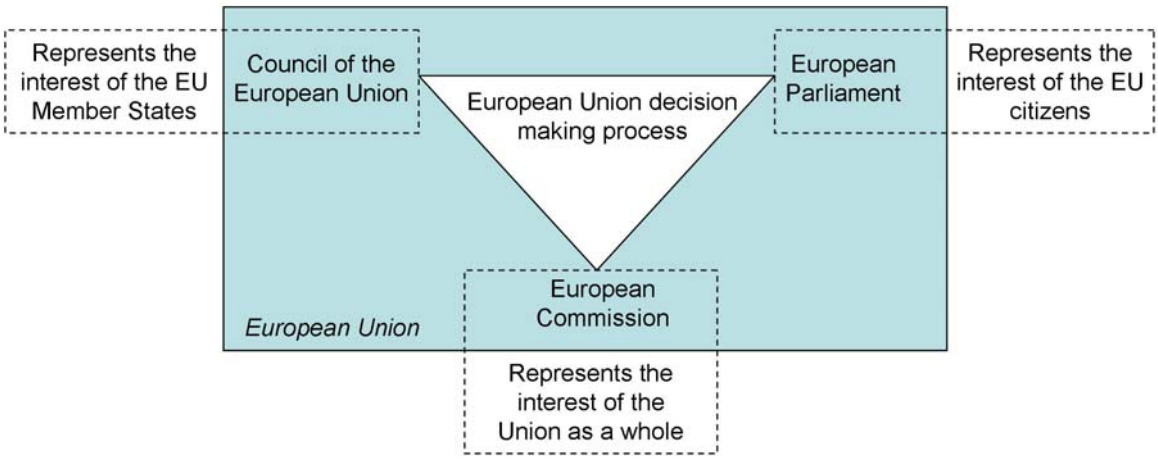


Figure 7: European Union decision making process before the Lisbon Treaty.

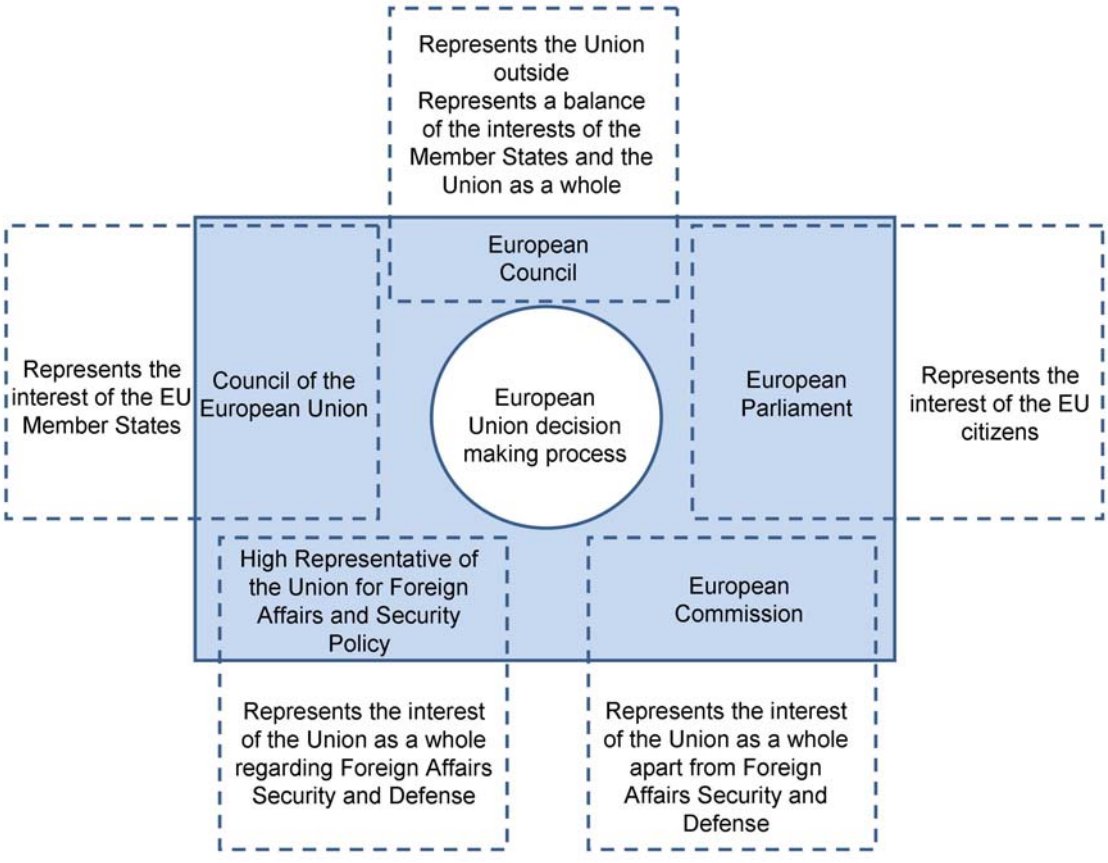


Figure 8: European Union decision making process after the Lisbon Treaty.

Historical Background

In 1951 six European countries, Belgium, France, Germany, Italy, Luxembourg and the Netherlands signed a Treaty establishing the European Coal and Steel Community which entered into force in 1952 and expired in 2002. In 1957 two Treaties were signed - the Treaties of Rome, whereby the European Atomic Energy Community (EURATOM) and the European Economic Community were formed. In 1965 the Marger Treaty provided a single Council and Commission for the three communities. In 1973 the first enlargement took place with Denmark, Ireland, and the United Kingdom becoming members of the Union. In 1981, Greece became the 10th member state and in 1986 Portugal and Spain were also included. In 1987 the Single European Act provided the foundations for achieving a single market in the Union which had by then 12 member states. In 1989 the fall of the Berlin Wall and the unification of East and West Germany highlighted the beginning of the fall of the communist regimes in central and east European countries making new borders in Europe. In 1992 with the Treaty of Maastricht that entered into force in 1993, the European Community was established replacing the European Economic Community. This treaty also provided the foundations for closer intergovernmental cooperation's opening ways for economic and political integration. It introduced the concept of European citizenship and launched the Economic and Monetary Union (EMU). The freedom of movement of goods, services, people and money dominate as concepts in the single market. In 1995 three more members, Austria, Finland and Sweden were added making the Union of 15. In 1997 the Treaty of Amsterdam was signed entering into force in 1999, amending the Treaty of Maastricht and the Treaty of Rome. In 2001 the Treaty of Nice was signed and entered into force in 2003 focusing on reforming the institutions of the Union for it to function more efficiently. In January 2002 the Euro as a single European Currency was circulated amongst the 15 member states. In 2004 the Union was enlarged to 24 states including 8 former communist countries: the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Slovakia and Slovenia as well as Malta and Cyprus, as a divided island. In 2007 Bulgaria and Romania also joined making a Union of 27 member states. Currently 16 of the 27 member states are part of the Eurozone: Austria, Belgium, Cyprus, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Malta, the Netherlands, Portugal, Slovakia, Slovenia and Spain. The latest treaty of the Union is the Lisbon Treaty which was signed in 2007 and entered into force in 2009. This treaty makes the European Council an institution and creates a new political figure, the President of the European Council, with the task to ensure the external representation of the Union on issues concerning the Union's common foreign and security policy. The President of the European Council will be assisted in this by the EU High Representative for Foreign Affairs and Security (often called the High Representative). The Treaty also sets the legal basis for a European External Action Service (EEAS) headed by the High Representative. The EEAS will be separate from the Commission and the Secretariat and will work in cooperation with the diplomatic services of the member states and will coordinate the Union's external actions. It will prepare policy proposals for the Council and European Parliament and, after approval, implement them. Furthermore, the EU's crisis management structures will be part of EEAS as well as the conduct of the European Security and Defence Policy.

The EU values and aims

The recently adopted Lisbon Treaty of the Union reaffirms the values of the Union for peace, democracy, respect for human rights, justice, equality, rule of law and sustainability. The aims of the Union according to the treaty are:

- offer people an area of freedom, security and justice without internal frontiers;
- work for sustainable development of Europe based on balanced economic growth and price stability, a highly competitive social market economy, aiming at full employment and social progress, with a high level of protection of the environment;
- combat social exclusion and discrimination and promote social justice and protection;
- promote economic, social and territorial cohesion, and solidarity among Member States;
- remain committed to economic and monetary union with the euro as its currency;
- uphold and promote the European Union's values in the wider world and contribute to peace, security, the sustainable development of the earth, solidarity and respect among peoples, free and fair trade, and the eradication of poverty;

- contribute to the protection of human rights, in particular the rights of the child, as well as the strict observance and the development of international law, including respect for the principles of the United Nations charter.

Member States of the EU

The EU has 27 member countries, each holding for six months in turn the rotating Presidency of the Council of the European Union. The member states are: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, and United Kingdom.

Languages of the EU

Under the motto "Unity in diversity" the European Union considers that many languages is an asset and promotes the freedom of speech of speech of its citizens in their own language. Currently there are 23 official languages which are: Bulgarian, Czech, Danish, Dutch, English, Estonian, Finnish, French, German, Greek, Hungarian, Irish, Italian, Latvian, Lithuanian, Maltese, Polish, Portuguese, Romanian, Slovak, Slovene, Spanish and Swedish. The number of languages is expected to increase as the Union will enlarge. Furthermore, there are more than 60 regional or minority languages in the Union. Nevertheless, the procedural languages or working languages of the European institutions are English, French and German, but the citizens may request official documents in any of the 23 official languages of the Union.

Organisation of the EU

The Union has an institutional framework in order to fulfil its aim to promote its values, advance its objectives, serve its interests, those of its citizens and those of the Member States, and ensure the consistency, effectiveness and continuity of its policies and actions. According to the Lisbon Treaty the Institutions of the European Union are as follow[12][13][14]:

- The *European Parliament* is elected directly by the citizens of the Union to represents their interests. It is one of the main decision making bodies in the union and it shares lawmaking powers with the Council of the European Union. The European Parliament also approves the EU budget.
- The *Council of the European Union* is made up of 27 government ministers representing each of the Member States. Together with the European Parliament it shares lawmaking and budgetary powers. It is a key decision making body coordinating EU economic policies and plays a central role in foreign and security policy. The EU presidency of the Council rotates every six months among the Member States and is chaired by the Minister of the country holding the presidency. Under the new Lisbon Treaty the Council of Foreign Ministers will be chaired by the High Representative for Foreign and Security Policy who is also the vice-President of the Commission. For other areas such as agriculture, energy etc. the Council is still chaired by the Minister of the country holding the Presidency.
- The *European Council* is made up of the most senior elected political representatives of the Member States with executive powers, together with the President of the Commission and the President of the European Council. Under the Lisbon treaty it has become one of the Union's institutions. The role of the European Council is to provide the Union's general political directions and priorities but it will not exercise legislative functions. The President of the European Council will chair the Council meetings and will represent the EU internationally at the highest level. The President will also ensure the representation of the Union on issues concerning foreign and security policy. The High Representative for Foreign Affairs and Security Policy will take part in its work.
- The *European Commission* represents the interests of the Union as a whole and is accountable to the Council and the European Parliament. The European Commission has the power to initiate proposals for legislation and it has the role of the executive of the Union to implement the decisions of the Parliament and the Council. The commission enforces the Union's policies, ensures that the budgets are implemented, manages EU programmes and makes sure that treaties are implemented properly.
- The *Court of Justice of the European Union* consists of at least one judge from

[12] Official Journal of the European Union, 2010/C8301, Volume 53, 30 March 2010, ISBN 1725-2423.
[13] European Commission, "Your Guide to the Lisbon Treaty", Belgium, July 2009, ISBN 978-92-79-12935-3.
[14] General Secretariat of the Council of the EU, "Background-President of the European Council", Press, Brussels, November 2009.

each EU country and the office is renewable every six years. Its role is to give legal judgment on cases brought before it and oversees that EU legislation is interpreted and applied in the same way in all EU countries, so that the law is equal fore everyone.

- The *European Central Bank*, together with the national central banks of the Member States whose currency is the euro, conducts the monetary policy of the Union. Its role is to maintain price stability in the euro area.
- The *Court of Auditors,* carries out the Union's audit. Its role is to check that EU funds are used properly.

In addition there are other bodies:

- The *Committee of the Regions* is one of the advisory bodies of the union which represents regional and local authorities. It has to be consulted before EU decisions are taken on matters such as regional policy, the environment, culture, education and transport.
- The European Economic and Social Committee is one of the consultative bodies of the union and represent organised and civil society. It gives representatives of Europe's interest groups such as employer organisations and trade unions, and other 'organised civil society' bodies, a formal platform to express their points of view on EU issues.
- The *European Investment Bank* has the task to contribute, by having recourse to the capital market and utilising its own resources, to the balanced and steady development of the internal market in the interest of the Union. Thus, its role is to finance economic development. Its budget is independent of the EU budget. It invests on projects for developing less-developed regions of common interest to several Member States, and for modernising or converting undertakings or for developing fresh activities called for by the establishment or functioning of the internal market.

There is also the European Ombudsman which has the task to investigate cases of complaints and uncover maladministration and propose solutions. The Ombudsman acts as an intermediary between the citizen and the EU authorities. It carries out investigations following a complaint or on its own initiative and operates independently.

The EU Commission

The European Commission has five years term in office and currently the 2010-2014 Commission is in place. The Members are:

- President: José Manuel Baroso (Portugal)
- High Representative of the Union of Foreign Affairs and Security Policy, Vice-President: Catherine Ashton (United Kingdom)
- Justice, Fundamental Rights and Citizenship, Vice-President: Viviane Reding (Luxembourg)
- Commissioner for Competition, Vice-President: Joaquín Alunia (Spain)
- Commissioner for Transport, Vice-President: Siim Kallas (Estonia)
- Commissioner for Digital Agenda, Vice-President: Neelie Kroes (Netherlands)
- Commissioner for Industry and Entrepreneurship, Vice-President: Antonio Tajani (Italy)
- Commissioner for Institutional Relations and Administration, Vice-President: Maroš Šefčovič (Slovakia)
- Commissioner for Environment: Janez Potočnik (Slovenia)
- Commissioner for Economic and Monetary Affairs: Olli Rehn (Finland)
- Commissioner for Development: Andris Piebalgs (Latvia)
- Commissioner for Internal Market and Services: Michel Barnier (France)
- Commissioner for Education, Culture, Multilingualism and Youth: Androula Vassiliou (Cyprus)
- Commissioner for Taxation and Customs Union, Audit and Anti-Fraud: Algirdas Šemeta (Lithuania)
- Commissioner for Trade: Karel De Gucht (Belgium)
- Commissioner for Health and Consumer Policy: John Dalli (Malta)
- Commissioner for Research, Innovation and Science: Máire Geoghegan-Quinn (Ireland)
- Commissioner for Budget and Financial Programming: Janusz Lewandowski (Poland)
- Commissioner for Maritime Affairs and Fisheries: Maria Damanaki (Greece)
- Commissioner for International Cooperation and Humanitarian Aid and Crisis Response: Kristalina Georgieva (Bulgaria)
- Commissioner for Energy: Günther Oettinger (Germany)
- Commissioner for Regional Policy: Johannes Hahn (Austria)
- Commissioner for Climate Action: Connie Hedegaard (Denmark)
- Commissioner for Enlargement and European Neighborhood Policy: Štefan Füle (Czech Republic)

- Commissioner for Employment, Social Affairs and Inclusion: László Andor (Hungary)
- Commissioner for Home Affairs: Cecilia Malmström (Sweden)
- Commissioner for Agriculture and Rural Development: Dacian Cioloş (Romania)

The European Commission has four main roles:

- to propose legislation to Parliament and Council;
- to manage and implement EU policies and budgets;
- to enforce EU law (jointly with the Court of Justice);
- to represent the European Union on the international stage.

The "Europe 2020"[15] Strategy

On the 17 June 2010 the Union adopted "Europe 2020"[15], the Union's new 10 year strategy. This strategy focuses on pulling the Union through from the global financial crisis and additionally puts forward three mutually reinforcing priorities:

- Smart growth: developing an economy based on knowledge and innovation.
- Sustainable growth: promoting a more resource efficient, greener and more competitive economy.
- Inclusive growth: fostering a high-employment economy delivering social and territorial cohesion.

The Commission has put forward seven flagship initiatives to catalyse progress under each priority theme:

- "Innovation Union" to improve framework conditions and access to finance for research and innovation so as to ensure that innovative ideas can be turned into products and services that create growth and jobs.
- "Youth on the move" to enhance the performance of education systems and to facilitate the entry of young people to the labour market.
- "A digital agenda for Europe" to speed up the roll-out of high-speed internet and reap the benefits of a digital single market for households and firms.
- "Resource efficient Europe" to help decouple economic growth from the use of resources, support the shift towards a low carbon economy, increase the use of renewable energy sources, modernise the transport sector and promote energy efficiency.
- "An industrial policy for the globalisation era" to improve the business environment, notably for SMEs, and to support the development of a strong and sustainable industrial base able to compete globally.
- "An agenda for new skills and jobs" to modernise labour markets and empower people by developing their skills throughout the lifecycle with a view to increase labour participation and better match labour supply and demand, including through labour mobility.
- "European platform against poverty" to ensure social and territorial cohesion such that the benefits of growth and jobs are widely shared and people experiencing poverty and social exclusion are enabled to live in dignity and take an active part in society.

The EU and AU Relationship

There is a permanent delegation of the EU to the AU. Their role is to

- help coordinate EU policy and action relating to the AU, and relations between the broader partner group and the AU
- maintain close contact with the AU Commission and its other institutions, and the representative offices of African sub-regional organisations and AU countries
- help, advise and support the AU on request, in all areas outlined in the Africa-EU strategy (e.g. development and implementation)
- monitor and report on all relevant AU developments
- assist the EU's political dialogue and broader relationship with the AU

Mr. Koen Vervaeke is the EU Special Representative to the African Union (until 31 August 2010)

Contact Information

Delegation-African-Union@ec.europa.eu

Tel: +251.11.416.37.00
Fax: +251.11.416.37.03/17

Press Officer
Christophe Boulierac
Christophe.boulierac@ext.ec.europa.eu

[15] European Commission, "Europe 2020. A strategy for smart, sustainable and inclusive growth", Communications from the Commission, COM(2010)2020, Brussels 3.3.2010

Secretaries to EUSR
Jayne Hill
Jayne.hill@ext.ec.europa.eu
Eva BLAKE
Eva.blake@ext.ec.europa.eu

5.2 EUMETSAT

The European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) was established in 1986 as an intergovernmental organisation with the objective to provide from space information that can be used for weather forecasting and climate applications. EUMETSAT has 26 Member States[16] and its headquarters are located in Darmstadt, Germany.

The main activity of EUMETSAT is to deliver weather and climate-related satellite data, images and products 24 hours a day, 365 days a year. This is supplied through the motoring of the surfaces of the earth, including the oceans by the system of meteorological satellites, in a high quality. EUMETSAT provides this information to the National Meteorological Services of the Organisation's Member and Cooperating States in Europe as well as private users for meteorological and climate related satellite data.

The information provided by the EUMETSAT systems also assists meteorologists in identifying and monitoring potentially dangerous weather conditions and issuing early forecasts and warnings to the relevant authorities and services. This is critical for the protection of human life, property, safety on air travel, shipping and road traffic, farming, construction and other industries.

EUMETSAT provides support in developing countries including Africa. There are various projects where EUMETSAT, together with other partners has provided support to African States. EUMETSAT has also established various cooperation agreements and signed memorandums of understanding with various African stakeholders. These collaborations are:

- In 2008, EUMETSAT signed a Memorandum of Understanding with the African Union Commission regarding the AMESD programme. EUMETSAT and EAC also collaborate under the GMES Africa initiative.
- In 1996, EUMETSAT and ACMAD established their cooperation aimed at the use of the METEOSAT satellite data. EUMESAT and ACMAD have also been cooperating under the PUMA project.
- In 1996, EUMETSAT and AGRHYMET established their cooperation to foster the use of METEOSAT data for operational and development activities, including rainfall estimates, training and agro-meteorological and hydrological applications. Collaboration with AGRHYMET was also achieved under the PUMA project.
- In 1996, EUMETSAT and ASECNA established their cooperation aimed at coordinating training activities on satellite meteorology for Africa, including the use of EUMETSAT data and products for the educational purposes of the Ecole Africaine de la Mètèorogie et de l'Aviation Civile (EAMAC) in Niamey, Niger which is operated by ASECNA.

As part of EUMETSAT recurrent tasks the following activities are implemented in support to its user in Africa:

- Data dissemination in Africa. Thanks to EUMETCast Africa (a component of GEONETCast operated by EUMETSAT), EO data and products are provided in real-time to various users in Africa. It encompasses EUMETSAT products but also third party product (VEGETATION, Marine, MODIS, etc)
- Training activities. In cooperation with the three Training centre (Niger, Kenya, South Africa) EUMETAST is conducting 3 to 5 training courses in Africa every year on satellite meteorology and is contributing to several other training events.
- User Forum in Africa. The purpose of the Forum is to reinforce the well established dialogue between EUMETSAT and the African user community to optimise the use of EUMETSAT satellite data and products throughout the continent..
- Help Desk. Through it Help Desk EUMETAST is providing support to its African user about their data acquisition infrastructure (EUMETCast station) and the EUMETAST data and products.

In addition to these core activities, EUMETAST is involved in various projects currently running in Africa. These projects are:

- Provision of Meteosat Second Generation in Africa (PUMA). This EDF funded project

[16] The 25 Member States of EUMETSAT are Austria, Belgium, Croatia, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia Luxembourg, The Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom. Romania will be 26th Member State. Its membership is currently pending approval.

is an international collaboration that provides the continent's National Meteorological and Hydrological Services with operational access to state-of-the-art satellite data, products and technology. The Meteosat Second Generation (MSG) of satellites has been the catalyst for the Preparation for Use of MSG in Africa (PUMA) project.

- User Forum in Africa. The purpose of the Forum is to reinforce the well-established dialogue between EUMETSAT and the African user community to optimise the use of EUMETSAT satellite data and products throughout the continent. This project is funded under the EC 7th Framework program.
- African Monitoring of the Environment for Sustainable Development (AMESD). This initiative takes the Preparation for Use of MSG in Africa (PUMA) project a stage further by extending the operational use of Earth observation technologies and data to environmental and climate monitoring applications. It is funded by the 9th EDF.
- Global Monitoring for Environment and Security (GMES) Africa initiative aims at strengthening of capacities and developing infrastructure for the intense and coherent exploitation by African users of Earth Observation (space and in-situ), data, technologies and services in support of the environmental policies for sustainable development in Africa.
- Other Projects (funded by FP7)
  - GEONETCast applications for and by Developing Countries (Devcocast)
  - Assessment of ICT for Disaster Management in Africa (AÏDA)
  - Mitigate and Assess risk from Volcanic Impact on Terrain and human Activities (MIA-VITA)
  - African-European Georesources Observation System (AEGOS)
  - EAMNet (European Marine Network)

Details of these projects can be found in Section 8.

5.3 European Space Agency

The European Space Agency (ESA) was established on 30 May 1975 to promote cooperation between European States in space research and technology and in space applications, with the aim of them being used for science and operational space application systems. ESA is an intergovernmental organisation, which today includes 18 members[17]. Most of them are also EU members, apart from Switzerland. Canada participates in various ESA activities under a Cooperation Agreement. Currently Hungary, Romania, Poland, Estonia and Slovenia are participating in the Plan for European Cooperating States (PECS), while other countries are in negotiation with ESA in order to join this initiative.

ESA's main activities involve drawing up a European Space Programme and carrying it out. The programmes of ESA are manifold and very broad from scientific and research projects and programmes in outer space to development of satellite-based technologies and services, and promotion of European industries.

ESA is broadly involved in international cooperation with other space faring nations like the United States, Japan, Russia as well as with developing countries like Africa. The Director of ESA's Earth Observation Programme has said that "Africa is relevant for Europe, and space is relevant for Africa."[18]Two important projects, which strength the cooperation between Africa and ESA, are the Global Monitoring for Environment and Security (GMES) and TIGER projects.

GMES is a European Union guided project in partnership with ESA.GMES responds to the African States' needs to "develop an integrated environmental monitoring capability" for the needs of the African citizen such as land, marine and emergency services.

The TIGER Initiative is an ESA founded project that was initiated in 2002 in response to the urgent need for action in Africa as stressed by the Johannesburg World Summit on Sustainable Development (WSSD) regarding water resources. TIGER aims to "help African countries overcome problems faced in the collection, analysis and dissemination of water-related geo-information by exploiting Earth Observation technology." TIGER includes several projects, and it contains two phases - 2002 and 2009. In TIGER different partners cooperate together with ESA such as the UNESCO Water Portal, CSIR, UNOOSA and the UN Economic Commission for Africa to fulfil the task of TIGER. The TIGER Stakeholder Group includes: the African Water Authorities; the African regional and national technical centres; the African regional economic and development institutions; the Committee of Earth Observation Satellites

[17] ESA Member States are: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, The Netherlands, Norway, Portugal Spain, Sweden, Switzerland and the United Kingdom.

[18] http://www.esa.int/esaEO/SEMRGE361AF_index_0.html

(CEOS); the international Agencies which sponsor Integrated Water Management; the international group of funding agencies for Integrated Water Resource Management; the private sector; and non-governmental organisations (NGOs).

6. International Actors

6.1 United Nations

United Nations and Africa

Various organisations within the United Nations are involved with activities in Africa but not all of them are actively involved using space and satellite applications. A selected list of UN Agencies, Departments, Programs, or Affiliates are: UN Sec.-Gen./Security Council, UN Special Adviser on Africa, UN Cartographic Africa Renewal, African Missions, Climate Change-Copenhagen, FAO, Sustainable Development, UNAIDS, UNCCD, UNCDF, UNCHR, UNCHS, UNCTAD, UNDP, POPIN, UNOWA, UN Peacekeeping, UNECA, UNECA-"NEPAD", PADIS, UNEP, UNESCO, UNFPA, UN-Habitat, UNHCR, UNICEF, UNIDO, UNMAS, UNOCHA, UNODC (Drugs & Crime), UNRISD, UNU. Details and contact information for each of the United Nations Agencies on Africa can be found in Appendix A. UNECA will be described in detail in Section 6.1.1 as it has a focal role in Africa and cooperates closely with all other agencies. It has a broad role through its mandate to promote the economic and social development of its member states, foster intra-regional integration and promote international cooperation for Africa's development

United Nations and Space

Regarding space activities there are also a number of organisations within the United Nations that are involved in space technologies and their terrestrial applications. Such agencies actively involved in using space technology are: the International Atomic Energy Agency (IAEA), the International Civil Aviation Organization (ICAO), the World Intellectual Property Organization (WIPO), the World Health Organization (WHO) and the World Bank. Other organisations using space technology and applications are the United Nations Secretariat, the United Nations Development Programme (UNDP), the United Nations Environment Programme (UNEP) and the United Nations International Drug Control Board (UNDCP). These organizations use space technology for the benefit of their Member States within the fields of remote sensing of Earth from space and Geographic Information Systems (GIS), communications and navigation, meteorology and hydrology. They also apply valuable space-related or space-aided activities to basic space sciences and improvement of safety, as well as other activities related to space science. Further to that they promote the United Nations aim of international cooperation in the peaceful uses of outer space. Space related activities within the United Nations system fall into four main categories:

- education and training;
- the provision of expertise to identify areas where space applications could be used;
- the dissemination of information on the status of space technology;
- pilot projects of applications to support economic and social development.

The use of remote sensing and GIS is being utilized particularly for capacity building in developing countries. Most effective efforts are directed towards the strengthening of operational capabilities of existing institutions. Mainly, these are involved in environmental management for sustainable development, natural resource monitoring or land-cover mapping. The application of satellite technology constitutes an important element of technical cooperation in meteorology and hydrology. In addition, satellite communication systems are considered vital tools for social and economic development, as demonstrated by the Regional African Satellite Communications System (RASCOM) project. They are also used to improve communications between the headquarters and the regional offices of the UN organizations and between the regional offices themselves, as demonstrated by the Mercure satellite telecommunications system. Meanwhile, applications of space technology for land, maritime and aeronautical mobile communications are being significantly used by the specialized agencies of the UN.

Through the provision of expert services, a marine and coastal zone data information system is being developed. It will improve management of water and coastal resources and also upgrade meteorological observations by installing networks of data collection plat-

forms (DCPs). It will also improve the meteorological telecommunication system by planning the implementation of a regional network. Reflecting the increasing awareness of global environmental changes, organizations within the United Nations system are cooperating in global schemes such as the Global Climate Observing System (GCOS), Global Terrestrial Observing System (GTOS), Global Ocean Observing System (GOOS), Global Atmosphere Watch (GAW), World Climate Research Programme (WCRP) and the World Climate Data and Monitoring Programme. Turning to the use of space applications to improve global safety, studies have been taking place to apply space technology to establish safety and early warning systems such as the Global Maritime Distress and Safety System (GMDSS) for the reduction of natural disasters.

The Committee on the Peaceful Uses of Outer Space (COPUOS) provides an effective mechanism for keeping states informed of the activities of the international, governmental and non-governmental organizations involved in this field. Within the United Nations family of specialized agencies, the FAO (Food and Agricultural Organization), the ITU (International Telecommunications Union), WMO (World Meteorological Organization), IMO (International Maritime Organization) and UNESCO (United Nations Education, Scientific and Cultural Organization) have been actively involved since the beginning of the space age with satellite remote sensing technology, space communications, satellite meteorology and space science.

The United Nations conduct a coordinated programme of activities in order to utilise space activities in the fullest[19]. In order to achieve this, a United Nations Inter-Agency Meeting on Outer Space Activities is conducted annually to discuss current and future activities, emerging technologies of interest and other related maters. The participating UN organisations are: ECA, ECE, ECLAC, ESCAP, ESCWA, FAO, IAEA, ICAO, ISDR, IMO, ITU, UN-DESA, UN-DPKO, UN-OCHA, UN-OOSA, UNDP, UNEP, UNESCO, UNHCR, UNIDO, UNITAR, UNODC, World Bank, WHO, WIPO, WMO.

The meeting issues a report on it deliberations for the consideration of the United Nations Committee on the Peaceful Uses of Outer Space (UNOOSA), the primary UN body that deals with peaceful uses of space and international cooperation in outer space. The meeting also produces, on behalf of the Secretary-General, a report on the coordinated space-related activities of the UN system[20]. UNOOSA will be described in detail in Section 6.1.2 as it is has the coordinating role for all space activities within the United Nations. Furthermore one of its programmes the UNSPIDER will be discussed in detail in Section 6.1.3.

6.1.1 United Nations Economic Commission for Africa (UNECA or ECA)

The United Nations Economic Commission for Africa (UNECA or ECA) was established by the Economic and Social Council (ECOSOC) of the United Nations (UN) in 1958 as one of the UN's five regional commissions. It is the UN's regional commission serving Africa. UNECA's mandate is to promote the economic and social development of its member states, foster intra-regional integration, and promote international cooperation for Africa's development.

The Commission is organized around six substantive programme divisions: Food Security and Sustainable Development; African Centre for Gender and Social Development; Trade, Finance and Economic Development; Information and Communications Technologies (ICT) and Science and Technology; Governance and Public Administration; and NEPAD and Regional Integration. It also has the African Centre for Statistics, the African Institute for Economic Development and Planning, the Office of Strategic Planning and Programme Management, and the Division of Administration. In addition, five sub-regional offices contribute their perspective to the work programme and support outreach activities. Drawing on this structure UNECA deploys several modalities and services to support its member states: policy analysis and advocacy; enhancing partnerships; technical assistance; communication and knowledge sharing; and supporting sub-regional activities. ECA plays a very active role in African development, particularly in the area of ICT and Science and Technology. However, ECA does not have a specific space program; space technologies and their applications are part of other programmes.

ECA's work programme now focuses on achieving results in two related and mutually supportive areas:

- *Promoting Regional Integration in support of the African Union vision and priorities.* ECA's support to the implementation of AUC's regional integration agenda focuses on undertaking research and policy analysis on regional integration is-

[19] http://www.un.org/events/unispace3/bginfo/activities.htm

[20] The website providing this information is http://www.uncosa.unvienna.org/uncosa/en/index.html.

sues, strengthening capacity and providing technical assistance to institutions driving the regional integration agenda, including strengthening and supporting the Regional Economic Communities (RECs), and working on a range of transboundary initiatives and activities in sectors vital to the regional integration agenda.

- *Meeting Africa's special needs and emerging global challenges.* ECA recognizes the importance of focusing attention on Africa's special needs, particularly within the context of achieving the MDGs. In this regard, ECA places emphasis on supporting efforts to eradicate poverty, placing African countries on the path of growth and sustainable development, reversing the marginalization of Africa in the globalization process, and accelerating the empowerment of women. It aims to provide significant technical support to the African Peer Review Mechanism (APRM) and also to promote peer learning and knowledge sharing in a range of development areas.

Within the two pillars, the commission focuses on the following thematic areas:

- Regional Integration, Trade and Infrastructure;
- Meeting the MDGs with a special emphasis on Poverty Reduction and Growth, Sustainable Development and Gender;
- Promoting Good Governance and Popular Participation;
- ICT, Science and Technology for Development; and
- Statistics and Statistical Development.

The choice of activities to be undertaken under each thematic cluster is based on the demand expressed by Member States and the RECs in various forums as well as on the specific competencies of ECA. In that context, the focus of ECA service delivery is at the regional and sub-regional levels with the Commission's 5 Sub-Regional Offices (SROs) each taking the lead in their respective sub-regions. At the regional level, the thrust of ECA activities is predominantly focused in two areas. First, ECA aims to play a significant role in monitoring and reporting on the progress being made by the continent on meeting global and continental commitments thus supporting progress in mutual accountability. Second, ECA aims to continue its significant contribution in the area of advocacy and consensus building, including developing common positions to give the continent a stronger voice in global forums. With regard to sub-regional activities, particular emphasis is now placed on advisory services and technical assistance aimed at building the capacity of the RECs to implement their agendas and, particularly, to promote regional integration. In particular the ECAs programmes are:

- *African Center for Statistics (ACS).* The objective is to strengthen capacities for statistical development at national, subregional and regional levels, including the promotion of the development of improved and integrated statistical databases in support of decision-making and policy formulation, monitoring and evaluation.
- *Food Security and Sustainable Development (FSSD).* The overall objective is to strengthen the capacity of member states to design institutional arrangements and implement national policies and programmes that reinforce the linkages within the nexus of food security, population, environment and human settlements in order to achieve sustainable development, and to contribute to building the capacity of African countries to utilize science and technology in achieving sustainable development.
- *African Centre for Gender and Social Development (ACGS).* The overall objective of the African Centre for Gender and Development (ACGD) is to inform and influence the content as well as the direction of economic and social development within Africa, with a view to making them more gender- responsive.
- *Governance and Public Administration Division (GPAD).* The main goal of the Governance and Public Administration Division (GPAD) is to improve governance and development management to enhance national capacity and capability in ECA's member states. It also focuses on support for the Africa Peer Review Mechanism (APRM) process.
- *ICT, Science and Technology Division (ISTD).* The objective is to strengthen the capacity for the development and the use of information and knowledge systems, such as spatial databases as support tools for decision-making and for socio-economic development at the local, national and sub-regional and regional levels.
- *Regional Integration, Infrastructure and Trade Division (RIITD).* The programme aims to support on-going efforts to achieve inter-REC free trade areas and common markets and monetary unions

among regional economic communities and their member states towards accelerating the process of Africa's economic cooperation, effective integration in the global economy and physical integration in accordance with the vision and mission of the African Union and its NEPAD program.

- *Economic Development and NEPAD Division (EDND).* The work of TFED focuses on issues relating to macroeconomic analysis, including monitoring of macroeconomic trends and macro-micro linkages. It also deals with international trade, international trade negotiations development, finance and financial sector policies, debt, aid, investment, and industrial policies.

UNECA provides technical assistance and policy advice to African countries and the regional economic communities (RECs). This assistance is delivered through a variety of tools, including on-demand regional advisory services, training workshops and seminars, and fellowship and internship programmes involving visiting scholars and researchers. Regional advisory services have focused on key areas, namely gender and development; ICT; the Millennium Development Goals (MDGs) and poverty reduction; public finance management; trade promotion and negotiation; water; and statistics.

UNECA's five sub-regional offices (SROs) serve as vital links between policy-oriented analytical work generated at headquarters and policy making at the sub-regional level. They also serve as hubs for the dissemination of UNECA's policy through hands-on workshops, training, data collection and knowledge sharing.

The headquarters of UNECA are located in Addis Ababa, Ethiopia and the five sub regional offices are the Central Africa SRO in Yaoundé in Cameroon, East Africa SRO in Kigali in Rwanda, North Africa SRO in Tangier in Morocco, Southern Africa SRO in Lusaka in Zambia, and the West Africa SRO in Niamey in Niger.

UNECA has 53 member states corresponding to the 53 members states of the UN that lie within Africa. The member states are the English Speaking Botswana, Cameroon (also French), Egypt (also French and Arabic), Eritrea, Ethiopia, Gambia, Ghana, Kenya, Lesotho, Liberia, Libyan Arab Jamahiriya (also Arabic), Malawi, Mauritius, Mozambique, Namibia, Nigeria, Seychelles, Sierra Leone, Somalia, South Africa, Sudan (also Arabic), Swaziland, Uganda, Tanzania, Zambia, Zimbabwe; the French Speaking Algérie (also Arabic), Angola, Bénin, Burkina Faso, Burundi, Cameroun (also English), Cap-Vert, Comores, Rep. du Congo, Dem. Rep. du Congo, Côte d'Ivoire, Djibouti, Egypte (also English & Arabic), Gabon, Guinée, Guinée-Bissau, Guinée équatoriale, Madagascar, Mali, Maroc (also Arabic), Mauritanie, Niger, Rép. Centrafricaine Rwanda, Sao Tomé-et-Principe, Sénégal, Tchad, Togo, Tunisie (also Arabic); and the Arabic Speaking Algeria (also French), Egypt (also English & French), Libyan Arab Jamahiriya, Morocco (also French), Sudan (also English), Tunisia (also French).

On 19 July 2010 in Addis Ababa, four European funding partners: Denmark, Norway, Sweden and the UK signed a memorandum of understanding (MoU) under the Joint Financing Arrangement (JFA) to provide substantial support to the ECA Business Plan 2010-2012. JFA is a step by the four partners towards fulfilling the Paris Declaration commitments to increase efforts to harmonize, align and manage aid for results. The commitment was in response to ECA's request for a long-term predictable partner support for projects and programmes in its Business Plan 2010-2012, and complements programmes funded through its regular UN budget.

Contact Information

Office of the Executive Secretary (OES):
Mr. Abdoulie Janneh
Executive Secretary
Ms Lalla Ben Barka
Deputy Executive Secretary
United Nations Economic Commission for Africa
P.O. Box 3005
Addis Ababa, Ethiopia

Office of the Executive Secretary
Tel: +251 11-544 3336
Fax: +251 11 551 4416

ICT and Science & Technology Division (ISTD):
Ms. Aida Opoku-Mensah, Director
United Nations Economic Commission for Africa
P.O. Box 3001
Addis Ababa, Ethiopia

Tel: +251 11 544 3167
Fax: +251 11 551 0512
Email: aopoku-mensah@uneca.org

Website: www.uneca.org

6.1.2 United Nations Office for Outer Space Affairs - UNOOSA

The United Nations Office of Outer Space Affairs (UNOOSA) was created on 13 December 1958 by resolution 1348 (XIII) of the

General Assembly, as a small expert unit within the Secretariat to serve the ad hoc Committee on the Peaceful Uses of Outer Space. In 1962 it became a unit within the Department of Political and Security Council Affairs and was transformed in 1968 to the Outer Space Affairs Division of the Department. Since 1993 the office has resided in Vienna. The Office is responsible for promoting international cooperation in the peaceful uses of outer space, to support the achievement of development goals for the benefit of humankind. The Office implements the decisions of the United Nations General Assembly and of the Committee on the Peaceful Uses of Outer Space and its two Subcommittees, the Scientific and Technical Subcommittee and the Legal Subcommittee.

The Office is headed by Ms. Mazlan Othman of Malaysia who has served as Director of the Office since December 2007 and has two sections: the Space Applications Section (SAS), which organizes and carries out the United Nations Programme on Space Applications (PSA), and the Committee Services and Research Section (CSRS), which provides substantive secretariat services to the Committee, its two subcommittees and its working groups.

Since its creation in 1971, the United Nations Programme on Space Applications (PSA), of the Space Applications Section (SAS), has had the mission to enhance the understanding and subsequent use of space technology for peaceful purposes in general, and for national development, in particular, in response to expressed needs in different geographic regions of the world.

It has made substantial progress in furthering knowledge and experience of space applications around the world. Provision of country capacity-building, education, research and development support and technical advisory services by the Programme have all helped to reduce the gap between the industrialized and developing countries. It is implementing programmes in the following thematic areas:

- Basic Space Science
- Use and Applications of Global Navigation Satellite Systems
- Natural Resources Management & Environmental Monitoring
- Satellite Communications
- Space Technology and Disaster Management

Contact Information

Office for Outer Space Affairs
Vienna International Centre
P.O. Box 500, A - 1400 Vienna
Austria

Phone: (43-1) 26060-0
Fax: (43-1) 26060 5830
E-mail: OOSA@unvienna.org

Website: www.oosa.unvienna.org/

6.1.3 United Nations Space-based Information for Disaster Management and Emergency Response (UN-SPIDER)

The United Nations Platform for Space-based Information for Disaster Management and Emergency Response (UN-SPIDER) programme was established pursuant to United Nations General Assembly resolution 61/110 on 14 December 2006. The mission of UN-SPIDER is to ensure that all countries and international and regional organizations have access to and develop the capacity to use all types of space-based information to support the full disaster management cycle.

UN-SPIDER has been involved in a number of initiatives that make space technologies available for humanitarian and emergency response. UN-SPIDER is the first to focus on the need to ensure access to and use of such solutions during all phases of the disaster management cycle, including the risk reduction phase, which will significantly contribute to reducing the loss of lives and property. UN-SPIDER is the gateway to space information for disaster management support, by serving as a bridge to connect the disaster management and space communities and by being a facilitator of capacity-building and institutional strengthening, in particular for developing countries. In this capacity UN-SPIDER has an active network in various African countries. Currently, in Africa there are four UN-SPIDER Regional Support Offices (RSO) and these are:

- UN-SPIDER RSO in Nigeria is domiciled in the National Space Research and Development Agency (NASRDA). The RSO in Nigeria was established in 2008 with the mandate to promote and support the use of space technology within and outside Nigeria for the management of the full disaster cycle including prevention and mitigation.

  Contact Information:

  Dr. Seidu Onailo Mohammed, Director General/Chief Executive,
  Phone: +234-8033139987

  Dr. Godstime James, Manager of Regional Support Office Remote Sensing and GIS,

Phone: +234-8022739154;
+234-7056195374
E-mail. godstimej@gmail.com

- UN-SPIDER RSO in Algeria is domiciled in the Algerian Space Agency (Agence Spatiale Algerienne, ASAL). It was established during the third African Leadership Conference on Space Science and Technology for Sustainable Development on 7 December 2009 in Algiers.
- UN-SPIDER RSO in South Africa is in the process of becoming established.
- UN-SPIDER RSO in Kenya is the Regional Center for Mapping of Resources for Development (RCMRD), which is an intergovernmental organization with 15 Member States in Eastern and Southern Africa Regions. (See Section 4.5.2). RCMRD provides GEO-Information and ICT products and services.

Contact Information

Mr. Jörg Szarzynski
UN-SPIDER
Hermann-Ehlers-Str. 10
53113 Bonn, Germany
Tel: +49 (0) 815 0677
Fax.: +49 (0)228 - 815 0699
Skype: un-spider-bonn
Email: Joerg.Szarzynski@unoosa.org

Mr. Michael Leitgab
UNOOSA / UN-SPIDER
Wagramer Strasse 5
A-1400 Vienna, Austria
Tel: +43 1 26060 4472
Fax: +43 1 26060 5830
Email: michael.leitgab@unoosa.org

UN-SPIDER
www.unspider.org
E-mail: un-spider@unoosa.org

6.1.4 WMO (and the WMO Space Programme)

The WMO Space Programme coordinates environmental satellite matters and activities throughout all WMO Programmes and provides guidance on the potential of remote-sensing techniques in meteorology, hydrology and related disciplines and applications.

It aims at continuously improving the provision of data, products and services from operational and R&D satellites contributing to the Global Observing System (GOS), as well as facilitating and promoting the wider availability and meaningful use of these data, products and services around the globe.

Space Programme activities are developed around the following four cornerstones:

- Satellite observation requirements from WMO and co-sponsored programmes
- Consolidating requirements, maintaining a Gap Analysis and promoting a vision of the future space-based component of the GOS
- Developing the space-based component of the GOS
- Supporting global planning and optimization, technical coordination, satellite inter-calibration and coordinated processing of environmental satellite observation data, in partnership with satellite agencies, CGMS and CEOS
- Enhancing the availability of satellite data, products and services
- Encouraging data sharing and interoperability, fostering an integrated global data dissemination strategy, promoting advanced dissemination systems, supporting data coding and metadata standardization within the WMO Information system
- Enhancing users' capability to take advantage of satellite applications
- Supporting user information activities, training through the Virtual Laboratory, monitoring satellite data use by WMO Members

WMO Space programme contribute also to E.R. 9: Enhanced capabilities of NMHSs in developing countries, particularly least developed countries, to fulfil their mandates

6.1.5 The Food and Agriculture Organisation of the United Nations (FAO)

The Food and Agriculture Organisation of the United Nations (FAO) is a specialised agency of the UN family which was founded in 1945 and is leading the international efforts to against hanger. FAO's mandate is to raise levels of nutrition, improve agricultural productivity, better the lives of rural populations and contribute to the growth of the world economy. It focuses special attention to developing rural areas which are home to 70% of the world's poor and hungry people.

FAO is composed of eight departments: Agriculture and Consumer Protection; Economic and Social Development; Fisheries and Aquaculture; Forestry; Human, Financial and Physical Resources; Knowledge and Communication; Natural Resources Management and Environment and Technical Cooperation.

FAO works in partnership with various institutions at all levels: private foundations, grassroots organizations, companies, professional

associations, other United Nations agencies, national governments and more. Some partnerships operate at national level or in the field, others are regional or global in nature. FAO works in partnership with others on food security, natural resource management, forestry and fisheries, early warning of food emergencies, disaster recovery, food safety, bioenergy and many other areas. Even though FAO does not have a specific space programme it uses satellite based information and applications to achieve its goals.

Contact Information

Director-General
Dr Jacques Diouf of Senegal
Regional Office for Africa
Maria Helena de Morais Semedo
Regional Office for Africa
FAO Building
Gamel Abdul Nasser Road
Box GP 1628,
Accra
Ghana

Tel. +233302675000
Fax. +233302668427
Email: fao-raf@fao.org

http://www.fao.org/Africa

6.1.6 United Nations Environnent Programme (UNEP)

The Unite Nations Environment Programme (UNEP) with a mandate stemming from 1972 is the UN's designated entity for addressing environmental issues at global and regional levels. UNEP mandate is to coordinate the development of environmental policy consensus by keeping the global environment under review and bringing emerging issues to the attention of governments and the international community for action. UNEP's global base in Nairobi, Kenya it is one of the only two UN programmes that has its base in the developing world the other one is HABITAT in Nairobi.

UNEP has five priority areas: environmental assessment and early warning, development of policy instruments, enhanced coordination with environmental conventions, technology transfer, and support to Africa. Even though UNEP does not have a specific space programme it uses satellite based information and applications to achieve its goals.

The member states of the UNEP are: Algeria, Indonesia, Angola, Islamic Republic of Iran, Antigua and Barbuda, Israel, Argentina, Japan, Australia, Kazakhstan, Austria, Kenya, Bahamas, Kyrgyzstan, Bangladesh, Mexico, Belgium, Monaco, Botswana, Morocco, Brazil, Netherlands, Bulgaria, Pakistan, Burkina Faso, Poland, Burundi, Republic of Korea, Cameroon, Romania, Canada, Russian Federation, Cape Verde, Saudi Arabia, Chile, Senegal, China, Somalia, Colombia, South Africa, Costa Rica, Sweden, Czech Republic, Thailand, Democratic Republic of the Turkey, Democratic republic of the Congo, Tuvalu, France, Uganda, Germany, United Kingdom of Great Britain, Ghana, and Northern Ireland, Haiti, United Republic of Tanzania, Hungary, United States of America, India, Uruguay

Contact Information

Division Headquarters
Cristina Boelcke
P.O. Box 30552
Nairobi 00100,
Kenya

Tel: +254 20 7623 727
Fax: +254 20 7624 270
Email: cristina.boelcke@unep.org

Regional Office for Africa (ROA)
Sekou Toure
P.O. Box 30552
Nairobi 00100,
Kenya

Tel: +254 20 7624 284
Fax: +254 20 7623 928
E-mail: sekou.toure@unep.org

Liaison Office with the African Union
ECA New Building 4th Floor,
No. 4NC4-4N13
P.O. Box 3001,
Addis Ababa,
Ethiopia

Tel: +251 1 443 431
Fax: +251 1 521 633
Email: unepoffice@uneca.org

Liaison Office with the European Union
14 rue Montoyer,
2nd floor B-1000 Brussels,

Tel: +32 2 213 30 50
Fax: 32 2 213 30 51
Email: info@unep.be

http://www.unep.org

6.1.7 World Food Programme

The World Food Programme (WFP) is the world's largest humanitarian agency which fights hunger around the world. It is part of the United Nations system working on a voluntary basis for funding. It was created in 1962 with a vision of the world in which every man, woman and child has access at all times to the food needed for an active and

healthy life. In order to achieve its targets WFP has a three year strategic plan 2008-2011.

The use of space based information and applications can assist the work of WFP. It has created networks of academic partnerships in order to take advantage of the in science and technology. In particular satellite and radar imagery and can provide WPF and its partners with valuable information regarding climate change, early warnings, disaster management and emergency response. It can also provide information for food security.

WFP works together with other sister organizations like FAO and the International Fund for Agriculture Development (IFAD) as well as with other UN agencies, governments, NGO's.

In Africa it works in the following countries: Algeria , Angola , Benin, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Congo, Democratic Republic Of Congo, Republic Of Côte D'Ivoire, Djibouti Egypt, Eritrea, Ethiopia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Rwanda, São Tomé And Principe, Senegal, Sierra Leone.

Contact Information

Via C.G.Viola 68
Parco dei Medici
00148 - Rome - Italy
Tel: +39-06-65131
Fax: +39-06-6590632

www.wfp.org

6.1.8 The United Nations Human Settlements Programme (UN-HABITAT)

The United Nations Human Settlements Programme (UN-HABITAT) is the UN agency for human settlements. It was established in 1978 and in 2002 the agencies mandates was strengthened and it became a fully fledged programme of the UN system. It contributes to the UN system to reduce poverty and promote sustainable development. The UN-HABITAT mandate is to promote socially and environmentally sustainable towns and cities with the goal of providing adequate shelter for all.

In order to achieve sustainable urbanisation and inclusive urban development it focuses on land and housing for all; participatory planning and governance; environmentally sound infrastructure and services; and innovative housing and urban finance. Even though UNEP does not have a specific space programme the use of satellite based information and applications would help to achieve its goals.

UNHABITAT has various partnerships at all levels including Africa

Contact Information

Headquarters
UN-HABITAT
P.O. Box 30030, GPO,
Nairobi, 00100, Kenya

Tel: (254-20) 7621234 (Operator)/ 7623120 (Information Services Section)
Tel: (254-20) 7623151/ 3153 (Media and Press Relations)
Fax: (254-20) 7624266/ 7624267/ 7624264/ 7623477/ 624060
E-mail: infohabitat@unhabitat.org

Regional Offices
Africa and the Arab States
UN-HABITAT Regional Office for
Africa and the Arab States
P.O. Box 30030, GPO,
Nairobi, 00100, Kenya

Tel: (254-20) 7621234/7623221
Fax: (254-20) 7623904/7623328 (Regional Office) 7624266/7 (Central Office)

http://www.unhabitat.org/

7. Space Activities in Africa by African Countries and Non African Countries

Space activities in Africa can be divided into two categories: space activities carried out by African countries and space activities remotely carried out in Africa by non-African countries.

7.1 Space Activities Carried Out By African Countries

7.1.1 Arab Republic of Egypt

The Egyptian Space Agency, The National Authority for Space Science and Remote Sensing, was established in 1994 to pursue, transfer, and provide the most advanced technology in the fields of remote sensing and peaceful application of space sciences, and build the self-capability to utilize these technologies to support national development activities.

Africa's first satellite was launched by Egypt on 28 April 1998. Nilesat 101, with a lifespan of 15 years, was manufactured by Matra Marconi Space (Astrium) and launched by the European Space Agency from Kourou, French Guiana. Similar to Nilesat 101, is Nilesat 102, except that it has an additional Receiving Horn Antenna to receive up-linking from London and Europe. Nilesat 102 was launched on 17 August 2000, by the European Space Agency.

EUTELSAT's Hotbird 4 was leased to the Egyptian Nilesat Company in 2005 and renamed Nilesat 103. Nilesat 101, Nilesat 102 and Nilesat 103 will provide Direct to Home digital TV channels and data transmission.

Egypt's first scientific satellite EgyptSat-1 was launched on 17 April 2007. EgyptSat-1 is an Earth Remote Sensing Satellite. This satellite was built by Egypt's National Authority together with the Yuzhnoye Design Bureau in Ukraine.

A fifth Egyptian Satellite, Nilesat 201, built by Thales Alenia of France, will be launched by Ariane in May 2010.

7.1.2 Ghana

Ghana as a member of the Africa Ministerial Conference on Science and Technology (AMCOST) and Commission on Science and Technology (COMSAT), through the Ministry of Environment, Science and Technology intends to collaborate with the African Regional Center for Space Science and Technology Education (ARCSST) at Ile-Ife on capacity development, and with the National Space Research and Development Agency (NASRDA) in Abuja on capacity utilization.

Specifically in the field of space, Ghana has no agreement with any European country. However, the following are some information concerning Ghana and International Obligations. For meeting International Obligations, Ghana has endorsed the agenda and goals of COPUOS and UNISPACE III and is a state party to the following treaties governing peaceful activities in outer space by signature only: Outer Space Treaty of 1967; Resource Agreement of 1968; and Liability Convention of 1972. Additionally, it has ratified four of the eleven UN Agreements, namely: Treaty banning nuclear weapon tests in the atmosphere, in outer space and underwater; Agreement relating to the International Telecommunications Satellite Organization; Convention on the International Mobile Satellite Organization; and International Telecommunications Constitution and Convention. Ghana has not accepted to or is not part to Registration Convention and Moon Agreement.

Ghana is in partnership with eight other African Countries to lead the continent's bid to host the world's largest telescope and other sub-stations in South Africa and the other countries. This has resulted into the African Square Kilometer Array Programme. The African Square Kilometer Array countries are South Africa, Ghana, Botswana, Zambia, Mauritius, Madagascar, Kenya, and Namibia. As part of the activities being undertaken to increasing Africa's chances of winning the bid, partner countries are establishing Earth Satellite Station on each country. Through the SKA programme, Ghana is proposing to expand the human capacity of the country in the field of Astronomy and Astrophysics by instituting programmes in the Kwame

Nkrumah University of Science and Technology.

7.1.3 Kingdom of Morocco

The Moroccan space agency, Centre Royal de Télédétection Spatiale (CRTS) was created by the government of Morocco to "promote, use and develop Remote Sensing applications in Morocco." Maroc-TUBSAT, Morocco's first remote sensing satellite was launched on 10 December 2001, as an international co-operation between CRTS Morocco and the Institut für Luft- und Raumfahrttechnik at TU Berlin. It was launched by the Indian Space Research Organization's Polar Satellite Launch Vehicle.

7.1.4 Kenya

Kenya is involved in the African Resource and environment Management satellite Constellation (ARMC) Initiative, which is a collaboration between the governments of Algeria, Kenya, Nigeria and South Africa. Regarding bilateral cooperation, the government of Kenya and the government of Italy have an agreement about the San Marco Launching and Tracking Station in Malindi, Kenya. The Agreement is coming to an end at the end of 2010. Currently the two countries are engaged in preparation and negotiations for the renewal of the Agreement which are in an advanced stage.

Furthermore, Kenya is involved in the following activities which require the use space based information and satellite applications:

- Square Kilometre Array (SKA).
- international radio telescope for the 21st century, and the San Marco Project, Malindi, Kenya
- Tripartite Agreement[21] between the Italian Republic, Republic of Kenya and the European Space Agency on the use of the San Marco Launching and Tracking Project in Malindi
- African Regional Centre for Space Science and Technology Education in English (ARCSSTEE (Nigeria)
- African Resource and environment Management satellite Constellation (ARMC)

7.1.5 Nigeria

The Nigerian space agency was established in 1999 for the purpose of establishing a fundamental policy for the development of space science and technology and to establish and coordinate the activities of space research centres in the country. Nigeria's first satellite, NigeriaSat-1, built by Surrey Satellite Technology Limited (SSLT), was launched on September 27, 2003. NigeriaSat-1 is a remote sensing satellite and is one of the satellites that make up the Disaster Monitoring Constellation.

Nigeria's NIGCOMSAT-1 was built by the China Great Wall Industry Corp. (CGWIC). NIGCOMSAT-1 had a projected life span of 15 years, however, in November 2008, NigComSat-1 failed due to a technical error on the satellite's northern solar array. A replacement for the satellite NigComSat-1R will be launched in 2011.

7.1.6 Republic of Algeria

Space activities in Algeria can be traced to 1965, when France used Algeria as a launch site. Space activities were suspended from that period until 2002, when Algeria launched her first satellite. The Algerian Space Agency Agence Spatiale Algérienne (ASAL), was established in 2003, and has as its objective designing and implementing the country's space programs, which must meet national needs (economic, social, and strategic) based on all relevant institutions.

Algeria's first satellite ALSAT-1 was launched on 22 November 2002. ALSAT-1 was built by Surrey Satellite Technology Limited (SSLT), United Kingdom, and forms part of the Disaster Monitoring Constellation (DMC).

In February 2006, four years after the launch of ALSAT-1, the Algerian National Space Technology Centre signed an agreement with EADS Astrium to help develop the ALSAT-2 program which has two satellites Alsat-2A and Alsat-2B. There are also planned Alsat 1B, Alsat-3, Alsat-4 and Alcomsat-1satellites.

Algeria is at 28º latitude North of the Equator, and Longitude 3º 00´ east of Greenwich. This makes it a good launch site. The city of Hammaguira at Latitude: 31.0995. Longitude: -2.8358 was used as an orbital launch site for France's nuclear weapons test and Reggane at Latitude: 26.7189. Longitude: 0.2769 has also been used as a suborbital launch site in the past. Tamanrassett, Algeria is an emergency landing site for space shuttles.

7.1.7 Republic of South Africa

South Africa was the first country in Africa to carry out any space related activity; this was with the establishment of the Royal Observatory at the Cape of Good Hope. Between that time and now, South Africa's activities in astronomy include the Southern African Large

[21] To note that this Tripartite Agreement has not yet been implemented

Telescope (SALT), the largest optical telescope in the Southern Hemisphere; participation in the High Energy Stereoscopic System (HESS), a gamma ray telescope, in neighbouring Namibia; and bidding for the Square Kilometre Array, destined to be the biggest radio astronomical telescope.

In 1999, South Africa launched her first satellite, SUNSAT, built by staff and students of the University of Stellenbosch. SUNSAT had a 2-year life span. Ten years later, on September 17 2009, South Africa's second satellite SumbandilaSat was launched. SumbandilaSat is a remote sensing satellite.

The South African Astronomical Observatory was established in 1820, and it "conducts fundamental research in astronomy and astrophysics by providing a world-class facility and by promoting astronomy and astrophysics in Southern Africa."[22]

In addition to this, there is also the Hartebeesthoek Radio Astronomy Observatory, which was originally built by the National Aeronautics and Space Administration (NASA) in 1961 as a station that would track U.S. unmanned space probes. Today, the centre has a radio telescope that can detect radio waves emitted by many different kinds of objects in the sky. There is also the High Energy Stereoscopic System, which consists of an array of four telescopes that can detect the light produced by gamma rays entering the atmosphere. Finally, there is the Friends of Boyden Observatory, which operates a unique science communication program for the public and schools.

The Centre for Scientific and Industrial Research (CSIR) provides "tracking, telemetry and command (TT&C) services for geosynchronous and polar orbiting spacecraft to the manufacturers, operators and users of satellites and launch vehicles."[23]

Hoedspruit, South Africa, is a location for emergency landing during shuttle launches.

7.1.8 Seychelles

Seychelles has since 1988 launched a geostationary satellite project called SEYSAT to fulfil its own satellite communications requirements. As such, Seychelles did not join RASCOM like the other 43 African countries have done. RASCOM is a consortium of 43 African states. They have pooled their space resources to provide satellite communications for the African continent.

DICT is the Government Department responsible for the Government's space programme. In addition, the President formed a special task force to also explore space resources and this was chaired by the Secretary of State. There were only 2 meetings held. There was no written policy but the core objectives of the SEYSAT were to:

- Provide Seychelles with reliable, robust and sustainable satellite telecommunications services.
- Commercialise the satellite orbital slots and the space/spectrum resources registered with the ITU.

Seychelles benefits from several agreements in respect to meteorological data. For example Seychelles being a signatory to the World Meteorological Organisation (WMO) allows the Seychelles Meteorological Office to exchange meteorological data obtained by satellite with many other countries that are members of the WMO. The Seychelles Meteorological Office has an agreement for the exchange of meteorological data via satellite with South Africa. The Seychelles Meteorological Office has been trying to get a similar agreement with the Indian Meteorological Department (IMD) but this has not been forthcoming.

Seychelles benefits from the Pan African e-Network project funded by the Indian Government but managed under the auspices of the African Union (AU). The Pan African e-Network consists of the Learning Centre, a Tele-Medicine facility and a VVIP node. The latter allows the 53 African Heads of States to have real-time video consultations with each other at no cost using VOIP over satellite.

The Seychelles Fishing Authority (SFA) has an agreement with Mozambique for the provision of the Vessel Monitoring System (VMS) for their vessels. Basically, SFA relays all the data for the vessels carrying the Mozambique flag.

The same type of agreement the Seychelles Meteorological Office has with South Africa, they also have one with France (via La Reunion) for the sharing of meteorological data obtained by satellite. There is also an agreement with the European Meteorological Satellite (EUMETSAT) for the same reason with EU States.

However, it should be noted that DICT on behalf of Seychelles has signed an MOU with the Korean Aerospace Research Institute (KARI) for the purpose of HR capacity development in the area of satellite network coordination and related areas. Seychelles was

[22] South Africa Astronomical Observatory Home Page http://www.saao.ac.za/

[23] Overview: CSIR Satellite Application Center (online) available at http://www.csir.co.za/SAC/index.html accessed on 27 September 2009.

also interested in having access to pictures obtained from the South Korean Low Earth Orbit (LEO) satellite, as that would be useful in the monitoring of Seychelles Exclusive Economic zone (EEZ). If pictures of activity in the EEZ can be obtained in Pseudo-real time mode, then this would be an effective tool to fight against high seas piracy. Furthermore, Seychelles was also going to benefit from the Indian Government with the launching of a very small (micro) multipurpose/remote sensing satellite system.

Currently, Seychelles, is exploring the setting up of a satellite tracking facility in collaboration with the University of Rome but this is still premature.

Through the Pan African e-Network Project Seychelles benefits from free education from several leading universities in India.

At this juncture, it must be clearly explained that Seychelles, similar to many African countries have been stuck with satellite communications, which comes at a premium cost (around $3000 per month for 1 MBPS Internet link), while the rest of the developed world have been receiving cheaper communications solutions/Internet at a fraction of a cost. In addition, while developed nations have enjoyed new satellite technology that make use of extremely small dishes (50cm), Seychelles and most African countries have no coverage of such technology and have to use large dishes which have always been an eye sore in our beautiful country. In Seychelles all telecommunications service providers (and most banks) make use of a VSAT (earth station) to link up a satellite for international connectivity. Now that African nations have submarine fibre optic cable at their doorsteps, the satellite business will face major market forces. Seychelles will shortly have a submarine fibre optic connection to the rest of the world but there will always be a need for satellite communications to link the outer islands with communications services.

7.1.9 Tunisia

Tunisia has established agreements with a number of African countries, especially member states of regional organisations of which Tunisia is also a member. These agreements include partnerships in the field of research and technology transfer and in particular projects concerning the use and exploitation of space data and information in various fields of applications. In these countries there are organisations of particular interest with which cooperation's is fostered. These are CNTS of Algeria, CRTS of Morocco, LCRSSS of Libya and NARSS of Egypt.

Tunisia is a member of a number of regional organisations which are active in the field of space. These are:

CRTEAN, the North African States Regional Centre of Remote Sensing, which has its headquarters in Tunisia and whose board of directors was chaired by Tunisia;

- OACT, the African Organization of Cartography and Remote Censing, whose board of directors is chaired by Tunisia.
- the regional center of science and space techniques teaching for French speaking Africa.
- the Ligue of the remote sensing centres in the Arab world, whose headquarters are in Lybia.
- ARABSAT by hosting a secondary monitoring station of its satellites.

Additionally, through its scientific institutions, Tunisia has a regional activity, such as the Association of Tunisian Young Scientists (AJST) which created in 1974 with the objective to promote and develop scientific spirit and techniques among young people. This association became a member of the Arab Union of spatial science and astronomy in 1998. The AJST, which comprises more than 20 scientific clubs (Space, Astronomy, Ecology, Computer science, Electronic, Robotic, Energy, etc) organizes aerospace activities for young people (aged 10 to 25). Those activities include launching micro rockets, mini rockets, experimental rockets and sounding balloons as well as sky observation as an initiation to astronomy.

Furthermore, the National Centre for Cartography and Remote Sensing (the national structure for the use and exploitation of space data) has a close relationship with various European space agencies for cooperating in activates in the field of space. These are the European Space Agency (ESA), the German space agency DLR, the French space agency CNES and IRD, and the Spanish INFOCARTO.

7.2 Space Activities Remotely Carried Out In Africa by Non African Countries

Kenya hosts the Regional Centre for Services in Surveying, Mapping and Remote Sensing for Eastern and Central Africa, and the Regional Centre for Space Science for Meteorology Services. Located in Kenya is also the San Macro Platform, developed by National Aeronautics and the Ricerche Aerospaziali, as a launch pad for Italian spacecraft. The Euro-

pean Space Agency also has a ground tracking station in Malindi, Kenya.

Dakar, the capital of Senegal was used as a site for transatlantic emergency landing, but had to be shut down for fear of terrorist attacks.

The Yundum Airport, Banjul, The Gambia, has facilities to support shuttle augmented landing during launch especially for launches into the low inclination orbit.

Amilcar Cabral, Cape Verde, is an emergency launch site for shuttle launches. The U.S. National Aeronautics and Space Administration (NASA) also has a tracking station in Cape Verde.

The Roberts International Airport (Monrovia), Liberia, is an emergency landing site for shuttle launches. The Liberian civil war made use of the emergency landing site unsafe.

Kinshasa, capital of Democratic Republic of Congo, is also an emergency landing site for shuttle launches. Political instability in DRC has made the use of this landing site unsafe.

7.3 European Member States Activities

Many European states have direct cooperation agreements or memorandums and are funding projects directly with various African countries. This cooperation is without the involvement of the United Nations, the European Union or any other intergovernmental organisation residing in Europe. In Table 1 the cooperation agreements between European states and African countries can be seen. In the following subsections the activities of different European states will be described.

Country	Agreement with African Countries/Agencies
Germany	German Aerospace Center DLR signed framework agreement with CSIR of South Africa (1995)
Italy	Government of Kenya and Government of Italy (1964). Agreement at San Marko Malindi, Kenya (1995, renewed 2010) on satellite tracking and launching station.
United Kingdom	MOU British National Space Center (BNSC) (since April 2010 UK Space Agency) and the Algerian Space Agency (ASAL) (2006)
France	Intergovernmental agreement between France and Algeria (2006). Implemented by the National Centre for Space Studies (CNES) and the Algerian Space Agency (ASAL). Three themes of cooperation: a) Telemedicine and tele-epidemiology; b) Hyper-Spectral space technologies; c) reduction of seismic vulnerability.

Table 1: European Member States- African Counties Agreements

7.3.1 Belgium

Through its membership of ESA, Belgium and the company Newtec are participating in the building of the first ever eLearning pilot platform on effective electoral assistance and electoral administration in the Democratic Republic of Congo (DRC). This project is called "SatElections" and other partners include Openet Technologies (I), SES Astra (L), ESA, UNDP, as well as the Independent Electoral Commission of the DRC. Via a cooperation agreement with the UNESCO-World Heritage Centre, Belgium also financed the SYGIAP Project and developed, in close cooperation with the Institut Congolais pour la Conservation de la Nature, a cost-efficient system allowing to map and monitor, by using satellite imagery, the 5 natural World Heritage sites and their buffer zones, as a basic decision supporting tool for improved site management with respect to nature conservation and sustainable development in the DRC.

Although Belgium does not have any bilateral agreement with an African country, many Belgian partners, such as institutes, universities, government and even commercial companies are involved in cooperative projects with African partners. A lot of these projects are conducted through EU, ESA, and EUMETSAT. Some examples are the Global Monitoring for Food Security, VGT4-Africa, and DevCoCast. But also via the French – Belgian SPOT-VEGETATION programme and the national STEREO-programme cooperative projects are executed in the domain of Earth Observation.

7.3.2 France

1) French space agreements in Africa

a) Intergovernmental agreement with Algeria

An intergovernmental agreement between France and Algeria was signed on 1 February 2006, which was implemented by the Na-

tional Centre for Space Studies (CNES) and the Algerian Space Agency (ASAL). It aims to support and promote the scientific, technical, industrial and commercial cooperation between the two states in the study and use of outer space for peaceful ends.

Three themes of cooperation were identified and will give rise to projects:

- Telemedecine and tele-epidemology to enable Algerian health public services to benefit from French knowledge;
- Hyper-spectral space technologies to know the environment better;
- Reduction of seismic vulnerability thanks to space data;

Two mixed committee meetings took place in April 2006 and February 2009. Also, the French Embassy of Algiers supports the training of Algerian students incurriculae in the space sector.

b) Discussions with South Africa

Since 2005, France, via CNES, has regularly met governmental actors from South Africa. In early 2010, cooperation in the domain of oceanography was discussed more thoroughly with the new South African National Space Agency (SANSA). Ocean observation from geostationary orbit is in fact a topic that could lead to a joint mission, as France and South Africa are on the same longitudes.

Cooperation in the telemedicine sector is also under study.

Additionally, the French Embassy in Pretoria was closely involved with the Franco-South-African Institute of Technology, the F'SATI, which welcomed its first students of space science in January 2009.

2) The development of space applications for health in Africa

In the framework of these activities of societal applications, in the area of development for health, France, through actions led by CNES, is supporting the development of surveillance and early alert systems for emerging or re-emerging diseases in Africa, with a 'tele-epidemiology' approach.

In concrete terms, the sanitary authorities of Senegal will validate an early alert system for the Rift Valley fever. This same approach is deployed in Burkina Faso to fight malaria and in some Mediterranean and North African countries to follow Vibrio diseases linked to the quality of coastal waters.

With respect to the area of sanitation, which groups telemedicine and tele-education activities, France has been involved in two projects led by CNES in Mali since 2008 and in Benin since 2009, which are financed by the Ministry of Immigration, Integration, National Identity and Solidarity Development (MIIINDS).

These projects could be extended to other French speaking nations.

3) The development of African competencies in teledetection

France is involved, via CNES, in the GEO-Net-CaB project. Supported by the European Commission since November 2009, this project integrates South Africa and the Regional African Centre of Space Sciences and Technologies in French Language (CRASTE-LF) as non-European partners. It will advance knowledge of the status of the needs and programmes on earth observation and will conduct a sensitising campaign on using these data and services for improving the adaptation capacity of nations facing climate change.

Furthermore, France participates in the SEAS Indian Ocean project that has established a satellite receiving station on the island of Réunion for the development of regional cooperation in terms of expertise and use of teledetection. The targeted nations are the members of the Indian Ocean Commission and in particular Madagascar, whose national committee of teledetection benefits from French expertise and financial support.

France also supports a regional Masters programme "Teledetection and Natural Risks" co-organised by the universities of the Reunion and of Antananarivo that should commence in October 2010.

4) The use of space for analysis of deforestation in tropical forests.

In December 2007,the UN Climate Change Convention in Bali decided to launch a process of awareness of the reduction of greenhouse gas emissions related to deforestation. For the nations of the Congo basin, this is about being able to actively participate in future negotiation mechanisms on climate change by producing the necessary information including national policies, measures and actions aimed at reducing greenhouse gases linked to deforestation and the degradation of forests.

In this framework, the *GEO Forest Carbon Tracking Task* (GEO FCT) has the specific aim of establishing a worldwide monitoring mechanism, in close partnership with governments, space agencies, scientific communities, the private sector and the relevant United Nations agencies. France participates

in this international initiative by encouraging the Congo basin nations to adhere to the GEO and by helping them insert themselves in this plan to contribute. Gabon should shortly join GEO.

At the Copenhagen summit, France committed itself to further financing the provision of SPOT satellite imagery for the monitoring of the Congo basin until 1025. The CNES participates in the implementation of this commitment, with numerous other French partners.

The SEAS Gabon project, resulting from a partnership between France and Brazil, provides a satellite reception antenna in Gabon, associated with a centre of expertise in teledetection for cartography and forest monitoring, as well as the establishment of environmental observatories. CNES has brought its expertise to the preliminary study of the implementation of this station.

Finally, since 2007 France has been associated with the "Open Initiative on the use of space technologies to support the World Heritage Convention", conducted by UNESCO. In this context, CNES has made an inventory, with IRD, of forest heritage sites in Latin America and Africa, from space data. Today Madagascar is the privileged partner of France and Africa on this topic.

7.3.3 Germany

The use of space technology in German-African research projects is still largely limited to remote sensing applications and the subsequent use of Earth observation (EO) data. On the other hand, EO data is now widely used to various extent in many research projects from biology, geography or agriculture to urban planning, without necessarily making such projects pure "space projects". In this sense, such a list of projects can not be considered complete or authoritative, but it gives good examples of the benefits of space-based research for African citizens and societies.

Projects at the German Aerospace Center (DLR)

DLR – German Remote Sensing Data Center – various projects

There is close cooperation with different institutions in Southern Africa:

- In the field of disaster management with Namibia, this includes emergency mapping provided by DLR's Center for Satellite Based Crisis Information after the floods in northern Namibia in 2008 and 2009. Moreover, capacity building and training activities have been performed together with the Namibian Ministry of Agriculture, Water and Forestry.
- In the field of Data and Information Management Systems (DIMS) an operational system has been installed at the CSIR Satellite Application Center (South Africa).
- With various universities and the South African National Biodiversity Institute (SANBI) in the context of biodiversity research.
- Currently a project proposal is being elaborated on a „Namibia Transboundary Flood-Disease Monitoring and Mitigation System – An International SensorWeb Pilot Project" in cooperation with UN-SPIDER, NASA/NOAA and other partners.

Furthermore, a project proposal is being developed in accordance with BMBF to establish Regional Science Service Centers in Southern Africa. Remote sensing components include Earth observation (EO) data processing (thematic products) and service provision, data management and archiving, EO applications for wet lands, extreme events, water management and satellite based crisis information. Section 2.1 above has more information concerning the establishment of Regional Centers of Excellence in Africa.

SAMUM (until 2008)

The first phase of SAMUM (Saharan Mineral Dust Experiment) was dedicated to the understanding of radiative effects in the source region. A first joint field campaign took place at Ouarzazate and Zagora, Morocco in August/September 2005 to create the experimental data set of surface and atmospheric columnar information. This data set will provide the base of the first thorough columnar radiative closure tests in Saharan dust. By the end of the first phase of SAMUM (end of 2006) regional radiative forcing and updated global estimates of climate effects of Saharan dust were provided. The second phase will quantify for the first time the radiative effects of the mixed plume of Saharan dust and biomass burning aerosols as it leaves the source region. For the characterization of the mixed plume a second joint experiment was envisioned on Sal, Cape Verde in spring 2007. As in the source region, closure efforts will be made in the complex aerosol mixture of the polluted dust plume. The feedback between diabatic processes and atmospheric dynamics will require major modelling efforts. The second phase of the SAMUM field campaigns started on 10 January 2008. The main objective of the measurements was the characteri-

zation of aged mineral dust. Therefore, after the first campaign in Morocco, the Islands of Cape Verde were selected as a measurement site. It is expected that a mixture of mineral dust, biomass burning aerosols and marine aerosols could be observed. The remote sensing instruments of the MIM include two lidar systems, a sun- and skyphotometer, a microtops-instrument for ozone columnar concentration and a UV-radiometer. The experiment ended on 15 February, 2008. At the end of SAMUM a realistic description of dust related aerosol processes and feedbacks will be provided with which global scale dust modelling will be improved.

The SAMUM Consortium

- Institute for Atmospheric Physics, Deutsches Zentrum für Luft- und Raumfahrt (DLR)
  - Airborne in situ and remote sensing studies on the vertical and areal distribution of microphysical and optical properties of Saharan dust
- Applied Remote Sensing Cluster, Deutsches Zentrum für Luft- und Raumfahrt (DLR)
  - Longwave Radiative Effects of Saharan Dust and the Impact of Particle Shape
  - Modelling and observation of Sahara dust transport, modification and radiative impact
- Leibniz-Institute for Troposheric Research Leipzig (IfT)
  - Modelling and observation of Sahara dust transport, modification and radiative impact
  - Multiwavelength lidar profiling of Saharan-dust/biomass-burning aerosol layers over the Cape Verde Islands
  - Optical properties of mixed mineral, biomass burning and marine aerosols at dry and ambient conditions
- Meteorological Institute, University of Munich
  - Multiwavelength LIDAR profiling of Saharan-dust/biomass-burning aerosol layers over the Cape Verde Islands
- Institute for Atmospheric Physics, University of Mainz
  - Longwave Radiative Effects of Saharan Dust and the Impact of Particle Shape
  - Modelling and observation of Sahara dust transport, modification and radiative impact
- Institute of Applied Geosciences, Technical University of Darmstadt
  - Physicochemical parameters of mixed mineral, biomass burning and marine aerosols
- Institute for Remote Sensing, University of Bremen
- Laboratoire de Physique de l`Atmosphère, Département de Physique, Faculté des Sciences, Université Mohammed Premier, Oujda, Morocco
- IFM-GEOMAR, University of Kiel
  - Flight operations (Partenavia) and technical support
- enviscope GmbH, Frankfurt am Main

Further Cooperation:

MISR-Satellite-Team (Multi-angle Imaging SpectroRadiometer), Jet Propulsion Laboratory (NASA), California

Study: Quantitative and robust discharge of soil parameters from hyper-spectral remote sensing data

The South African "Thicket Biome" with its dense bush vegetation stores in pristine condition very large amounts of carbon for a semi-arid region. Currently the region is subject to a fundamental change from dense bush to an open savannah-like system that is triggered in particular by overgrazing by goats and is reflected in addition in the degradation of vegetation including reduction of soil fertility. Local institutions working on the reforestation of the biome are interested in the quantification of above and belowground carbon storage. In this context, the most important physical parameters of the soil (organic carbon, iron, clay) are being derived and quantified using hyperspectral remote sensing.

Cooperation:

Deutsches Zentrum für Luft- und Raumfahrt – German Center of Applied Remote Sensing

Living Lands Foundation – Presence Network, South Africa

University of Stellenbosch, Department of Soil Science, South Africa

Bilateral Agreements

Country:	South Africa
Partners:	BMBF – Federal Ministry of Education and Research, Department of Arts, Culture, Science and Technology, RSA
Type of Agreement:	Contract
Purpose:	Cooperation in the fields of science, research and technology
Signed on:	12.06.1996
Country:	South Africa
Partners:	DLR, CSIR
Type of Agreement:	Framework Agreement
Purpose:	
Signed on:	27.07.1995

The German Aerospace Center DLR signed the Framework Agreement with CSIR of South Africa in 1995, which has been used since then as a basis for different forms of cooperation.

Initiatives of the Federal Ministry for Education and Research (BMBF)

Establishment of Regional Centers of Excellence in Africa

The Federal Ministry for Education and Research (BMBF) together with partners from Southern and West Africa has established a new initiative called “Climate Change and adapted land management in Southern Africa”. A key element of this initiative is to set up regional Centers of excellence (Regional Science Service Centers) as a long-term African research infrastructure. The RSSC is an initiative of the BMBF, in cooperation especially with South Africa, Namibia, Angola, Zambia and Botswana. The RSSC will focus on three thematic areas: climate change, water and land management. The initiative consists of:

- a research programme
- a capacity development programme
- a regional advisory and information service.

A series of workshops is planned to prepare the RSSC initiative. There will be workshops in Angola, Botswana, Namibia, South Africa and Zambia. These mainly serve to discuss and scope the RSSC from a national point of view.

To this end the BMBF is going to invest 5.1 million EUR in the next two years. The investment for the further build-up phase may later be up to 100 million EUR. The key question of this research cooperation is how agriculture can adapt to the increasing water scarcity. Another central issue for researchers is how people can better protect themselves against floods.

The Center for Development Research, University of Bonn, directs currently the construction of one of the two regional Centers supported by the BMBF – the “WASCAL” (West African Science Service Center on Climate and Adapted Land Use) in Ouagadougou, Burkina Faso. The aim is to collect, store and analyze the data gathered by African and international researchers on climate change and land use. The second Center will be built in South Africa.

German University in Cairo

The German University in Cairo (GUC) is one of the largest single projects within the Federal Government's programme to export German study programmes to other countries. The GUC is of exceptional political importance and is a reflection of the outstanding collaboration between the two countries. The most recent development was the introduction of new subject priorities.

Key areas of cooperation

The current priority topics with potential relevance for space research and technology are materials research, environmental research and climate research. Further areas of cooperation are physical and chemical technologies, biotechnology, aquaculture and health research. The so-called Lake Nasser Project is a thematically wide-ranging and extensive project of scientific and technological cooperation (STC) with Egypt. Its aim is to find ecologically acceptable ways of developing new settlement and economic areas for approximately 1.5 million people in the region of the Aswan dam. The project is to be continued under the future BMBF funding priority Integrated Water Resources Management (IWRM).

Biodiversity Monitoring Transect Analysis in Africa (BIOTA Africa)

The Remote Sensing Unit deals with local and regional model generation of land cover distribution for further analysis in the BIOTA project. Besides land cover and land use mapping over a wide area, current research

is targeted at a new spectral unmixing approach for more precise land cover detection. Through this, more sensitive temporal change detection and more accurate estimation of species distribution dependence on specific land cover types is possible. Further research topics are modelling regional biomes distribution for conclusions about species occurrence, and analysing fire patterns in Western Africa

- Scientific support for sustainable use and conservation of biodiversity in Africa.
- A continental observation network in Africa, contributing to GEOSS (Global Earth Observation System of Systems).
- A network for observing land degradation and for developing measures to combat desertification in Africa.
- A network for capacity development and rural development in Africa.

Core Participants

Austria , Bénin, Burkina Faso, Côte d'Ivoire, Finland , Germany, Kenya, Morocco, Namibia, Senegal, South Africa, The Netherlands, Uganda .

BIOTA-West Africa

BIOTA-West[24] is an interdisciplinary cross-national subproject of the project network BIOTA AFRICA. Therein, German and West African partners are engaged in close collaboration in biodiversity research in the Ivory Coast, Burkina Faso and Benin. The goal is to identify processes leading to a species decrease and to develop methods for their conservation. Building up a continental monitoring network of African biodiversity and land degradation will deliver to decision makers scientifically established and socioeconomic basics for efficient and sustained environmental management.

- The contemporary status of biodiversity in West Africa and concepts for its sustainable conservation
- Land use changes: agriculture and restoration[25]
- Tools and products for decision making and practical application[26]
- Capacity building[27]

[24] For further information see www.biota-africa.org
25 http://www.biota-africa.org/West_Phase3_WP.php?subcode1=0&maincode=B
26 http://www.biota-africa.org/West_Phase3_WP.php?subcode1=0&maincode=C

BIOTA East Africa

The BIOTA East Africa consortium works in the third funding phase for the completion of its mission, which received a sharper profile during phase II. Especially during the past two years, all efforts have focused on five overarching themes with their well-defined deliverables. All work packages contributed to this goal and adapted their programs to achieve BIOTA East Africa's objectives.

- Monitoring and assessment of biodiversity in space and time
- Understanding natural processes of change
- Understanding human use, value and impact in space and time
- Recommendations and measures for sustainable use of biodiversity
- Capacity building and information policy at local, national, and international levels
- Cross-cutting Themes

BIOTA Southern Africa

BIOTA Southern Africa decided to develop a structure that will support the integrated generation of user-oriented products. The new structure consists of 28 work packages funded by BMBF and 11 projects funded by the DS&T. Each of the work packages integrates various disciplinary and interdisciplinary activities, always defined by one or several joint products, generated for one or several specific users. Strong interest in various products has been expressed explicitly in letters of support.

Existing structures based on individual disciplines will be maintained where appropriate, however, the project structure reflects the product-oriented approach.

- Drivers of change
- Monitoring of Change
- Land use, Impact and Value
- Process Analysis
- Interventions and Restoration
- Transformation and Capacity Development

BIOTA Morocco

BIOTA Maroc was set up to use the biodiversity monitoring results to create suitable tools for sustainable land use and resource management under changing environmental and socio-economic conditions. In selected mu-

27 <http://www.biota-africa.org/West_Phase3_WP.php?subcode1=0&maincode=Dg

nicipalities, we will initiate pilot projects for which, together with our Moroccan partners, it will develop adequate intervention schemes and knowledge transfer formats for participative resource planning processes at the communal level.

Local communities shall be enabled to integrate external scientific information and indigenous system knowledge about resource availability and fragility in modern planning processes and management concepts, in order to strengthen their collective identity. In the long run, an enhanced capacity for autodetermined local resource management seems the best way of optimising ecosystem services and of initiating endogenous development processes that help to conserve the fragile environment of arid mountain ecosystems.

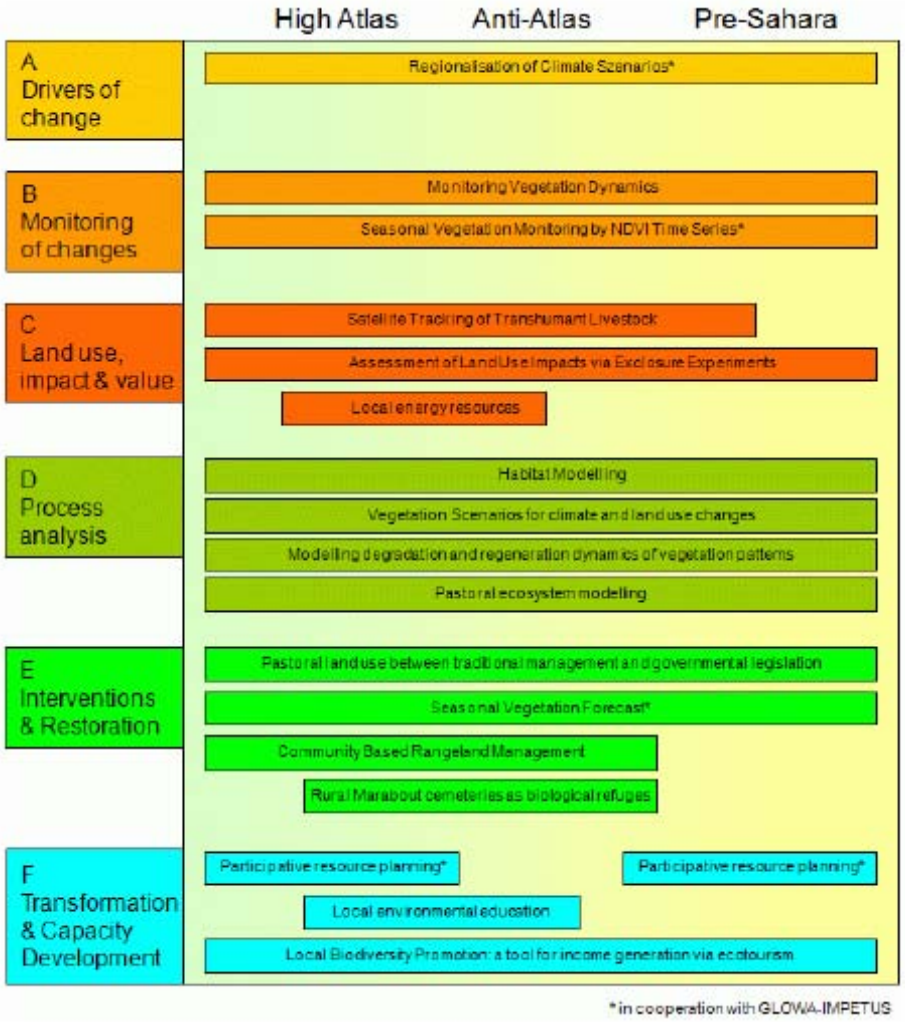
BIOLOG Africa: BIOTA Congo Project

The Cuvette Centrale is a reservoir of medicinal plants. In 1999, the Ministry for Education and Research (BMBF) invited project applications for a new interdisciplinary program with focus on biodiversity and global change (BIOLOG) in Africa. It started in 2001, with the establishment of personnel and infrastructural foundations in Kinshasa and Salonga National Park. Since 2002, the field site LuiKotal, in the southwest of Salonga National Park, has been operating continuously.

7.3.4 Italy

Under the terms of several Agreements between the Government of Kenya and the Government of Italy, (the first of them concluded in 1964), Italy is present in Malindi, in the location where the San Marco Base has been set up.

The Agreement under which the cooperation is undertaken at present is the "Agreement between the Government of the Republic of Kenya and the Government of the Republic of Italy concerning the Satellite Tracking and Launching Station at the San Marco Malindi, Kenya", signed in Nairobi on 14 March 1995. The Agreement has been renewed until 31 December 2010 and is currently under renegotiation. Under Italian Law Decree 128/03, the responsibility of managing the



base has been entrusted to the Italian Space Agency (ASI) since 1 January 2004. Previously, that responsibility accrued to the University of Rome "La Sapienza", through the CRSPM (Centro Ricerche Progetto San Marco).

The location of the San Marco Base, which has been recently re-named "Luigi Broglio - Space Center (BSC)", on the Indian Ocean is ideal to launch and support equatorial satellites, provide TT&C activities and acquire Earth Observation (EO) data over the Central and Eastern Africa region.

The "Broglio Space Center (BSC)" is divided into two main segments:

- the Sea Segment, consisting of 5 offshore platforms for the rocket launch activities (activity suspended);
- the Land Segment, known as Base Camp, covering an area of about 3.5 hectares. It contains a TT&C Complex, with two ground stations that provide real time acquisition, recording, and transmission of satellite data, and a Remote Sensing Center, dedicated to Earth Observation satellite data acquisition and processing; it also includes a geophysical facility.

The space activities are currently concentrated in the two Ground Stations of the land Segment and both are currently active in: 1) the TT&C and acquisition of the scientific data of the NASA SWIFT and the ASI AGILE astrophysics satellite missions; 2) tracking from and hosting support at Malindi for ESA and CNES, (tracking support to the satellites launched by the Ariane launchers, carried out from the Center Spatial Guyanese in Kourou; in the future, these services will be extended also to the Soyuz and Vega launches). Cooperation activities with ESA are undertaken pursuant to a tripartite Agreement among ESA, the Government of the Kenya Republic and the Government of the Italian Republic, signed in 1995, which is going to be extended and amended to take into account the new services that will be provided from the BSC. The BSC has provided support to the Chinese human missions Shenzou.

The Broglio Space Center also hosts experimental receiving antenna for the GALILEO constellation installed at the request of ESA, which are devoted to testing the reception of the navigation message transmitted by the first two satellites of the constellation: Giove-A and Giove-B. The data collected are then relayed to the European Space Operation Center (ESOC) in Darmstadt (Germany) for analysis. An ESA service for a geodetic GPS receiver is also in operation.

In the frame of the Inter-governmental Agreement, education programmes are also provided for. ASI supports the technical education and professional training of Kenyan students. Scholarships have been offered by the Italian Government to Kenyan researchers for PhDs in Italian Universities, complemented with a one year traineeship in Italian space industries and/or research Centers.

The Remote Sensing stations are currently on stand-by in the acquisition of Earth observation satellites data, but considering the important role that space applications may play in that area, the Joint Steering Committee set up for the management of the bilateral Italian-Kenyan Agreement has appointed a joint Working Group with the task of producing a Feasibility Study for the realization of a Regional Center for Earth Observation in Kenya. The study foresees a distributed facility: acquisition, pre-processing, archiving and cataloguing would take place at the BSC in Malindi; processing, specific products generation and distribution to users would take place in Nairobi.

In the new draft Agreement between the Government of the Republic of Kenya and the Government of the Italian Republic currently under negotiation, there is a specific mention of the two main European space applications programmes, in particular:

1. The Lisbon Declaration on Global Monitoring for Environment and Security (GMES) and Africa, dated 7 December 2007, and the Maputo Declaration, of 15 October 2006, both recognizing the important role that Earth Observation Services play in the sustainable development and security of Africa;
2. The participation of the Government of the Italian Republic in the Galileo Program and the important benefits that may accrue from this program to African Countries.

Moreover, since August 2008, ASI has started to study the possibility of launching stratospheric balloons and sounding rockets from the Broglio Space Center.

Furthermore, ASI has recently negotiated with the Egyptian National Authority for Remote Sensing and Science (NARSS) a Memorandum of Understanding for collaboration in the field of Earth Observation, Space Applications and Technologies, which is expected to be signed in the forthcoming months.

7.3.5 Netherlands

The Netherlands has no bilateral agreement with any African country or space agency. The Dutch consultancy bureaus and value adders have a large number of cooperative projects with African parties, both national and via ESA, EU, UN etc., some also on a private commercial basis. Some well-known examples are WACMOS and TIGER projects via ESA, the PUMA/AMESD projects via the EU, UN projects like RiskView and Agromet-Shell, several Water projects and the FESA Food project. Next to that the well known ITC University Faculty at Enschede has educated and trained several thousands of African students in geo-informatics and Earth sciences in the past decades.

7.3.6 Switzerland

Switzerland does not have any specific agreements with African partners in the space sector. The numerous bilateral agreements of scientific and technical cooperation with African partners could, however, include some projects in the space sector. As Switzerland does not have a national space programme it conducts the majority of its space activities through ESA.

In terms of partnerships or projects in Africa making use of space applications, there is a certain number of them whose nature, location, concerned actors, and origin of spatial data are very varied. An example is the establishment in east Chad of a hydrogeological geographical information system by Swiss specialists seconded at UNHCR and at UNITAR/UNOSAT. There are also other projects in the Horn of Africa involving Swiss academic centres in cooperation with local entities and often international agencies, aiming at the realisation of data banks, notably 'mapping' using satellite imagery (either commercial or through other channels).

7.3.7 Spain

Spain has no direct cooperation agreements between Africa and Spain in the field of space. Indirectly, this cooperation is channelled through the ESA programmes in which Spain participates, particularly in the case of the GMES programme and specifically through the GMES and Africa Action Plan. Furthermore, the European Space Agency (ESA) has contracted Indri, a Spanish company, which works in remote sensing systems and applications, to conduct a characterization study of Mediterranean Africa (Algeria, Egypt, Libya, Morocco and Tunisia) on Earth Observation (EO) sector. This project, named GAZELLE PLAN, belongs to the EOMD programme (Earth Observation Market Development), which is devoted to fostering the usage of geoinformation products and services within new markets and sectors. The aim of the GAZELLE PLAN is to analyse the current situation of the EO sector in this region in order to draw a Strategic Plan to steer activities during the following 5 years. The study includes a market characterization supported by the execution of mini-projects (or service trials) in different countries involving local stakeholders and European & Canadian companies, and a final action plan reflecting the recommendations for the future.

Furthermore, there is some Spanish space companies working in consortiums that develop international activities with Africa based on space technologies. The projects are listed below.

Projects Based on Navigation

SIRAJ

In the context of the 7º Frame Programme, three Spanish companies are taking part in the SIRAJ consortium (Pildo Labs as coordinator, and Navya Solutions and SENASA as partners). SIRAJ (SBAS[28] Implementation in the regions of ACAC and ASECNA), is a project recently approved in the 7th FP 2nd GALILEO Call Theme Transport (including Aeronautics) under the International Activities & EGNOS Service Extension. The project duration will take 18 months.

The main objective of SIRAJ is to perform concrete actions in support of the EGNOS service extension into areas covered by the Arab Civil Aviation Commission (ACAC) and the Agence pour la *Sécurité de la Navigation aérienne en Afrique et à Madagascar* (ASECNA).

The activity aims at promoting and demonstrating the benefits to a critical sector such as civil aviation in a real environment and also attacking the necessary actions to develop a suitable framework towards regional EGNOS service extension with a direct strategy.

It implies formulating the main guidelines and recommendations for a possible Euro-ACAC Global Navigation Satellite System in order to evaluate a potential investment in EGNOS Service Extension in the ACAC and ASECNA regions. SIRAJ intends to:

- Achieve a definitive action regional plan focusing on involving the major players in the process.

[28] Satellite-Based Augmentation Systems

- Demonstrate to the major regional stakeholders the benefits of EGNOS.
- Create the necessary solid collaborative framework between institutions to ease service extension and future operations in the Region.

In this activity, Pildo is in charge of the management of the whole activity and the promotion of the activity. In addition, the company is leader of the concept proof and flight demonstrations and the related conclusions and recommendations for cooperation with the region.

The Consortium: Coordinator & Partners

Partners	Country
Pildo Labs (coordinator)	Spain
Telespazio	Italy
Egis Avia	France
Navya Solutions	Spain
ACAC	International Organization
ASECNA	International Organization
SENASA	Spain

METIS

The Spanish company Pildo Labs was involved in the consortium for the project METIS, which was led by the Italian company Telespazio.

The main objective of METIS was to prepare the introduction of GNSS services in the Mediterranean region aiming at the enhancement of transport safety, reliability and security.

Three main activities were performed in light of this goal in the frame of METIS:

- The first was the elaboration of a Euro-Mediterranean Policy to prepare the introduction of the EGNOS service, and the future Galileo, in the MEDA area, for the period 2009-2019, taking into account the particular needs of the area and the social and economic benefits that their use will generate.
- A set of GNSS Service demonstrations was then conducted to remark the benefits of the use of EGNOS for Civil Aviation and freight transport (maritime, road and rail).
- A complete training and awareness programme complemented all the activities performed within the project and created a solid collaborative framework between institutions.

Within the service demonstrations, Pildo Labs acted as technical support for the set up and installation of the MedaCoN network, continuously monitoring the EGNOS signal through a network of six data collection stations located in Tunis, Morocco, Turkey, Israel, Palestine and at Pildo Labs' premises. The results obtained were stored and presented through a website also developed by Pildo Labs and the main objective of the network, which was to demonstrate the possibilities offered by the EGNOS system to Civil Aviation over the region, was fully accomplished.

Moreover, and also in the frame of the METIS demonstrations, Pildo Labs gave support to Telespazio in the frame of the civil aviation demonstrations. In addition to the help on the activity coordination, Pildo Labs designed and encoded an LPV[29] procedure for the EGNOS flight trials at Çanakkale airport that demonstrated that EGNOS will enable APV[30] operations in the Mediterranean countries reducing the need of the conventional and expensive ground navigation aid infrastructure such as ILS[31] to perform precision approaches.

The Consortium: Coordinator & Partners

Partners	Country
Telespazio (coordinator)	Italy
Al-Akhawayn University	Morocco
ESSP	France
FDC	France
Thales Alenia Space	France
Pildo Labs	Spain

Projects Based on Earth Observation

The MalarSat Project

Malaria is one of the major public health challenges undermining development in the world. The aim of the MalarSat Project is to provide malaria risks infection maps on a global scale using Earth Observation data to support and prevent epidemic episodes. The proposed service for creating malaria risk maps would be critically useful to improve efficiency in insecticide programs, vaccine

[29] Localizer performance with vertical navigation
[30] Approach Procedure with Vertical guidance
[31] Instrument Landing System

campaigns and the logistics epidemic treatment.

Different teams have already carried out studies in order to exploit the use of Earth Observation (EO) data with epidemiology purposes. In the case of malaria risk maps, it has been shown that meteorological data is not sufficient to fulfil this objective. In particular, being able to map the malaria mosquito habitat would increase the accuracy of risk maps. The malaria mosquitoes mainly reproduce in new water puddles of very reduced dimensions (about 1 meter wide). Different teams have already carried out studies in order to exploit the use of Earth Observation (EO) data with epidemiological purposes. Nowadays it has been demonstrated that there is no instrument that could detect such small patches of water unless there are many of them spread in an area of several hundreds of meters. MalarSat aims at using the radar altimeter data from the EnviSat, RA-2, to try and build indicators of mosquitoes' existence.

Radar altimetry measurements have been traditionally used to monitor ocean surfaces and ice caps. Previous studies have shown the potentialities of radar altimetry to monitor land surfaces at the global scale. However, radar altimetry applications over continental surfaces remain relatively underdeveloped. The major innovation of this study is the use of EnviSat RA-2 data to detect water puddles of very reduced dimensions (about 1 meter wide) where typically malaria mosquitoes reproduce. The interest of RA-2 relies on its ability to discriminate a particular echo shape. Due to the flatness of these water surfaces (and therefore very specular reflection), these areas will produce a particular echo shape even if the water surface is only a small part of the radar footprint. Despite RA-2 spatial resolution (about 350m) being rather poor when compared to other radar systems like ASAR (20m), its ability to discriminate a particular echo shape can enable detection of water surfaces much smaller than the RA-2 spatial resolution. In this study, five-year-measurements of the EnviSat radar altimeter are studied over Mozambique. The temporal and spatial stability of the parameters from the different nominal retracking procedures is assessed.

The results show, among other things, the capabilities of radar altimeter to map flooded areas. These results are very promising and they open ground for new applications in the future. The Spanish company Pildo performed the data management tools and data visualization of the project, in strong collaboration with the Spanish International Health Centre Hospital Clinic (CRESIB, Barcelona) and the Spanish companies GTD Information Systems and IsardSAT -leader of the project-.

CBERS-2B General Information

The CREPAD[32] Program of the Remote Sensing Area of INTA[33] is aware of the constant increase and improvement of the Earth Observing Satellite network. Therefore a constant effort is being made to incorporate data from those new satellites and sensors. During the last year the implementation of CBERS 2B data have been carried out.

The CBERS-2B satellite, successfully launch in 2007, is a low earth polar orbit satellite aimed at the observation of the Earth. Onboard it carries three cameras but only two of these cameras will be acquired by CREPAD antennas in Maspalomas Space Station (located in the island of Gran Canaria, Spain). In March there was a problem with the Attitude Control System and the satellite lost attitude. At the moment CBERS 2B is not working. It is planned to be replaced by CBERS-3 in 2011.

The setting up of the Maspalomas Space Stations for receiving CBERS 2B images has included among its tasks the distribution of data free of charge to all interested Spanish, African, Brazilian and Chinese users[34]. This is an important initiative of wide spreading availability of remotely sensed data to African countries, promoting the use of geoinformation for the sustainable development of Africa.

The image catalogue accessible via Internet will be accessible by means of the CREPAD web site (http://www.crepad.rcanaria.es/en/index-en.html).

7.3.8 Sweden

There is no agreement between Sweden or SNSB and an African counterpart in the field of space. There is no other partnership or program between Sweden and Africa but SNSB does sponsor a few minor projects that are performed in cooperation between Swedish researchers/companies and African counterparts and that rely on space technologies for their implementation. Under this framework some projects in Sweden are:

[32] Center for REception, Processing, Archiving and Dissemination of Earth Observation Data.
[33] (Spanish) National Institute for Aerospace Technology.
[34] For this purpose, a Memorandum of Understanding (MoU) between INPE (The Brazilian National Institute for Space Research), INTA and CRESDA (China Center for Resources Satellite Data and Applications) has been signed by Brazil, China and Spain in 2008.

- The project "Choosing the right way in the Mtwara Development Corridor" aims to demonstrate the benefits of combining satellite data with other relevant geographical information to facilitate the planning of sustainable development in developing countries in Africa. It is undertaken by the World Wildlife Fund (WWF) together with the company Metria.
- The project performed by the company Forsway concerns bringing Internet access to the African people using satellite based triple-play. This is now being developed further in an ESA-project (in Artes-3).
- The project "Remote Sensing of Carbon Dynamics in the Sahel" is performed by the University of Lund.

7.3.9 United Kingdom

In 2006 the United Kingdom entered into a Memorandum of Understanding between the British National Space Center (BNSC) and the Algerian Space Agency (ASAL). The aim of this MOU[35] is to provide a framework of collaborative activities and to review areas of common interest in the civil aspects of space; to facilitate the interchange of information, technology and personnel in areas of mutual interest. The United Kingdom had previously cooperated in connection with Algeria's national satellite AISAT-1.

[35] The link for the MOU can be found in the following link: http://www.ukspaceagency.bis.gov.uk/assets/channels/about/MoU_Algeria.pdf

8. Projects in Africa

8.1 Alphabetic list of projects in Africa

In the following subsections a list of projects that have been competed and are ongoing in Africa using satellite applications are listed. These projects are developed by European, African and International actors. The list is not exhaustive but provides an overview of the types of projects developing.

8.1.1 Africa Health Infoway

Name

Africa Health Infoway

Objectives

Support health workers with high quality data to improve patient care, assist health managers with collecting real-time data from health facilities, help policy-makers with reliable health data for evidence-based decision-making, support health managers with information to monitor and evaluate programmes and resources, support the general public by providing all the necessary information for making informed choices about health programmes, and support partners with sustainable feedback on the programmes to which they are committed.

Description

Africa Health Infoway is an ICT-based network of e-health solutions, to support health at districts in 53 African countries. It is planned for implementation over 5 years, delivering connectivity to global information, strengthening the capture, use and exchange of health information.

Consortium

This project is conducted under the World Health Organisation.

Contact Information

Mr. Sahlu Getachew
Email: sahlug@who.int

http://www.who.int/africahealthinfoway/en/

8.1.2 African Monitoring of the Environment for Sustainable Development (AMESD)

Name

African Monitoring of the Environment for Sustainable Development (AMESD)

Objectives

AMESD aims to extend the operational use of Earth Observation technologies and data to environmental and climate monitoring applications in order to provide all African nations with the resources needed to manage their environment more effectively and ensure long-term sustainable development in the region.

Description

AMESD builds on the success of the Preparation for the Use of Meteosat Second Generation in Africa (PUMA) programme and paves the way for GMES Africa.

It has received funding from the European Union's European Development Fund and is being implemented by the African Union's Commission, with international technical assistance and the support from five Regional Implementation Centres, EUMETSAT and the European Commission Joint Research Centre.

The programme is being executed in close cooperation with five Regional Economic Groupings – the Economic Community of West African States (ECOWAS), Communaute Economique et Monetaire de l'Afrique Centrale (CEMAC), the Intergovernmental Authority on Development (IGAD), the Southern African Development Community (SADC), and the Indian Ocean Commission (IOC) – the African, Caribbean and Pacific Group of States (ACP) Secretariat. The World Meteorological Organization, UN Economic Commission for Africa (UNECA), UN Environment Programme (UNEP) and Food and Agriculture Organization (FAO) are also overseeing the programme.

AMESD's focus areas are

- Providing the African user community with better access to Earth Observation, field and ancillary data;
- Setting up five operational regional information services to support and improve decision-making in environmental management
- Establishing national, regional and continental environmental information processes, frameworks and activities to enable African governments to meet international environment treaty obligations
- Organizing specialized training and staff exchange programmes to maintain the technical capability of African AMESD stakeholders in the long term.

CEMAC THEMA: Management of water resources

ECOWAS THEMA: Water management for cropland and rangeland management

SADC THEMA: Agricultural and environmental resource management

IGAD THEMA: Land degradation mitigation and natural habitat conservation

IOC THEMA: Coastal and maritime management

Funding Body

The European Union's European Development Fund

Consortium

The responsible body for this project is the African Union Commission. The consortium consists of Thales Alenia Space; IRD, Research Institute for Development, France; BRL Ingénierie and Conzortio ITA, Italy; AGRYMET, Regional Implementation Center for the ECOWAS region; CICOS, Regional Implementation Centre for the CEMAC region; BDMS, Regional Implementation Centre for the SADC region; ICPAC, Regional Implementation Centre for the IGAD region; MOI, Regional Implementation Centre for the IOC region; EUMETSAT; EC-JRC. This project started in October 2007 and will have a duration of 48 months.

Contact Information

AGHRYMET
admin@aghrymet.ne
Tel: (227) 2031 5316 / (227) 2031 5436
Fax: (227) 2031 5435
www.agrhymet.ne
CICOS
Cicos_int@cicos.info
Tel: (243) 81 555 36 76 / (243) 81 555 36 75
www.cicos.info
BDMS
meteo@gov.bw
Tel: (267) 395 6281
Fax: (267) 395 6282
www.weather.info.bw
ICPAC
director@icpac.net
Tel: (254) 20 387 83 40
Fax: (254) 20 387 83 43
www.icpac.net
MOI
moi@intnet.mu
Tel: (230) 427 4434
Fax: (230) 427 4433
www.moi.gov.mu
info@amesd.org
www.amesd.org

8.1.3 African Monsoon Multidisciplinary Analyses (AMMA)

Name

African Monsoon Multidisciplinary Analyses (AMMA)

Objectives

African Monsoon Multidisciplinary Analyses (AMMA) is an international project to improve our knowledge and understanding of the West African monsoon (WAM) and its variability using a multi-scale approach: time-scales, space scales, etc.

AMMA's main goal is to provide African decision makers with improved assessments of similar rainfall changes which are likely to occur during the 21st century due to natural fluctuations and as a result of anticipated global climate change. An essential step in that direction is to improve our ability to forecast the weather and climate in the West African region.

Consortium

Based on a French initiative, AMMA was built by an international scientific group and is currently funded by a large number of agencies, especially from France, the UK, US and Africa. It has been the beneficiary of a major financial contribution from the European Community's Sixth Framework Research Programme.

The Senegalese scientific teams involved in AMMA are CSE, UCAD, IPD and DIREL. Medias-France is working with the Senegalese scientific teams. The study site is Barkedji, located in the Ferlo area.

CSE, UCAD and Medias-France are working together on environmental issues (climate, meteorology, hydrology, water quality, re-

mote sensing (land use and land cover)…). IPD is in charge of all aspects related to entomology and DIREL is involved in veterinary investigations.

Contact Information

http://www.redgems.org/spip.php?rubrique61

8.1.4 African Resource Management Constellation (ARMC)

The African Resource Management Constellation (ARMC), a collaboration currently involving Nigeria, South Africa, Kenya and Algeria, aims to develop a constellation of satellites to provide real time, unrestricted and affordable access to satellite data to support effective environmental and resource management in Africa. The first workshop was held in May 2005 in Algeria. There is already a constellation design that would have each satellite equipped with a 2.5m resolution panchromatic imager and a 5m resolution multispectral imager in 6 multispectral bands. The program would also include capacity building initiatives and the development of low-cost multi-source ground receiving stations to aid the less privileged countries who can gain access through these stations to remote sensing and meteorological satellite data. The ARMC Space Agreement was signed in Algiers in December 2009.

Contact Information

Prof. Ade Abiodun
Tel: +1-516-333-5474
Cell: +1-516-884-0559
E-mail: aaabiodun@optonline.net

8.1.5 African Satellite Communication & Galileo Applications (AFSAGA)

Name

African Satellite communication & Galileo Applications (AFSAGA)

Objectives

The purpose of AFSAGA is to develop awareness of the latest telecommunication technologies and their potential use in the southern African region. The project analyses the potential applications available from Galileo and satellite communications in South Africa as well as in the Southern African Development Community (SADC) region.

Description

South Africa and its neighbouring countries have a combination of well-equipped telecommunication urban centres with a large area of less-equipped telecommunication infrastructure density. It is this combination that makes the use of satellite applications a clear advantage in national development. Galileo combined with other satellite applications is clearly an enabler towards sustainable development in the SADC region.

AFSAGA will demonstrate to the SADC community that satellite applications can not only efficiently support economic sectors, such as industry, farming, marine, mining, tourism, construction, road infrastructure and civil aviation, but also events, such as the 2010 FIFA World Cup.

Informing key users

The possibilities offered by combined Galileo and satellite-based telecommunication applications, as well as what has already been carried out in other regions, will be communicated to selected user communities.

Analysing user needs

The second step is an analysis and consolidation of identified user needs. This consolidation process will include the cross-checking and justification of a proposed satellite-based solution. This process will be used to analyse the regional barriers to the implementation of such applications as well as the establishment of an action plan to be discussed with regional stakeholders.

Funding Body

Co-financed by EC under FP6

Consortium

The Council for Scientific and Industrial Research (CSIR), South Africa and Thales Alenia Space, France who was the coordinator. The project started on 01/03/2007 with a duration of 12 months.

Contact Information

Yannick Lefebvre
THALES ALENIASPACE, FRANCE
Navigation Business Unit
Building P, Door 265 26,
Ave J. F. Champollion,
31037 Toulouse, France
Tel: +33 5 34 35 71 95
Fax: +33 5 34 35 55 18
Email:
yannick.lefebvre@thalesaleniaspace.com
http://www.afsaga.org

http://ec.europa.eu/enterprise/policies/space/files/afsaga.pdf

8.1.6 African-European Georesources Observation System (AEGOS)

Name

African-European Georesources Observation System (AEGOS)

Objectives

The AEGOS project aims at setting-up the preparatory phase for the building of an information system containing and making accessible data and knowledge on African geological resources including mineral resources, raw material, groundwater and energy (georesources).

Description

AEGOS is a Support Action of the European Union 7 th Framework Programme. It is actually the preparatory phase of a pan-African observation system including the information and knowledge so far collected and future acquisitions regarding georesources in Africa. The information system will be based on a distributed architecture with local and Internet access. The management of the intellectual propriety rights on the data sets will be addressed.

The main objectives of AEGOS project are the definition of:

- operational procedures for data management (spatial data infrastructure, metadata and data),
- user-oriented products and services including the preparation of innovative spin-off projects,
- the African-European partners network: strengthening and development,
- a geoscience contribution to GEOSS, in the context of the Infrastructure for Spatial Information in Europe (INSPIRE).

Consortium

In this project consortium was: Beak Consultants GmbH, Germany; BGR Bundesanstalt für Geowissenschaften und Rohstoffe, Germany; BGS British Geological Survey, United Kingdom; BRGM Bureau de Recherches Géologiques et Minières, France ; CIFEG Centre International pour la Formation et les Echanges en Géosciences, France; CGS Council for Geoscience, South Africa; CzechGS Ceska Geologicka Sulzba, Czech Republic; DNG Ministère des Mines et de l'Energie - Direction Nationale de la Géologie, Guinea ; GD-SOM-UNZA Geology Department, School of Mines, University of Zambia, Zambia; GSD Geological Survey Department, Ghana; GSE Geological Survey of Ethiopia, Ethiopia; GSM Geological Survey and Mines Department, Uganda; GTK Geologian Tutkimuskeskus, Finland; INETI Instituto Nacional de Engenharia, Tecnologia e Inovação, Portugal; IRA Institute for Resource Assessment, Tanzania; IRD Institut de Recherche pour le Développement, France ; JRC European Commission/ DG Joint Research Centre/ Institute for Environment and Sustainability, International Organisation; MMI Ministère des Mines et de l'Industrie - Direction des Mines et de la Géologie, Sénégal ; PGI Panstwowy Instytut Geologiczny, Poland; RBINS Royal Belgian Institute of Natural Sciences, Belgium, Partner; SEAMIC Southern and Eastern African Mineral Centre, International Organisation; TNO Netherlands Organisation for Applied Scientific Research, Netherlands; UEMOA Union Economique et Monétaire Ouest Africaine, International Organisation; DGS Department of Geological Survey, Botswana; DNG M Direcção Nacional de Geologia, Mozambique; EUMETSAT European Organisation for the Exploitation of Meteorological Satellites, International Organisation; MAWF Ministry of Agriculture, Water and Forestry, Namibia; RMCA Royal Museum for Central Africa, Belgium; SGU Sveriges Geologiska Undersökning, Sweden. The project was co-funded by EC FP7 Environmental theme. The project started on 09/02/2009 with a duration of 30 months.

Contact Information

AEGOS Project - Work Package 9 Co-leader.
Ms Cindy Petersen (Council for Geocience)
cnr. Oos & Reed Streets
Bellville, Cape Town, South Africa 7535
Telephone: +27 21 943 6730
Mobile: +27 78 233 4071
Fax: +27 86 615 5858
http://www.aegos-project.org

8.1.7 Africover

Name

Africover

Objectives

To strengthen the capabilities of African countries for sustainable planning and management of natural resources, by producing reliable land cover databases and strengthening national capacities in operational applications of remote sensing.

Description

The use of remote sensing and geographic information system technologies for natural resource management. It also seeks to establish a digital georeferenced database on land cover for Africa. Africover reinforces national and subregional capacities for the establishment, update and use of land cover maps and spatial data bases and produces geographic information required for decision making, planning and natural resources management in African countries. Based on the Landsat Thematic Mapper (TM), Africover collects and produces data for 10 African countries: Burundi, Democratic Republic of the Congo, Egypt, Eritrea, Kenya, Rwanda, Somalia, Sudan, Uganda and United Republic of Tanzania.

Consortium

This project is under the Food and Agriculture Organization (FAO), United Nations Environmental Program (UNEP). The funding is provided by the Italian Cooperation, FAO and UNEP

Contact Information

c/o FAO Representative in Kenya
P.O. BOX 30470 - 00200 City Sq.
Nairobi, Kenya
Tel: +254 - 20 - 3743454
Fax: +254 - 20 – 3743498
Email: info@africover.org
http://www.africover.org/index.htm

8.1.8 Assessment of ICT for DM in Africa – Best practices and information sharing (AIDA)

Name

Assessment of ICT for DM in Africa – Best practices and information sharing (AIDA)

Objectives

The objectives of the AÏDA project are :

- to reduce the risk of natural disasters in Africa
- to improve the capacity to respond to disasters
- to fill the ICT information gap
- to promote existing successful ICT solutions;
- to open GEONETCast for alerting purposes;
- to strengthen the European leadership in ICT solutions;
- to pave the road for long-term ICT cooperation with Africa.

Description

Disaster occurrence and loss has increased rapidly over recent decades. In the past 20 years, over 3 million people have lost their lives around the world in large-scale natural disasters, the majority of them in developing countries. The situation is particularly problematic in Africa, a continent which is very prone to the negative effects of climate change. In some countries, years of development gains have suddenly been lost, deepening poverty for millions and leaving them even more exposed to future natural hazards. The African communities are very vulnerable economically, socially and environmentally to natural hazards and the situation is aggravated by the rapid growth of the population. Many of these fragile economies and communities do not have sufficient capacity to resist the disaster. Out of sheer necessity, many African people live at high risk. The increasing number of natural hazards strikes those who are already the most vulnerable.

“Reducing the risk of natural and human-induced disasters and improving the capacity to respond to emergencies” was deservedly identified as one of the key-societal challenges to be addressed within the “ICT for sustainable growth” work programme. better prevention (monitoring), detection (early warning) and management (planning and deploying forces when a crisis has been triggered) of these disasters could have saved many of these 3 million lives. ICT-based solutions can play a crucial role here, as they can reduce the risks the African population is exposed to. As such, ICT can save many lives.

On top of these considerations, we see that there is a huge demand for ICT related information in Africa at the moment. During the World Conference on Disaster Reduction in Kobe/Japan in January 2005, all countries agreed to implement the Hyogo framework for action. This action plan builds upon the resilience of nations and communities to disasters and is translated into concrete actions at all levels. Many countries are setting up National Disaster Action Plans and require decent ICT related information to select the most appropriate ICT tools.

In spite of the urgent need, many African countries have no access to or cannot afford the most innovative or modern ICT solutions. Nevertheless, some ICT solutions have already successfully proven their value in Africa while other ICT solutions might look promising but need to be investigated and potentially tailored to the African situation. On the African side, many stakeholders could

benefit from the knowledge and experience that's already available to other communities or in other countries within Africa. Therefore, sharing knowledge, experience and best practices about affordable and successful ICT solutions theatre used in support of DRM (Disaster and Risk Management) in Africa is of utmost importance.

Multiple sources (e.g. the UN/ISDR website) already provide disaster information for Africa. Little to nearly no information is available about ICT solutions in support of the mitigation and management of the disaster. There is a lack of ICT information available and it forms a gap.

The AÏDA project wants to fill this gap, and collect and share the ICT for DRM knowledge and solutions in Africa. As such it will advance the use of ICT solutions by the stakeholders involved in disaster management across multiple hazards categories and across country boundaries within Africa. This activity does not compete with existing initiatives from the UN. At the contrary, AÏDA fully complements those initiatives by focusing on the role of ICT within DRM, an aspect that has hardly been addressed.

Consortium

The coordinator is the Flemish Institute for Technological Research (VITO), Belgium. The partners are: Ardhi University Disaster Management Centre (ARU), United Republic of Tanzania; Council for Scientific and Industrial Research (CSIR), South Africa; European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), International Organisation; GEOSAT Technology, France ; Institut d'Economie Rural du Mali (IER), Mali; International Institute for Geo-Information Science and Earth Observation (ITC), Netherlands; Federal University of Technology (FUTM), Nigeria; Switzerland; Technology Sans Frontière (TSF), Luxembourg. This project has been funded under FP7. It started on 01/06/2008 with a duration of 24 months.

Contact Information

Project Coordination
Gilliams Sven
Tel: +32-14-336827
Fax: +32-14-322795
Email: sven.gilliams@vito.be
www.aidaonline.info

8.1.9 City Mapping Project (SIM) in Sudan

Name

City Mapping Project (SIM) in Sudan [36]

Objective

The objective of the City Mapping Project (SIM) is the updating and complementing of available geographic information for the support of aid delivery in South Sudan, which is grounded on the physical infrastructure.

Description

The SIM project focuses on the development and establishment of "two-sided A1 foldable maps" in Sudan by very high resolution satellite imagery and vector data. The tools for the maps, satellite imagery and vector data, have been planned by a series of assessments of some months duration. The requirement and the use of the maps is related to humanitarian workers, who are in need of accurate mapping to locate agency offices, logistical assets (fuel points, warehouses, transport hubs), guest houses and other accommodation, and major roads and streets. Moreover, these maps can be used as a basis for longer term urban planning and monitoring by local authorities. The main activities in this project are an inter-agency working group, data collection/management training, and to compile updated info and training GoSS.

Consortium

This project was funded by the United Nations and is in cooperation of Central Bureau of Statistics, UNDP, UNMIS, NSSE, EUSC,UNJLC, FAO, UNICEF, UNHCR,WHO, UNFPA,UNOCHA.

Contact Information

Joe Crowley
Information Management Unit
House 23, block 11
Riyadh
Khartoum, Sudan
Tel: +249 (0) 912160272
Email: simadmin@unsudanig.org

[36] http://www.unsudanig.org/sim/index.php?fid=projects

8.1.10 Coordination and Advancement of Sub-Saharan Africa-EU Science & Technology Cooperation Network (CAAST-Net)

Name

Coordination and advancement of sub-Saharan Africa-EU science & technology co-operation network (CAAST-Net)

Objectives

CAAST-Net serves as a platform aiming to advance bi-regional cooperation in science and technology.

Description

CAAST-Net is a four-year joint Africa-Europe platform dedicated to advancing bi-regional cooperation in science and technology.

The Joint Africa-EU Strategic Partnership, endorsed at the December 2007 Summit in Lisbon, is a political vision and roadmap for future cooperation between the two continents in existing and new areas. Cooperation between Europe and Africa in the field of science and technology, with its long and multifaceted history, is one of the pillars supporting the increasingly diverse relationship between our two continents and is a strand running both explicitly and implicitly through the joint strategy and its first action plan.

Against the background of a global consensus that indigenous capacity in science and technology is an essential pre-requisite to economic competitiveness, sustainable development and poverty reduction, the Network for the Coordination and Advancement of sub-Saharan Africa-EU Science & Technology Cooperation (CAAST-Net) has been developed as a high-level platform, financed by the European Union's Seventh Framework Programme (FP7), to advance international cooperation between Europe and Africa for mutual benefit.

CAAST-Net's goal is an increase in the quality and quantity of bi-regional cooperation in science and technology between Europe and Africa, targeting areas of mutual interest and benefit through greater use of instruments under the FP7, as well as through other instruments of international cooperation, and through lobbying for greater synergy between R&D and development instruments.

In support of its goals, CAAST-Net has a range of activity clusters aiming:

- To support and inform existing Europe-Africa S&T policy dialogue and cooperation processes.
- To identify and prioritise common research areas of mutual interest and benefit, especially for inclusion in the thematic priorities of the FP7 Cooperation Programme, harnessing cooperation to address specific problems faced locally and regionally within Africa, as well as common problems of a global nature.
- To promote synergy between Europe-Africa S&T partnerships and development cooperation instruments in order to enhance the creation and application of new knowledge in support of achieving the MDGs, poverty alleviation and economic growth.
- To undertake specific activities dedicated to strengthening of the participation of African countries in the Framework Programme through a series of events aimed at raising awareness, providing information, brokering partnerships and optimising synergies.
- To monitor the performance and impact of Europe-Africa S&T cooperation under the Framework Programme to inform future cooperation policy.

Consortium

African partners are the Ministry of Higher Education, Science and Technology (MoHEST), Kenya; Research (Africa) (Pty.) Ltd (RA), South Africa; Ugandan National Council for Science and Technology (UNCST), Uganda; Ministry of Education (MINEDUC), Rwanda; Ministry of National Education and Scientific Research (DR-MENRS), Madagascar; Science and Technology Policy Research Institute, Council for Scientific and Industrial Research (CSIR-STEPRI), Ghana; Direccao Geral de Ensino Superior e Ciencia (DGESC), Cape Verde; Ministry of Scientific Research and Innovation (MINRESI), Cameroon; Ministère de la Recherche Scientifique (MRS), Senega;
Department of Science and Technology (DST), South Africa. The European partners are the Africa Unit, Association of Commonwealth Universities (on behalf of the UK Department of Universities, Innovation and Skills) (AU-ACU), United Kingdom; International Bureau and National Contact Point Life Sciences of the German Federal Ministry of Education and Research (PT-DLR), Germany; Institut de Recherche pour le Développement (IRD), France; The Research Council of Norway, Division for Strategic Priorities (RCN), Norway; University of Jyväskylä/ Finnish Universities' Partnership for International Development (on behalf of the Finnish Ministry of Foreign Affairs) (JyU-UniPID), Finland; Swedish Governmental Agency for Innovation Systems (VINNOVA), Sweden; Centre de Coopération Internationale en Recherche

Agronomique pour le Développement (CIRAD), France; Fundação para a Ciência e Tecnologia (FCT), Portugal. This project was funded under the FP7 and expected to be completed in 2013.

Contact Information

Africa Region Coordinator
Dr Eric Mwangi

CAAST-Net Africa Region Coordinator
Principal Research Officer Department of Research Development
Ministry of Science and Technology
Harambee Avenue, 9583-00200 Nairobi
Kenya
Tel: +254 (20) 241369
Email: emwangi23@yahoo.com

EU Region Coordinator
Dr Andrew Cherry

CAAST-Net Coordinator
Science and Technology Adviser
The Africa Unit
The Association of Commonwealth Universities
Woburn House, 20-24 Tavistock Square
London WC1H 9HF, United Kingdom
Tel: +44 (0)20 7380 6746
Email: a.cherry@acu.ac.uk

Administrative Officer
Ms Sara Calamassi

CAAST-Net Administrative Officer
The Africa Unit
Association of Commonwealth Universities
Woburn House, 20-24 Tavistock Square
London WC1H 9HF
United Kingdom
Tel: +44 (0)20 7380 6723
Email: s.calamassi@acu.ac.uk

http://www.caast-net.org

8.1.11 Climate for Development in Africa Programme (ClimDev-Africa)

Name

Climate for Development in Africa Programme (ClimDev-Africa)

Objective

ClimDev-Africa aims to enhance economic growth and progress towards the MDGs through mitigating the vagaries of climate variability and climate change, and to ensure that development achievements already gained are climate resilient in the longer term.

Description

ClimDev is an African development programme to integrate Climate Risk Management (CRM) into pertinent policy and decision processes throughout the continent. It will be implemented under the direction of the Joint AUC-ECA-AfDB Secretariat with support from a number of sources. It is envisaged as a 3-phase programme over an 11-year period. Climate Risk Management is both: a) A rational way of coping with the effects of climate variability today, and b) A practical, 'no regrets' way of incorporating longer-term climate uncertainty and risk, into policy and decision making tomorrow. Currently, climate data is little used in development processes in Africa because of weaknesses in both demand for, and supply of, pertinent climate services. In order to strengthen resilience to short-term climate variability, ClimDev-Africa will assess primary stakeholder needs in relation to climate information and build capacity in institutions (public, private and civil) and people to use climate knowledge more effectively in planning and decision-making. It will also build capacity in climate and weather institutions, to be able to provide the many information services required.

Consortium

This project is implemented under the direction of the Joint AUC-ECA-AfDB Secretariat.

Contact Information

Email: yadeyemi@uneca.org

8.1.12 Crisis Monitoring and Response Technologies (CriTech)

Name

Crisis Monitoring and Response Technologies (CriTech)

Objectives

Crisis Monitoring and Response Technologies-CriTech aims to develop and verify solutions to enhance decision-making in crisis management including experimenting with multiple-source information retrieval techniques, real-time data stream prioritisation/visualisation, numerical modelling systems, and visualisation/display systems.

Description

CriTech has been developed under the policy Theme "Europe as a World Partner", Agenda "Global Security"

It is based on the Strategy of the Global Security and Crisis management Unit of JRC, which aims at risk reduction, crisis prevention, crisis preparedness, crisis response and recovery.

Through the development and verification of solutions, Cri-Tech strengths the EU's partners and its capacity in crisis prevention, preparedness and response, focusing especially on humanitarian natural disasters, public health crises, conflict and political crises. Instruments include the Community Instrument for Stability, the Community Humanitarian Aid Instrument, the Community Disaster Risk Reduction Action Plan, the Community Action Plan on Reinforcing the European Union's Disaster Response Capacity, the Community Health Programme, and the external security dimension of GMES.

Its objectives are:

- to develop reliable and up-to-date Geospatial data infrastructure and the maintenance of existing geospatial data infrastructure, including updating pinning essential datasets and upgrading underpinning software and hardware to maintain its high quality, efficiency and reliability.
- to develop field-based data gathering tools to support crisis situation awareness and post crisis needs assessments. This project is run in cooperation with ISFEREA Action.
- to develop situation and secure room models
- to host critical servers and a secure working environment to handle/store sensitive data on a restricted access basis in a collaborative crisis management portal. This project is run in cooperation with ISFEREA Action
- to model tsunamis and tropical cyclones
- to develop and implement a Global Flood Detection System
- to model epidemiological events and quantitative risk assessment of emerging health threats

Consortium

This project is under the leadership of JRC.DG.G Institute for the Protection and Security of the Citizen, Ispra (Italie) and is currently ongoing.

Contact Information

Mr. Alessandro Annunziato, leader
alessandro.annunziato@jrc.ec.europa.eu

8.1.13 Cross-cutting project "Application of remote sensing for integrated management of ecosystems and water resources in Africa"

Name

Cross-cutting project "Application of remote sensing for integrated management of ecosystems and water resources in Africa"

Objectives

The target of the cross cutting project is to promote biodiversity conservation, to aid sustainable development and to help alleviate poverty in Africa.

Description

The development of processes, which give the possibility to African countries to access and use satellite data and information and communication technologies, including the Internet, Remote Sensing (RS) and Geographic Information Systems (GIS) to monitor, assess and manage Ecosystems and Water Resources.

Consortium

This project is under UN sponsorship and UNESCO assemblies

Contact Information

Mr. K F Seddoh, UNESCO Project leader:
Email: k.seddoh@unesco.org
Mr. J Ahanhanzo, UNESCO Project Manager
Email: J.Ahanhanzo@unesco.org

8.1.14 Distribution of Vegetation Data in Africa through EUMETCAST (VGT4-AFRICA)

Name

Distribution of vegetation data in Africa through EUMETCAST (VGT4-AFRICA)

Objectives

The VGT4AFRICA project aimed at setting up an operational and timely distribution system of VEGETATION data from the SPOT satellites and creating high level derived products to all African countries.

Description

The first task ('get it there') was to make the production and near-real time distribution of 10 products fully operational. The data is provided, free-of-charge, to all non-commercial users all over Africa. In order to reach out to remote areas and areas lacking internet connectivity, the EUMETCast satellite

telecommunication system, operated by EUMETSAT, complements the deliveries via the web and on fixed media. EUMETCast has been used in various research projects and is rapidly becoming the global GEONETCast data dissemination system.

Delivering state-of-the-art environmental data on the desks of the experts in the area of earth observation in Africa has, however, not been enough. That is why the second objective of the project was to 'get the data used' in Africa. The project has achieved this by organizing 4 international training workshops, as well as several user support activities and follow-on projects.

Data provided to more than 100 users

VGT4Africa operationally produces and provides data to 11 EUMETCast receivers (another 5 are in preparation) to over 100 users in more than 20 different countries, including international organizations like the Food and Agriculture Organization (FAO) and the World Food Programme (WFP), and 4 regional environment research and education centres. The data is now used, for example, for a GIS web portal in South Africa, food security early warning systems in the Sahel and crop monitoring and livestock research in Kenya.

Follow-up research

VGT4Africa inspired a number of follow-on projects. Its user support and training activities will continue through an FP7 project: “GEONETCast for and by developing countries” that will extend this approach to South America and China. Certain follow-up projects focus on international cooperation and application development, like Endeleo and VGT@Work, and further pave the way for the African Union’s African Monitoring of the Environment for Sustainable Development (AMESD) project and other similar activities in Europe and Africa.

Consortium

The coordinator is Flemish Institute for Technological Research, Belgium and the partners are the Joint Research Centre – Institute for Environment and Sustainability, Italy and MEDIAS-France, France The project started on 01/01/2005 with a duration of 36 months.

Contact Information

Geert Borstlap
Flemish Institute for Technological Research NV
Teledetection and Earth Observation (TAP) division
Boeretang 200, BE-2400 Mol, Belgium
Tel: +32 14 33 68 39
Fax: +32 14 33 55 99
Email: geert.borstlap@vito.be
http://www.vgt4africa.org

8.1.15 Earth Observation and ENVironmental modelling for the mitigation of HEAlth risks (EO2HEAVEN)

Name

Earth Observation and ENVironmental modelling for the mitigation of HEAlth risks (EO2HEAVEN)

Objectives

The project aims at monitoring changes induced by human activities, with emphasis on atmospheric, river, lake and coastal marine pollution.

Description

EO2HEAVEN follows a multidisciplinary and user-driven approach involving public health stakeholders who work closely with technology and service providers in both the earth observation and in situ environmental monitoring domain. The result of this collaboration aims at the design and development of a GIS based upon an open and standards-based Spatial Information Infrastructure (SII) envisaged as a helpful tool for research of human exposure and early detection of infections.

The key factors of the EO2HEAVEN system are:

- enhanced integration of remotely sensed and in-situ environmental measurements, and
- the development of models to relate these environmental data to exposure and health data. Both factors will directly address current goals of GEOSS such that the resulting system will be integrated into the GEOSS infrastructure after successful validation already during the course of the project. Throughout the life span of the project the stakeholder requirements from three different use cases (in Europe and Southern Africa) will be assessed and the technical solutions proposed by EO2HEAVEN will be evaluated through an iterative process, thus ensuring that the solutions can be applied on a global scale. EO2HEAVEN will specify and implement the SII as an open architecture based upon international standards and adaptive geospatial Web services in alignment with the large-scale initiatives INSPIRE and GMES. The SII will include bridging capabilities at

the syntactic and semantic levels to and between environmental and health systems. Ongoing and recently completed research projects in the ICT, environmental and health domains will be studied and used in an integrative approach.

Consortium

Atos Origin Sociedad Anonima Espanola, ES; Technische Universität Dresden, DE; 52°North Initiative For Geospatial Open Source Software GmbH, DE; Umweltbundesamt, DE; Spot Image (Si) SA, FR ; Nevantropic, FR ; Commission of the European Communities - Directorate General Joint; Research Centre – Jrc, BE; Council for Scientific And Industrial Research, ZA; Open Geospatial Consortium (Europe) Limited, UK; Fraunhofer-Gesellschaft zur Förderung der Angewandten Forschung e.V., DE; Bureau de Recherches Geologiques et Minieres, FR; Stichting International Institute for Geo-Information Science and Earth Observation, NI; University of Kwazulu-Natal, ZA; ICT Policy Implementation Technical Unit, MZ. The project is funded under EC FP7 Environmental theme. The project started on 01/02/2010 with a duration of 36 months.

Contact Information

http://www.eo2heaven.org

8.1.16 Europe-Africa Marine EO Network (EAMNet)

Name

Europe-Africa Marine EO Network (EAMNet)

Objectives

EAMNet aims to construct a network linking Earth Observation (EO) information providers, user networks and centres of excellence in Europe and Africa in the area of coastal and marine observations towards sustainable development in Africa.

Description

The network will undertake capacity building and maintenance and build upon existing infrastructure and expertise in Africa. The overall aim is to improve the exploitation of EO data for coastal and oceanic monitoring towards an Africa-wide observation system (GOOS-Africa). It will provide an interface between European GMES-related core and downstream services and R&D projects (notably MyOcean) and African initiatives (e.g. AMESD) with the emerging GMES-Africa initiative. It will also provide further links with GEO. EAMNet is coordinated by PML with 9 other partners in Africa and Europe.

EAMNet will:

- Promote exploitation of existing and new marine EO data streams produced in Europe and Africa (e.g. MyOcean, OSI-SAF and DevCoCast) and disseminate the data using fast and reliable systems (GEONETCast)
- Harmonize and contribute to the evolution and improvement of application of these data. Improve the coverage of the marine GEONETCast reception network
- Contribute to existing training courses undertaken in AMESD, DevCoCast and other initiatives
- Develop EO specific modules within the courses of three African Universities Coordinate R&D activities in Africa with GMES projects in Europe and promote best practices through targeted exchange of personnel and open fellowships
- Connect the African and EU marine communities by co-organizing yearly symposia (in parallel with the alternating AARSE and AMESD meetings), maintaining a web site and publishing a regular newsletter
- Identify the requirements of the African stakeholders for the provision of GMES services
- Contribute to the implementation of the Action Plan for GMES and Africa Partnership resulting from the Lisbon Process on “GMES and Africa” and planned to be endorsed at the next EU-Africa Summit.

Consortium

The coordinator is Plymouth Marine Laboratory, UK and the partners are the University of Cape Town, South Africa; Institute of Marine Research, Portugal; Danish Meteorological Institute, Denmark; EUMETSAT; Natural Environment Research Council – National Oceanography Centre Southampton, UK; Institute of Marine Sciences University of Dar-es-Salaam, Tanzania; University of Ghana, Ghana; MeteoFrance - Centre de Météorologie Spatiale, France; National Institute of Oceanography & Fisheries, Egypt. The project is funded under EC FP7 Cooperation and the starting date was the end of 2009 with a duration of 36 months.

Contact Information

Plymouth Marine Laboratory
Prospect Place
The Hoe
Plymouth
United Kingdom
PL1 3DH

Tel: +44 (0)1752 633 100
Fax: +44 (0)1752 633 101
forinfo@pml.ac.uk
http://www.eamnet.eu

8.1.17 European Geostationary Navigation Overlay Service (EGNOS)

Name

European Geostationary Navigation Overlay Service (EGNOS)

Objectives

The European Geostationary Navigation Overlay Service (EGNOS), aiming to improve the open public service offered by the USA's Global Positioning System (GPS).

Description

The development of EGNOS arose from a tripartite agreement between the European Space Agency (ESA), the European Commission (EC) and Eurocontrol, the European Organisation for the Safety of Air Navigation.

EGNOS is Europe's first venture into satellite navigation and makes GPS suitable for safety critical applications such as flying aircraft, navigating ships through narrow channels, or road transport.

EGNOS provides both correction and integrity information about the GPS system, delivering opportunities to use the more accurate positioning data for improving existing services or developing a wide range of new services. In terms of correction of data, EGNOS improves the accuracy of GPS by providing a positioning accuracy to within three metres. By comparison, someone using a GPS receiver without EGNOS can only be sure of their position to within 17 metres. Regarding integrity of message, EGNOS also provides verification of the system's integrity, which relates to the trust that can be placed in the correctness of the location information supplied by the navigation system. EGNOS' infrastructure consists of three geostationary satellites over Europe and a network of ground stations.

- EGNOS will offer three services. These are:
- Open Service: The service is provided free of charge without any guarantee or resulting liability. It is open for use to anyone with an EGNOS-enabled satellite navigation receiver.
- Safety-of-life Service: EGNOS will provide a valuable integrity message warning the user of any malfunction of the GPS signal within six seconds.
- Commercial Service: EGNOS provides a terrestrial commercial data service called the EGNOS Data Access Service (EDAS).

EGNOS' infrastructure comprises a ground network of 34 ranging and integrity monitoring stations (RIMS), six navigation land earth stations (NLES) and four mission control centres and signal transponders on three geostationary satellites (Artemis, Inmarsat AOR-E and Inmarsat IOR-W). The signal coverage area includes most European countries.

The possibility of extending EGNOS coverage to Africa is being evaluated. The European Commission has invited experts from Africa and Europe to contribute further analysis on 11 specific areas aimed at the preparation of an implementation plan.

Consortium

EGNOS was developed by the EGNOS Operator and Infrastructure Group (EOIG) with the support of the European Commission (EC), the European Space Agency (ESA) and Eurocontrol. ESA had overall responsibility for the design and development of the EGNOS system. The European Commission has assigned the operational management and maintenance of EGNOS to the European Satellite Services Provider (ESSP), a company founded in 2001 by seven European air navigation service providers: AENA (Spain), DFS (Germany), DSNA (France), ENAV (Italy), NATS (UK), NAV (Portugal), Skyguide (Switzerland). The funding is form EC, ESA, Eurocontrol.

Contact Information

European GNSS Supervisory Authority (GSA)
Rue de la Loi, 56
1049 Brussels, Belgium
Tel: + 32 2 297 16 16
Fax: + 32 2 296 72 38
http://www.egnos-portal.eu/

8.1.18 FAO African Water Resource Database

Name

FAO African Water Resource Database[37]

Objectives

The African Water Resource Database (AWRD) focuses on a dataset for inland aquatic resource management, especially inland fisheries and aquaculture, and improvement of food security through a geographic information system (GIS) analytical framework.

[37] http://www.fao.org/fishery/collection/awrd/en

Description

Through the recommendations of the Committee on Inland Fisheries for Africa (CIFA), the AWRD was established. The AWRD is successor operation to the earlier project led by the Aquatic Resource Management for Local Community Development Programme (ALCOM) entitled the "Southern African Development Community Water Resource Database" (SADC-WRD). The project itself is subdivided in two parts. The first part treats the description and overview of the AWRD and separates two application types. One is written for administrators and managers and the other for professionals in technical fields. The second part more practically responds to the question "how to" supply and include a technical manual for spatial analysts and a workbook for university students and teachers. The storage and collection of the AWRD covers the whole African continent, especially the surface water bodies, watersheds, aquatic species, rivers, political boundaries, population density, soils, satellite imagery and many other physiographic and climatological data. The spatial analysis improves understanding for aquatic resources management and answers related questions. Thanks to the visual nature of the database it is possible to integrate different types of information in an interrelated program.[38]

Consortium

This project is developed under FAO and is currently ongoing.

Contact Information

Email: Jose.AguilarManjarrez@FAO.org

8.1.19 Food Security Assessment (FOODSEC)

Name

Food Security Assessment (FOODSEC)

Objectives

FOODSEC's target is crop monitoring and assessment of agricultural production in regions stricken by food insecurity.

Description

FOODSEC supports the Food Security and Food aid/assistance policies of the European Union by monitoring crops and assessing agricultural production. It particularly supports the 1st Action Plan.

This technical support contains research and scientific development in different contexts of yield forecast models, combining NDVI indicators; Water satisfaction models; methodology of vulnerability analysis combining socio-economical and physical indicators; making use of spatial analysis and of various survey datasets, calibration and validation of rainfall estimates products (2nd Generation METEOSAT); validation and assessment of the MODIS medium resolution products and integration as crop masks in the crop monitoring systems. FOODSEC has extended its expertise in other fields of food security like vulnerability analysis, post harvest losses, market prices.

Objectives are:

- to monitor crop yield by delivering crop yield monitoring and yield forecasting on a regular basis (national/ regional Bulletins) over the Horn of Africa, Central Asia, and South America. Start future extension of the activities in Sub-Saharan Africa according to existing infrastructures.
- to provide ad-hoc assessment on regions at acute risk (DPRK, others).
- to improve the efficiency of the crop monitoring system through the integration of new products and develop yield forecast modules (from qualitative to quantitative EWS).
- to acquire complementary expertise on some of the other components of food security; vulnerability assessment, post-harvest losses and market analysis, by building on present studies (NRI, IFPRI) and partnerships (7 agencies participating with JRC in the implementation of the Integrated food security Phase Classification-IPC) to carry out research in these fields.
- to build capacity of national and regional FS information systems in Africa, through provision of technical support and training in the various components of FS, in particular crop monitoring and yield assessment.
- to improve scientific cooperation with African universities and networks of research (FARA, ASARECA) and CGIAR organisations in crop monitoring and food security.
- to contribute to the ACP Thematic program in the field of Food Security and Agriculture, in particular in the common research topic "competition on land", by playing an increased role in the GMES "Global Land" (Food Security and Agriculture component)
- to provide relevant inputs to projects such as GEOLAND, AMESD or GMFS.

[38] http://www.fao.org/fishery/gisfish/servlet/CDSServlet?status=ND1saXRlcmF0dXJlX2NmLjI4MjImNj1lbiYzMz1pbmxhbmRfZG9jJjM3PWluZm8~

- to ensure technical support in Food Security matters to the EC Delegations and in particular to the EC Delegation to the UN organizations in Rome, by providing technical monitoring and advice on EC funded Food Security Projects and programmes.
- to define with RELEX family DG's the contribution of JRC to the Food Security Thematic Programme (2008-2013), including priorities in term of geographic and thematic extensions and identifying synergies with the Component 1 of the FSTP (ARD - Agricultural research for Development).

Consortium

This project is developed under JRC.DDG.H Institute for environment and sustainability, Ispra (Italie) and is currently ongoing.

Contact Information

Mr. Olivier Leo, leader
Email: olivier.leo@jrc.ec.europa.eu
http://mars.jrc.ec.europa.eu/mars/About-us/FOODSEC

8.1.20 GMES and Africa: Regional Network for information Exchange and Training in Emergencies (GARNET-E)

Name

GMES and Africa: Regional Network for information Exchange and Training in Emergencies (GARNET-E)

Objective

This project aims to address the implementation of a sustainable emergency response service in Africa, using satellite Earth Observation data.

Description

The objective of GARNET-E is to contribute to the partial re-alignment of the “GMES Emergency Response in Africa” agenda, from technical activities focused purely on risk and poverty reduction and response using European capacities, to those more directed to building sustainable local capacities, leading to real wealth creation in Africa. GARNET-E focuses to enable and enhance the ability of African states to use satellite Earth Observation for the management of natural and man-made humanitarian emergencies and to develop a network of EU and African organisations and African users, in order to build economic, technical and commercial capacity within African states, along the priority lines being identified in consultation with the African Union under the ‘GMES and Africa’ initiative. The project has two technical objectives:

- To exchange information, through workshops on GMES Emergency Response Core Service; and the International Charter - Space and Major Disasters.
- To improve the quality of the current GMES Service, for African emergencies, through local requirements gathering exercises; and the ingestion of in situ data in product generation.

Consortium

This project is co-ordinated by Infotera Ltd and it has 17 African and European Partners. The European partners are: Infoterra Ltd in the UK, Metria (Sweden), Keyobs (Belgium), Vito (Belgium), PLUS (Austria), EDISOFT (Portugal), ITC (The Netherlands), DMCii (UK), SYSECO (Belgium). The partners in Africa are: African Association for Remote Sensing of the Environment (AARSE), EIS-Africa, Regional Centre for Mapping of Resources for Development (RCMRD), Regional Centre for Training in Aerospace Surveys (RECTAS), Centre Royal pour Teledetection Science (CRTS), Centre for Science and Industrial Research (CSIR), Satellite Operations Centre in South Africa, Societe Nacionale Protection Civile in Cape Verde (SNPC-CV), GeoSAS Ltd. It works close with global partners: UNITAR/UNOSAT, UNSPIDER, UNEDRA. This project is funded under FP7 and is currently ongoing.

Contact Information

Jo Lambert
GARNET-E Project Manager
Infoterra Ltd.
Europa House, Southwood Crescent, Southwood, Farnborough, Hampshire, U.K. , GU14 0NL
Tel: +44 (0)1252 362053
Fax: + 44 (0) 1252 375016
Email: Jo.Lambert@infoterra-global.com

8.1.21 GEO Network for Capacity Building (GEONetCab)

Name

GEO Network for Capacity Building (GEONetCab)

Objectives

The purpose of the GEO Network for Capacity Building (GEONetCab) project is to create the conditions for the improvement and increase

of the GEO capacity building activities and framework, with special emphasis on developing countries, new EU member states (and EU neighbouring states) and climate monitoring, and will serve the bigger goal of improved effectiveness and efficiency of GEO capacity building for application in the GEO societal benefit areas

Description

The project will deliver the following output:

- Capacity building needs in earth observation are identified (at a generic and global level, but with emphasis on the target regions).
- Specifications for earth observation capacity building are described.
- Resource providers are identified.
- Sustainable brokerage between stakeholders (including resource providers) is established.
- A mechanism to facilitate cooperation between stakeholders and providers is established.
- A global base of technical expertise for education and training in earth observation is established (with emphasis on developing countries, new EU member states and climate monitoring).
- Monitoring and evaluation mechanisms for determining the efficacy of GEO capacity building efforts are established.

To achieve maximum impact demonstration projects will be carried out in Southern Africa, the French-speaking African region, the Czech Republic and Poland, with spin-offs to EU neighbouring countries and Latin America and Asia. The project (with a duration of three years) will be carried out by a strong consortium of partners from the Netherlands, France, South Africa, Morocco, the Czech Republic and Poland, supervised by an advisory board with worldwide representation and strong connections to GEO.

Consortium

Stichting International Institute for Geo-Information Science and Earth Observation, Nl; Centre National d'Etudes Spatiales, Fr ; Institut de Recherche pour le Developpement, Fr; Centrum Badan Kosmicznych Polska Akademia Nauk, Pl; Karlova Universita, Přírodovědecká Fakulta V Praze, Cz ; Council For Scientific And Industrial Research, Za; Centre Régional Africain des Sciences et Technologies de l'Espace – LF, MA ; Umvoto Africa pty ltd, ZA. The project is co-funded by EC FP7 Environmental theme. It started in October 2009 and the duration is 36 months.

Contact Information

Mark Noort
International Institute for Geo-Information Science and Earth Observation (ITC)
PO Box 6, 7500 AA Enschede, the Netherlands
Tel: +31(0)53 4874 221
Fax: +31(0)53 4874 436
E-mail: m.noort@itc.nl
http://www.itc.nl

8.1.22 Geographic Information Support for Treatment Programmes (GISA)

Name

Geographic Information Support for Treatment Programmes (GISA)

Objectives

The objective of this project was to study the potential role of geographic and space techniques related to HIV/AIDS and ARV issues in sub-Saharan Africa. Geographic information systems (GIS) technology makes it feasible for example to track the HIV epidemic in defined areas.

Description

In support of local health centres and humanitarian organisations, precise local mapping of AIDS cases, HIV infection and related risk factors can:

- provide demographic and environmental perspectives on the current status of the epidemic,
- portray historical and long-term trends,
- evaluate the delivery and impact of prevention and care services,
- test hypotheses on social, political and environmental determinates of transmission,
- target outreach efforts to locations where transmission is likely to be highest.

GISA will:

- Support local health centres and work towards improved information management related to the treatment of patients, reporting with up-to-date information from the geo-spatial context
- Link to ongoing initiatives as well as national and local authorities

Consortium

In this project ESA, World Health Organization (WHO), Medecins Sans Frontieres (MSF) were involved. UNAIDS, Telecoms Sans Fron-

tieres (TSF) had an advisory role. This project has been completed.

Contact Information

Didier Schmitt
Email: didier.schmitt@esa.int
http://gisa.web.cern.ch/gisa/en/

8.1.23 GEONETCast applications for and by developing countries (DevCoCast)

Name

GEONETCast applications for and by developing countries (DevCoCast)[39]

Objectives

The DevCoCast project aims to involve developing countries in the GEONETCast initiative and is setting up a strong user and provider base.

Description

GEONETCast, provides reliable, worldwide and continuous access to Earth Observation information and is a core infrastructure and early success of the Global Earth Observation System of Systems (GEOSS). DevoCoCast aims involve developing countries and ultimately to introduce and embed GEONETCast based data in various local and national planning and decision-making processes in a systematic manner.

Many developing countries have no reliable access to real-time environmental information and are exposed to serious environmental risks. The GEONETCast concept overcomes existing telecommunication limitations to provide reliable access to this vital environmental information.

DevCoCast uses GEONETCast to disseminate existing environmental added-value data (both in-situ and satellite-based) from various sources in Africa, South and Central America, Asia and Europe to a broad range of end-users in developing countries.

This project is building on the work from previous and ongoing projects like PUMA, AMESD, VGT4AFRICA.

It will focus on

- Sharing cross-cutting Earth Observation products, produced by and for African countries.
- Building on African capacity
- Embedding EO and GEONETCast in everyday African applications

Furthermore, EUMETCast satellite receivers are established or upgraded to operational status at 4 large marine ecosystem monitoring services in Ghana, Namibia, Tanzania, Senegal. These 4 marine end-users are supported to embed the marine data they receive into their applications. Additional EUMETCast receivers are also built up in 3 pilot national Desert Locust Centres in Eritrea, Mali and Sudan, to support UN-FAO's fight against the Desert Locust.

Consortium

The project coordination is the Flemish Institute for Technological Research NV (VITO), Belgium. Partners are the African Centre of Meteorological Application for Development (ACMAD), Niger; Regional centre for training and operational applications in agrometeorology and hydrology (AGRHYMET), Niger; Food Supply Agency of the Ministry of Agriculture (CONAB), Brazil; Centre for Surveying and Assessment of Agriculture and Natural Resources (CREAN), Argentina; Council for Scientific and Industrial Research (CSIR) - Meraka Institute, South Africa; Danish Meteorological Institute (DMI), Denmark; Brazilian Agricultural Research Corporation (EMBRAPA), Brazil; European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), Germany; National Institute for Space Research (INPE), Brazil; National Institute of Agriculture Technology (INTA), Argentina; Technology International Institute for Geo-Information Science and Earth Observation (ITC), The Netherlands; European Commission - DG Joint Research Centre - Institute for Environment and Sustainability (JRC-IES), Italy; University of Cape Town (UCT), Marine Research Institute (MA-RE), South Africa; Institute Natural Environment Research Council (NERC), National Oceanography Centre Southampton (NOCS), UK; Plymouth Marine Laboratory (PML), UK. This project started on 01/05/2008 with a duration of 36 months and is funded under the FP7 Environment-Group on Earth Observations (GEO) theme.

Contact Information

Flemish Institute for Technological Research NV
Teledetection and Earth Observation (TAP) division
Boeretang 200BE-2400 Mol, Belgium
Tel: +32 (0)14 33 55 11
Fax: +32 (0)14 33 55 99
http://www.devcocast.eu

[39] http://www.eumetsat.int/groups/cps/documents/document/pdf_br_fs01_en.pdf

8.1.24 Geo-Spatial Information Analysis for Global Security and Stability –Information Support for Effective and Rapid External Action (EU) (ISFEREA)

Name

Geo-Spatial Information Analysis for Global Security and Stability – Information Support for Effective and Rapid External Action of the EU (ISFEREA[40])

Objectives

The aim of Geo-Spatial Information Analysis for Global Security and Stability (ISFEREA) is to provide scientific support to EU external relations policies, specifically those concerned with international crisis management, and especially the Community Instrument for Stability, the Community Humanitarian Aid Instrument, the Community Action Plan on Disaster Risk Reduction Initiative, the Community Action Plan on Reinforcing the Union's disaster response capacity, and the external security dimension of Global Monitoring of Environment and Security (GMES) policy.

Description

ISFEREA supports the Strategy of the Global Security and Crisis Management Unit of JRC in risk reduction, prevention, crisis preparedness and crisis response and recovery (reconstruction). In the future ISFEREA will develop methodologies focusing on newly-defined and further improved procedures based on textural and morphological processing, and multi-temporal description by multi-source comparison (archived satellite images, existent cartography).

The specific relation of this project with Africa can be found in its contribution in the development of the GMES and Africa Action Plan 2008-2010 in international crisis management, in risk reduction, prevention and crisis preparedness.

Its objectives are:

- to provide scientific support to EU external relations policies, specifically those concerned with international crisis management, and especially the Community Instrument for Stability, the Community Humanitarian Aid Instrument, the Community Action Plan on Disaster Risk Reduction Initiative, the Community Action Plan on Reinforcing the Union's disaster response capacity, and the external security dimension of Global Monitoring of Environment and Security (GMES) policy;
- to improve automatic information extraction from VHR satellite data;
- to contribute to methodological improvement for computer-enhanced visuals;
- to develop tools for interpretation, image information mining and automatic feature extraction, patterns;
- to improve recognition and image understanding using new-generation satellite data including optical and radar new sensors;
- to optimise satellite data collection by testing and benchmarking the new generation of satellite sensors in both the optical and radar domains;
- to define new image structural descriptors based on multi-scale morphological and textural image characteristics, extension to spatially uncertain reasoning;
- to improve automatic image matching procedures and 3D volumetric information extraction;
- to define, implement and test new inferential engines able to handle spatial uncertainty for integrating different partially non-consistent geo-information sources in the final assessment, as satellite VHR images for multi-temporal change detection;
- to study the neuro-cognitive processes involved in satellite image data interpretation tasks and improvement of the possibility to integrate them by machine learning processes;
- to foster scientific collaboration with high-level European and international research centres involved in similar topics by means of results oriented bilateral agreements and activities including the implementation of shared experiments, building of shared reference datasets for testing and validation of new algorithms and systems, organisation of thematic workshops, exchange of senior researchers and coordination of PhD and post-doc activities to activate synergies and share newly-developed tools and systems;
- to improve image data processing systems, spatial data infrastructure, and the validation protocol and digital map production;
- to develop a Global Crisis Atlas; procurement, processing and standardisation of global continental and regional datasets within a GIS framework;
- to support crisis risk assessment in policy areas like Disaster Risk Reduction, Crisis Preparedness capability through content information at the local scale (at city or settlement, high information input), local city scale earth observation EO based information and regional scale remote

40 http://.jrc.ec.europa.eu/jpb_public/act/publicsimplesearch.html?activityId=2195&viewAll=N&searchtext=Africa, satellite&wpId=100&yearSearch=100

sensing only for crisis hotspots with the aim to test a global approach for assessing the element at risk focusing on the building stock and population;
- to quantify potential losses for well defined hazardous events in two research components; extraction of built-up stock and definition of the vulnerability of the built-up stock;
- to model in a GIS context crisis risk based on hazard events, built-up stock and its vulnerability, based on the results of satellite data and GIS in the analysis of crisis risk for both local decision makers/policy makers;
- to monitor temporal changes in the built-up stock to quantify the increased crisis risk due to urban sprawl especially when related to informal and vulnerable settlements;
- to develop regional built-up layers focusing on high risk, hazard, prone areas;
- to support human settlement analysis in policy areas like disaster Risk Reduction;
- to improve crisis preparedness capability by information about (e.g. villages, cities) and informal settlements (e.g. slums, refugee/IDP camps);
- to focus on developing information layers on built-up areas including location and characterisation of settlements from satellite data and geo-spatial analysis;
- to support post-disaster damage assessment for needs assessment/recovery planning and reconstruction monitoring Policy areas;
- to support conflict resources monitoring and conflict risk modelling.

Consortium

The project is run by JRC.DG.G Institute for the Protection and Security of the Citizen, Ispra (Italie). European Partners are: ESA - European Space Agency; EUSC - European Union Satellite Centre European Partners; COC - Competence Centre in Information Extraction and Image Understanding for Earth Observation; TRSL PV - University of Pavia, Department of Electronics, Telecommunication & Remote Sensing Laboratory; RETINA - Bilkent University, Department of Computer Engineering, RETINA Vision and Learning Group; RSLab - University of Trento, Department of Information Engineering and Computer Science, Remote Sensing Laboratory. The following institutions are involved: UN OCHA - United Nations, Office for the coordination of Humanitarian Affairs; UN DP - United Nations, Development Programme; UN HABITAT - United Nations, Human Settlement Programme; UN DPKO - United Nations, Peacekeeping Operations; UN SC - United Nations, Security Council; WB - The World Bank; CEODE CAS - Chinese Academy of Sciences, Center for Earth Observation and Digital Earth. This project is funded by the European Commission and is currently ongoing.

Contact Information

Mr. Martino Pesaresi, leader
Email: martino.pesaresi@jrc.ec.europa.eu
http://isferea.jrc.ec.europa.eu/

8.1.25 Global Forest Resource Monitoring – TREES-3

Name

Global Forest Resource Monitoring – TREES-3

Objectives

The Global Forest Resource Monitoring – TREES-3 aims is to have an overview on deforestation in a global perspective, focused on Tropics and Eurasia forest and boreal areas.

Description

Changes in forest resources is one of the relevant EU policies related to global environmental and forestry issues. TREE-3 maps and measures these changes inforest and boreal resources in Eurasia and Caribbean and Pacific regions.

The objectives are:
- To continue the development of a system for documenting forest cover changes in the Tropics and boreal Eurasia. Particular emphasis will be placed on the finalisation of a semi-automatic processing chain for producing forest cover change maps from Landsat-type imagery for tropical forests ecosystems and on the feasibility of acquisition of a satellite imagery database for the year 2010. This is performed in the framework of the TREE-3 in collaboration with the UN-FAO, and MONDE and FOREST Actions (methodology).
- To estimate forest cover changes in tropical regions (South & Central America, Africa and South East Asia). This project is performed in the framework of TREE-3 in contribution to JRC's sub-Priority Area "Climate Change" and in collaboration with UN-FAO and MONDE Action (methodology).
- To organise validation workshops at JRC, Ispra or in tropical countries for the estimation of forest cover changes in tropical regions. This project runs in collaboration with UN-FAO and MONDE action

and aims to aid in the framework Commission project "Support to the Global Forest Resources Assessment Process" (DG-AIDCO).
- To support FAO in the implementation of a series of sub-regional workshops, in which national counterparts will be trained on land use change assessment. This is performed in the framework of TREE-3 and in collaboration with the UN-FAO and MONDE Action and in the framework of Support to the Commission project "Support to the Global Forest Resources Assessment Process" (DG-AIDCO).
- to develop a data distribution mechanism, allowing the provision of the data to FAO for its network of national partners. This project is run in collaboration with the UN-FAO and MONDE Action and in the framework of Support to Commission project "Support to the Global Forest Resources Assessment Process" (DG-AIDCO).
- To link through a Web interface TREES-3 forest change interpretations to other global forest data sets, e.g. in-situ forest measurements from the FAO National Forest Monitoring and Assessment. This is performed in the framework of TREE 3 in collaboration with the JRC-IES FOREST and MONDE actions and in the framework of the EuroGEOSS project.
- To extend the thematic analysis of the database of the radar satellite mosaic at 50 m resolution from the Advanced Land Observing Satellite (ALOS) over Sub-Saharan Africa using the sample site results of land cover assessments from the JRC-IES MONDE Action. This project is run in collaboration with the MONDE Action and in the framework of the collaborative agreement with the Japanese Space Agency (JAXA).
- To develop a method for forest cover change assessment in the Tropics from the integration of recent medium resolution satellite data from radar sensors (e.g. ALOS PALSAR mosaics at 50 m resolution) with historical satellite data from optical sensors (e.g. Landsat-type of data).
- To refine knowledge on the evolution of forest resources in Central Africa.
- To contribute to the sub-Priority Area "Climate Change" (deliverable 11.1) and the Thematic Programme "Africa Caribbean Pacific Observatory" (deliverable 11.2). This project is run in the framework of support to the Commission project "FORAF" (DG-AIDCO).
- To contribute to the Forest Carbon Tracking task of GEO (Group on Earth Observation). This project is run in collaboration with the MONDE Action. Contribution to the Thematic Programme "Africa Caribbean Pacific Observatory"
- To provide scientific and technical support concerning deforestation, carbon emissions from deforestation and sustainable forest management to Commission DGs and delegations dealing with international environmental conventions and agreements, in particular as a follow-up to EC Communication on deforestation issued in October 2008,in collaboration with JRC-IES GHG-AFOLU Action. This project is run in contribution to the JRC's sub-Priority Area "Climate Change" and the Thematic Programme "ACP Observatory" (Global Climate Change Alliance)

Consortium

This project is coordinated by JRC.DDG.H Institute for environment and sustainability, Ispra (Italy) and is currently ongoing.

Contact Information

Mr. Frederic Achard
Email: frederic.achard@jrc.ec.europa.eu

8.1.26 Global Ocean Observing System in Africa (GOOS-AFRICA)

Name

Global Ocean Observing System in Africa (GOOS-AFRICA)

Objectives

The GOOS affords a global view of the ocean system and the support of operational services worldwide by observation, modelling and analysis of marine and ocean areas[41] and also the establishment of the Regional Ocean Observing and Forecasting System in Africa.[42]

Description

As part of the GEOSS, the Global Earth Observing System of Systems, the UN and UNESCO governing bodies have ensured that international cooperation takes up the first priority of GOOS- Africa.[43] In order to have a global view of the ocean system by observa-

[41] http://www.ioc-goos.org/component/option,com_oe/task,viewGroupRecord/groupID,39/
[42] http://www.ioc-goos.org/index.php?option=com_oe&task=viewDocumentRecord&docID=2737
[43] http://www.ioc-goos.org/component/option,com_oe/task,viewGroupRecord/groupID,39/

tion and the support of operational ocean services, modelling and analysis of marine and ocean areas, GOOS Africa presents accurate description of the state of the "oceans, including living resources, continuous forecasts of the future conditions of the sea for as far ahead as possible, and the basis for forecasts of climate change."[44] Additionally, through the establishment of the Regional Ocean Observing and Forecasting System in Africa, GOOS Africa can contribute to the sustainable use and protection of its marine and coastal environments and the activation of an improved management of the impacts of climate change. To reach these goals, GOOS Africa has a system of programmes and platforms. The programmes are designed to serve oceanographic researchers, coastal managers, parties to international conventions, national meteorological and oceanographic agencies, hydrographic offices, marine and coastal industries, policy makers and the interested general public in the following activities: monitor, understand and predict weather and climate; describe and forecast the state of the ocean, including living resources; improve management of marine and coastal ecosystems and resources; mitigate damage from natural hazards and pollution; protect life and property on coasts and at sea; enable scientific research. The programmes include "different and complementary aspects of establishing an operational ocean observation capability for all of the world's nations."[45]

Consortium

Member states through government agencies, navies and oceanographic research institutions working together in a wide range of thematic panels and regional alliances. The funding for this project some from IOC, UNEP, WMO and ICSU and GOSS-Africa is currently ongoing.

Contact Information

Mr. Geoff Brundrit (GOOS-AFRICA Chairman)
Oceanography Department
University of Cape Town
Private Bag, Rondebosch 7700
Cape Town, South Africa
Tel: +27 (21) 650 3278
Fax: + 27 (21) 650 3979
Email: oceangeoff@iafrica.com

Mr. Keith Alverson
Global Ocean Observing System,
Intergovernmental Oceanographic Commission
1, rue Miollis, 75732 Paris Cedex 15, France
Tel: +33 (0)1 45 68 40 42
Email: k.alverson@unesco.org

[44] http://www.ioc-goos.org/component/option,com_oe/task,viewGroupRecord/groupID,39/
[45] http://www.ioc-goos.org/component/option,com_oe/task,viewGroupRecord/groupID,39/

8.1.27 GMES Space Component

Name

GMES Space Component

Objectives

The provision of Earth-observation data to turn into services for monitoring the environment and supporting civil security exercises.

Description

The GMES Space Component comprises five types of new satellites called Sentinels that are being developed by ESA specifically to meet the needs of GMES, so-called Contributing Missions and the associated Ground Segment infrastructure.

All the dedicated Sentinels, Contributing Missions and Ground Segment infrastructure will be part of the overall GMES Space Component architecture managed by ESA. ESA is therefore responsible for establishing a mechanism to integrate, harmonise and coordinate access to all the relevant data from the multitude of missions contributing to GMES. This is carried out in close cooperation with national space agencies, EUMETSAT and, where relevant, with owners of non-European missions contributing to the GMES objectives.

The Sentinel missions include radar and super-spectral imaging for land, ocean and atmospheric monitoring. The first three Sentinels are currently under industrial development, with Sentinel-1 planned to launch in 2012.

When identifying the necessary elements to provide GMES services to Africa, the following thematic areas aspects should be specifically evaluated:

- Capacity building framework for GMES and Africa
- Policy and institutional framework for GMES and Africa
- Infrastructure framework for GMES and Africa
- Thematic framework for GMES and Africa
- Natural disasters
- Food security and rural development
- Long-term management of natural resources

- Water resource management
- Impacts of climate variability and change
- Marine and coastal areas
- Conflicts and political crises
- Infrastructures and territorial development

Consortium

The European Space Agency is the coordinator and actors involved are ASI, BNSC, CDTI, CNES, CSA, DLR Applied Remote Sensing Cluster, EC, EUMETSAT. The funding is provided by the European Commission and the project is currently ongoing.

Contact Information

Josef Aschbacher
Head, GMES Space Office
ESA ESRIN, Frascati/Italy
Email: josef.aschbacher@esa.int
http://www.gmes.info

8.1.28 Identifying Natural Mosquito Habitats

Name

Identifying Natural Mosquito Habitats

Objectives

The objectives are the development and demonstration of EO Technology for Identifying Natural Mosquito Habitats and Predicting Malaria Risk in Africa.

Description

The project is part of Kenya's National Malaria Control Program that involves an interdisciplinary team in insect ecology, human health and tropical diseases, and ecologists. The objectives of this project are to provide detailed maps, monitor changes to mosquito habitats (wetlands), and develop malaria risk maps using data on topography, larva ecology, human health, households, and habitats. This information will be an integral part of an overall larva-control strategy to identify larvicide-application practices that would reduce the mosquito population.

Detailed space-based data on wetlands and mosquito control are not available for the region but can be obtained for reasonable cost by high-resolution Earth-observation sensors. Data acquired over time can monitor the change and provide a more accurate resource map. Continuous cloud cover during wet seasons and the cumulus clouds caused by heat during the dry season means that an all-weather satellite with high spatial resolution modes, like RADARSAT, will provide a reliable source of multi-temporal imagery.

Consortium

The prime contractor for this project is Noetix Research Inc. (Ontario, Canada) and the partners are C-Core (Canada) and the Kenya Medical Research Institute (KEMRI). This project is funded by the European Space Agency and the Canadian Space Agency.

Contact Information

Diego Fernandez
Email: Diego.fernandez@esa.int
Tom Hirose
Email: tom.hirose@noetix.on.ca

8.1.29 Information Society Technologies-IST-Africa (2010-2011)

Name

Information Society Technologies-IST-Africa (2010-2011)

Objectives

IST-Africa aims at supporting the 8th Africa-EU Strategic Partnership regarding Information Society Technologies.

Description

IST-Africa is supported by the European Commission under FP6 and FP7, IST-Africa (Regional Impact of Information Society Technologies in Africa) is a multi-stakeholder initiative focused on:

- Supporting JEG8 Activities and Implementation of the 8th Africa-EU Strategic Partnership
- Skills Transfer to Support Research Capacity Building & STI Development in Africa
- Community building to support EU-African Research Cooperation.

Like other regional initiatives, IST-Africa focuses on capacity building by organising conferences and training of workshop participants on eHealth, eLearning, and eGovernment projects. IST-Africa is open to international collaboration particularly with the EU, and is interested in having space projects.

Consortium

The coordinator of the project is IIMC International Information Management Corporation Limited ("IIMC", Ireland). Partners are the Department of Science and Technology (South Africa); Ministry of Communications, Science and Technology (Botswana); Ministry of Communications, Science and Technology (Lesotho); ICT Policy Implementation Techni-

cal Unit (Mozambique); Ministry of Education (Namibia); National Computer Board (Mauritius); COSTECH - Tanzania Commission for Science and Technology; Uganda National Council for Science and Technology; Ministry of Higher Education, Science and Technology (Kenya); Ministere de l'Enseignement Superieur et de la Recherche Scientifique (Burundi); Ministry in President's Office in charge of Information and Communication Technology (Rwanda); Agence Nationale des Technologies de l'Information et de la Communication (Cameroon); Ministère de la Recherche Scientifique (Senegal) and Ministry of Communications and Information Technology (Egypt). IST-Africa is currently funded under FP7 but has also been running under FP6 before.

Contact Information

Paul Cunningham / President/CEO/IST-Africa Initiative Coordinator
IIMC International Information Management Corporation Limited
13 Docklands Innovation Park, 128 East Wall Road, Dublin 3, Ireland
Tel: +353-1-8170607
Fax: +353-1-8170606
Email: paul@iimg.com
http://www.ist-africa.org/home/

8.1.30 Integrated Water Resource management for Zambia (IWAREMA)

Name

Integrated Water Resource management for Zambia (IWAREMA)

Objectives

IWAREMA is aiming at generating a variety of environmental maps to provide local policy makers with the necessary tools for effective water resource management.

Description

Zambian water authorities are integrating information based on satellite imagery to alleviate water shortages. Due to inadequate information many water-related problems are faced. The IWAREMA project is an ESA project which focuses on generating a variety of environmental maps to provide local policy makers with the necessary tools for effective water resource management.

IWAREMA was one of the projects initiated under ESA's TIGER initiative, launched in 2002 to assist African countries to overcome water-related problems and to bridge Africa's water information gap using satellite data.

As part of the IWAREMA (Integrated Water Resource management for Zambia) project, funded through ESA's Data User Element, data from ESA's multispectral MERIS sensor aboard Envisat was used to create maps depicting existing water resources, suitable dam locations and land cover. The project is carried out by the Belgium Company GIM (Geographic Information Management) in partnership with the University of Zambia and the Zambian water authorities.

Consortium

This project is conducted under the European Space Agency and the Zambian Water Authorities.

Contact

health@esa.int

8.1.31 Integrating Morocco in the European Research Area (M2ERA)

Name

Integrating Morocco in the European Research Area (M2ERA)

Objectives

The M2ERA project aims to reinforce the S&T cooperation and partnership between Morocco and Europe in order to foster the integration of Morocco into the European Research Area.

Description

The activities foreseen in this project will complement previous initiatives implemented at the regional level, such as ERA-MED (Strengthening the European Research Area in Mediterranean Countries) and MIRA (Mediterranean Innovation and Research Coordination Action). They will target the specific needs of Morocco, such as supporting capacity building, promoting partnership and reinforcing policy dialogue. The activities planned within M2ERA project include:

- Study of Moroccan-European S&T cooperation.
- Capacity building for implementing thematic National Contact Point (NCPs)
- Promotion of partnership between Europe and Morocco
- Implementation and dissemination activities

Consortium

For this project the following entities are involved: Association Research and Develop-

ment (R&D) Morocco (R&D, Morocco); Ministry of Higher Education and Scientific Research (MENESFCRS, Morocco); National Centre for Scientific and Technological Research (CNRST, Morocco); Aster S.C.P.A. (Aster, Italy); Open University of the Mediterranean Region (TETHYS, France); CIRPS Consortium (CIRPS, Italy). This project started on 01/12/2008 for a duration of 36 months and is funded under EC FP7 Capacities.

Contact Information

Mohamed Smani
Tel: +212 5 22 22 44 66
Email: rdmaroc@menara.ma
http://www.pin.edunet.ma/m2era.php

8.1.32 International Boundary Information System (UNIBIS) project.

Name

International Boundary Information System (UNIBIS) project.

Objectives

The UNIBIS objective is the creation and the maintenance of a worldwide geodatabase on international boundaries, including Africa.

Description

Under the UNIBIS project a geodatabase is being created. This database collects geometry, status, claims, dates of treaty agreements, and reference materials for international frontiers. The use of authoritative sources of information, such as treaty maps, coordinates, rulings, treaty text, and interpreted satellite imagery is included in the UNIBIS in consultation with other international experts on boundaries. Thereby potential conflicts can be discovered and UNIBIS assists the state on conflict prevention activities and readiness for assistance on boundary demarcation issues.

Consortium

This project is funded by the United Nations and various UN entities are involved. This project is currently ongoing.

Contact Information

Dr. Wilber Ottichilo
Director General
Regional Centre for Mapping of Resources for Development
Kasarani Road, PO BOX 18118 -00500,
Enterprise Road, Nairobi, Kenya
Tel.: +254 20 860654 or 803320/2/9
Fax: +254 20 861673 or 802767
Email: ottichilo@rcmrd.org or rcmrd@rcmrd.org
Web: http://www.rcmrd.org
Dozie Ezigbalike
Senior Geographic Information Systems Officer, DISD
Knowledge Management Coordinator, ISP
UN Economic Commission for Africa P.O. Box 3005, Addis Ababa, Ethiopia
Tel: +251-1-51-02-75 or +251-1-44-45-69
Fax: +251-1-51-05-12
Email: ezigbalike.uneca@un.org

8.1.33 ITU Projects

In line with the C7 line of Action of the World Summit on the Information Society (WSIS), ITU implemented different projects in the African continent that would make use of telecommunications infrastructure.

These include the e-government projects in Benin, Burkina Faso, and Cameroon, and e-health projects in Zambia, Senegal, Mozambique, Mauritania, Lesotho, Kenya, and Ethiopia.

Contact Information

Mr Souheil Marine, Head ICT Applications and Cybersecurity Division ITU
Tel: +41 22 730 5323
Fax: +41 22 730 5484.
Email: souheil.marine@itu.int.
http://www.itu.int/ITU-/cyb/app/projects/afr/

8.1.34 Land Degradation Assessment in Drylands project (LADA)

Name

Land Degradation Assessment in Drylands project (LADA)

Objectives

The Land Degradation Assessment in Drylands project (LADA) is focused on the creation of a basis for informed policy advice on land degradation at the global, national and local level.

Description

To achieve this target, LADA used the assessment of land degradation at different spatial and temporal scales and the creation of a baseline at global level for future monitoring by applying a variety of technologies, from satellite images to digital databases to soil and vegetation sampling, linked with biophysical and socio-economic issues. This

assessment of LADA could be only realised by the definition and mapping of different Land Use Systems, the involvement of stakeholders at all levels and contributive approaches, expert consultations and capacity building. LADA projects were set up in Senegal and in South Africa.

Consortium

This project is funded by the Global Environment Fund (GEF). The FAO and UNEP are involved in this project which is due to be completed by the end of 2010.

Contact Information

Email: Lada-Secretariat@fao.org

8.1.35 Mitigate and Assess Risk from Volcanic Impact on Terrain and Human Activities (MIA-VITA)

Name

Mitigate and assess risk from volcanic impact on terrain and human activities (MIA-VITA)

Objectives

Following UN International Strategy for Disaster Reduction recommendations and starting from shared existing knowledge and practices, the MIA-VITA project aims at developing tools and integrated cost effective methodologies to mitigate risks from various hazards on active volcanoes (prevention, crisis management and recovering).

Description

In EU countries, volcanic risks assessment and management are tackled through scientific knowledge and monitoring, although there is still a need for integration between all risk management components. For international cooperation partner countries (ICPCs), the risk management depends on local situations but is often less favourable.

The methodology will be designed for ICPC contexts but will be helpful for European stakeholders to improve their experience in volcanic risk management. The project multidisciplinary team gathers civil defence agencies, scientific teams (earth sciences, social sciences, building, soil, agriculture, Information Technologies and telecommunications) and an IT private company. The scientific work will focus on:

- risk assessment methodology based on a multi-risk approach developed at Mt Cameroon by one of the partners in cooperation with Cameroonian institutions
- cost efficient monitoring tools designed for poorly monitored volcanoes (satellite & gas analysis & volcano-seismology)
- improvement in terms of vulnerability assessment (people, buildings and biosphere)
- socio-economic surveys to enhance community resilience
- integrated information system (data organisation and transfers, communications) taking advantage of GEONETCast initiative Results will be achieved with help from local scientists and stakeholders in Africa (Cameroon, Cape Verde), in Asia (Indonesia, Philippines) and will be validated on a European volcano (Montserrat). The objectives will be reached through sharing/transfer of know-how, through scientific and technological developments, and through dissemination/training.

Consortium

The consortium of the project is: Bureau de Recherches Geologiques et Minieres (BRGM), France; Institute Nationale Geophysica Volcanologia (INGV), Italy; Instituto Superior Tecnico (IST), Portugal; Laboratoire de Géographie Physique (CNRS), France; Norwegian Institute for Air Research (NILU), Norway; KELL, Italy; Instituto de Engenharia de Sistemas e Computadores Investigação e Desenvolvimento (INESC-ID), Portugal; Hohenheim University, Germany; University of Cambridge, United Kingdom; French Civil Defence (DDSC), France; Italian Civil Defence (DPCI), Italy; National Meteorological and Geophysical Institute (INMG), Cape Verde; Ministry of Industry, Mines and Technological Development (MINIMIDT), Cameroon; Center for Volcanology and Geological Hazard Mitigation (CVGHM), Indonesia; Philippine Institute of Volcanology and Seismology (PHIVOLCS), Philippines. This project is funded under the EC FP7 Environmental theme. It started on 01/10/2008 and is expected to run for 48 months.

Contact Information

Pierre Thierry
Bureau De Recherches Geologique et Minieres (BRGM)
Quai Andre Citroen - Tour Mirabeau
Paris, France
Tel: +33-2-38643264
Fax: +33-2-38643689
http://miavita.brgm.fr/default.aspx

8.1.36 MOnitoring Natural resources for DEvelopment Co-operation (MONDE)

Name

MOnitoring Natural resources for DEvelopment Co-operation (MONDE)

Objectives

The establishment of dedicated monitoring techniques for promoting the sustainable management of natural resources in African, Caribbean and Pacific Group of States (ACP) countries is the main target of MOnitoring Natural resources for DEvelopment Co-operation - MONDE.

Description

Water, forests, agricultural/grazing land, coastal resources and their biological diversity have high economic and social value. Dynamic information on the location, condition and evolution of environmental resources is needed for Europe (both the EU and its Member States) to properly define, target, deliver and evaluate its development Aid strategies and programmes. African countries' decision-makers in national governments and regional organisations need ready-to-use information on environment on which to base appropriate policy responses.

The overall objectives of the monitoring of natural resources in Africa are:

- to improve the knowledge of natural resources dynamics, availability and trends and their correlation with the economies of beneficiary countries and livelihoods;
- to contribute to the ACP Observatory for Sustainable Development which is being built by the European Commission Services and the ACP Countries and Regions.

Consortium

The JRC.DDG.H Institute for environment and sustainability, Ispra (Italie) is coordinating the efforts for this project. Active collaboration in Europe with scientific institutions in the EU Member states (e.g. the Royal Society for Protection of Birds in UK, the IRD and CIRAD in France, the Université Catholique de Louvain in Belgium, the CNR in Italy). Collaboration outside of Europe with institutions from two African regions for the terrestrial part (the COMIFAC in Central Africa or the AGRHYMET centre in West Africa); with Southern Africa for the marine activities and national institutions in Africa (the Centre de Suivi Ecologique in Senegal or the University of Cape Town in South Africa). Collaboration with international organisations such as the African Union Commission and several UN bodies (e.g. UNEP and FAO).

Contact Information

Mr. Andreas Brink
Email: andreas.brink@jrc.ec.europa.eu
http://gem.jrc.ec.europa.eu/index.php/pages/MonitoringNaturalResourcesinAfrica/1

8.1.37 Moroto High School Telecentre Project

Name

Moroto High School Telecentre Project

Objective

Providing broadband access to the Moroto region of Uganda for teleducation and telemedicine.

Description

In Uganda's rural communities like most parts of Africa, there is general lack of basic ICT infrastructure. By the end of 1998, Multipurpose Community Telecenters (MCT) pilot projects had been launched with the support of IDRC/ITU/UNESCO at three different sites in Uganda. The broad mission of the MCTs was to study the efficacy of use of ICTs to promote rural community development.

Consortium

This project is run with the World Bank involvement. The project contributors are Education Program of the World Bank Institute World Links, Bill and Melinda Gates Foundation, Uganda Ministry of Education and Sports, Schools On-line, and participating schools in Uganda and Zimbabwe. The funding is provided by World Bank's Energy for Rural Transformation (ERT) program working in conjunction with WorLD and World Links.

Contact Information

Mr. Meddie Mayanja
Email: mayanjameddie@yahoo.com,
or mayanja@hotmail.com

8.1.38 New Technologies to Avoid Digital Division in e-Divided areas (Net-ADDeD)

Name

New Technologies to Avoid Digital Division in e-Divided areas (Net-ADDeD)

Objectives

NeT-ADDeD aims at developing and validating technical features for improved performance of operation of hybrid satellite-wireless technologies, in line with the growing demand for broadband communication within the 'International Cooperation' (INCO) countries, as well as with their evolution in Europe.

Description

The following axes contribute to improved conditions for use of the hybrid satellite-wireless technologies:

- The specification, implementation and validation of self-installation processes usable by customers;
- The implementation of compact transportable terminals, easy to deploy and adapt to the conditions in the INCO countries;
- The development and validation of remote satellite control capabilities for reduction of operating costs;
- The development and validation of mechanisms of service differentiation on the satellite component.

The above features were validated under real conditions in more than ten validation sites in North and Central Africa (Morocco, Burkina Faso, Benin), Turkey, Cambodia and Europe, which were selected with the support of national and regional public authorities. The sites correspond to different profiles of rural communities and cover domains, such as healthcare, education, agriculture, tourism, corporate and residential services.

NeT-ADDeD contributes in the e-Learning efforts in Africa. It provides innovative applications that meet the specific requirements of rural user communities in specific countries. Project activities in Central Africa aim at sustaining two main e-learning actions: basic education for illiterate populations in the most isolated areas of Burkina Faso and specific medical training in Benin. For this, the Groupement des Retraités Educateurs sans Frontières (GREF) is working on the development of specific contents for alphabetization and improving socio-economical activity dedicated to the rural communities. Furthermore, the Université Médicale Virtuelle Francophone (UMVF), an institutional medical training organisation, is adapting French medical university content to African conditions (hygiene, prevention of venereal diseases, nutritional education, public health, etc.).

Consortium

The coordinator for this project is Astrium SAS, France. Partners involved are: CNES, France; Cemagref, France; Université Blaise-Pascal, France; Sabanci University, Turkey; Telemedicine Technologies SA, France; Ministère de l'Industrie, du Commerce et des Nouvelles Technologies, Département de la Poste des Télécommunications et des Technologies de l'Information, Morocco; Groupement des Retraités Educateurs sans Frontières, France; Université Médicale Virtuelle Francophone, France; Istanbul Medical Faculty – Istanbul University, Turkey; Faculté de Médicine et de Pharmacie – Université Hassan II, Morocco; FORTHnet SA, Greece ; Connected Schools, France. This project started on 01/04/2007 with duration of 24 months. The project was funded under EC FP6.

Contact Information

María Zapata Margelí
EADS ASTRIUM
Telecom Systems Department
Telecommunications Business Development
31 Rue des Cosmonautes
31402 Toulouse Cedex 4, France
Tel: +33 5 62 19 94 59
Fax: +33 5 62 19 94 94
Email: maria.zapata@astrium.eads.net
http://www.netadded-project.eu

8.1.39 Quantification, understanding and prediction of carbon cycle, and other GHG gases, in sub-Saharan Africa (CARBOAFRICA)

Name

Quantification, understanding and prediction of carbon cycle, and other GHG gases, in sub-Saharan Africa (CARBOAFRICA)

Objectives

The CARBOAFRICA network develops and strengths the enhancement of an Earth observations system, the capacity building activities of Europe to understand global change process and the promotion of the integration of the environmental dimension in the social and economic context by supporting Sub-Saharan African countries on the path of a sustainable development.

Description

CARBOAFRICA aimed to reach its target by the establishment for the first time of a GHG fluxes monitoring network in Africa.

The network has been implemented to quantify, to understand and to predict greenhouse

gas emissions in Sub-Saharan Africa by a multi-disciplinary integrated approach and associated spatial and temporal variability. For the network, CARBOAFRICA used and expanded carbon observing systems and established new infrastructures, improving the required monitoring systems such as geo-spatial data sets and metadata systems.

CARBOAFRICA specific objectives are:

- to consolidate and expand the terrestrial carbon and other GHG fluxes monitoring network of sub-Saharan Africa;
- to provide an analysis of the requirements in order to establish a terrestrial GHG monitoring system for sub-Saharan Africa;
- to understand, quantify and predict the GHG budget of sub-Saharan Africa and its associated spatial and temporal variability;
- to assess the current land use change and evaluate the potential for carbon sequestration in sub-Saharan Africa in the context inter alia of the Kyoto Protocol.

Consortium

The consortium consists of 15 European and African organisations: University of Tuscia (IT), Max Planck Institute - BGC (DE), Lund University (SE), FAO, GTOS programme (International), CIRAD (FR), NERC-CEH (UK), CNR-IBIMET (IT), IAO (IT), 2nd University of Naples (IT), CSIR (ZA), UR2PI (CG), ARC (SD), CEA-LSCE (FR), King's College of London (UK), University of Leicester (UK)

Contact Information

Mr. Dario PAPALE
Department of Forest Science and Environment (DISAFRI),
University of Tuscia
via S. Camillo de Lellis, 01100 Viterbo, Italy
Tel: +39 0761 357251
Fax: +39 0761 357389
Email: darpap@unitus.it
http://www.carboafrica.net/objectives_en.asp

8.1.40 Remote detection contribution for update of urban maps in sub-Saharan Africa

Name

Remote detection contribution for update of urban maps in sub-Saharan Africa

Objectives

The remote detection contribution for update of urban maps in sub-Saharan Africa project goal lies in the development of urban maps for Cameroon, Senegal and Togo, which can be used for efficient decision making and the production of a software platform consisting of satellite remote sensing, a database of urban space-maps in digital format and hard copies of updated space maps of a number of target towns.

Description

The remote detection contribution for update of urban maps in sub-Saharan Africa project is part of the ARAPKE Action Plan and it works in partnership and in cooperation with African governments. The software platform is subdivided into three parts: „the development of urban maps", "consultation of urban cartography", and "the updating of urban maps". The remote detection contribution for update of urban maps in sub-Saharan Africa project fulfils the project by setting the following objectives: to acquire and process images for the setting up of the initial database; to digitalise old maps on 14 sites; to produce initial spatio-maps cities; to implement maps on test zones with geo referencing, classification of specific entities, extracting useful geographical information (communication methods, land demarcation limits, integration into the test zones).

Consortium

Partners are National superior school of technology of Yaounde, University of Yaounde I (Cameroon), Polytechnique school of France-Ville, University Marien Ngouabi, University of Bangui, CFA - STEPHENSON

Contact Information

Professeur Emmanuel TONYE,
Laboratoire d'Electronique et de traitement du signal
Département des Génies Electrique et des Télécommunications
Ecole Nationale Supérieure Polytechnique
Université de Yaoundé I, B.P. 8390 Yaoundé Cameroun.
Tel: +237 222-45-47
+237 994-47-53
Fax: +237 222-45-47
Email: tonyee@hotmail.com
www.lets.africa-web.org

8.1.41 Rural Multipurpose Community Telecentres Pilot Projects in Africa

Name

Rural Multipurpose Community Telecentres Pilot Projects in Africa

Objectives

The purpose is to try out and test new technologies and applications, and to demonstrate the economic, social and cultural development impact of provision of such tools to populations in rural and remote areas.

Description

The telecentres will offer low-cost means of providing library services through the provision of access to national and worldwide electronic information banks as well as providing information support for literacy campaigns, basic and non-formal education, information on government programmes, etc. They will also provide facilities for the generation and exchange of community-based information.

Consortium

This project is conducted under UNESCO with the involvement of ITU, IDRC, and African Information Society Initiative (AISI) programme.

Contact Information

J. Rose
Email: j.rose@unesco.org
http://www.unesco.org/webworld/telematics/telecentre.htm

8.1.42 Support Action to GMES-Africa on Earth Observation (SAGA-EO)

Project

Support Action to GMES-Africa on Earth Observation (SAGA-EO)

Objectives

The objective of SAGE-EO is to study an organisational model and its associated technology, and to prove a concept is viable for Earth Observation (EO) in five African countries.

Description

In order to strengthen the foundations of GMES in Africa, SAGA-EO is a comprehensive feasibility study focusing on the drafting of one dimension of possible future framework for African GMES. The SAGA-EO's target is bridge the gab that there is no project that tackles the EO field as a global approach. In order to implement capacity building is to set up dedicated EO networks of users belonging to the same African Country and link them together with existing initiatives in Africa and in Europe. The African national EO user network in each country will contribute in building thematic networks developed at national, regional or continental levels.

Consortium

Thales Alenia Space France (France), Geosat Technology SARL (France), GAF AG (Germany), Agence Nationale De La Meteorologie Du Senegal (Senegal), Bureau National d'Etudes Techniques et de Développement / Centre de Cartographie et de Télédétection (Cote d'Ivoire), INAM (National Institute of Meteorology) (Mozambique), Ghana Meteorological Agency (Ghana), Agence Nationale de l'Aviation Civile (Republique du Congo), Comafrique (Ivory Coast). This project is currently ongoing.

Contact Information

Jean-Paul Raoust
Thales Alenia Space France, France
Tel: +33 (0)4 92 92 77 01
Email:jeanpaul.raoust@thalesaleniaspace.com

8.1.43 SCenarios for Hazard Emergency Management (SCHEMA)

Name

SCenarios for Hazard Emergency Management (SCHEMA)

Objectives

The SCHEMA project explores the use of earth observation techniques to elaborate a generic method helping experts in building vulnerability and hazard impact maps associated with tsunamis and disaster scenarios.

Description

The vulnerability map creation technique takes into account the vulnerability characteristics of the systems facing a hazard (types of buildings, categories of inhabitants, etc.), of the environment (location of buildings in old areas, access conditions, etc.) and of crisis organisations that support efficient rescue operations.

The hazard scenario prepared by SCHEMA will be reviewed by end-users and stakeholders in countries recently impacted by coupled earthquake-tsunami events.

The resulting work flow will be embedded within existing hazard and exposure analysis techniques to provide risk evaluation of tsunami and related phenomena.

The project goal has been split into six sub-objectives:

- To draw, from the recent Asian tsunami post disaster studies, the input and output data required for hazard modelling, vulnerability/damage assessment and emergency management;
- To specify rules that provide hazard intensity, vulnerability and damage scenario descriptions to support relief managers, rescue planners and policy makers;
- To design and develop a scenario elaboration methodology;
- To propose tsunami-based disaster scenarios in five selected test sites;
- To validate the generic scenario development methodology based on a review of the results obtained for the test sites.

Consortium

GEOSCIENCES CONSULTANTS sarl (France) is the coordinator. The partners of the consortium are: ALGOSYSTEMS (Greece), HIDROMOD (Portugal), University of Bologna, Geophysics Laboratory (Italy), University of Coventry -Centre for disaster management- (UK), National Observatory of Athens- Institute of Geodynamics- (Greece), Centre Royal de Télédétection Spatiale (Morocco), ACRI-ST (France), Bulgarian Academy of Science -Space Research Institute- (Bulgaria), Joint Research Centre -Institute for the Protection and Security of the Citizen- (International Organisation), TUBITAK–MRC-EMSC (Turkey). The project started on 01/08/2007 with a duration of 36 months.

Contact Information

Richard Guillande
GEOSCIENCESCONSULTANTS sarl
157 rue des Blains
92220 Bagneux, France
Tel: + 33 146 646 060
Fax: + 33 146 646 161
Email: rg-gsc@wanadoo.fr

8.1.44 Second Administrative Level Boundaries data set project (SALB)

Name

Second Administrative Level Boundaries data set project (SALB)[46]

Objectives

The target of the Second Administrative Level Boundaries data set project (SALB) is to provide the international community with a global standardized GIS layer containing the delimitation of administrative boundaries. This is possible through the working platform for the collection, management, analysis, visualisation and sharing of sub-national data for the long-term and conserving this from the national to the global level.

Description

The SALB was initiated in 2001, with the founding of the UN Geographic Information Working Group (UNGIWG). Since the initiation of SALB, it has accrued different sub national levels for the data. The first one was created in 1990, the 2nd sub national level for the data in 2000. All of the databases are standardised as regards the international border, metadata profile, spelling, coding scheme and editing protocols used. The digital maps follow a thematic mapping in order to achieve precise location or modelling based on the differences between the documents compiled. All of the data of the platform contain National Mapping Agencies' contact information, coded historic changes starting from 1990 for the 1st sub national level and from 2000 for the 2nd sub national level, and seamless GIS maps corresponding to the different situations observed since January 2000.

Consortium

UN Geographic Information Working Group (UNGIWG) in collaboration with and validated by the National Mapping Agencies (NMA) of each UN Member State. This project is funded by the UN.

Contact Information

World Health Organisation
Office 3139
20, av Appia
1211 Geneva 27
Switzerland
Tel: +41 22 791.47.44
Fax : +41 22 791.48.06
e-mail: SALB@un.org

8.1.45 Service Availability Mapping (SAM)

Name

Service Availability Mapping (SAM)[47]

Objectives

SAM's goal is the collection and presentation of basic information in health services such as health infrastructure, human resources

[46] http://www.unsalb.org/

[47] http://www.who.int/healthinfo/systems/serviceavailabilitymapping/en/index.html

and services offered and to support decision making by providing sub-national planners with the skills and tools required to map and monitor service and resource availability on a regular basis.

Description

This project aims at the sub-national or district level. It consists of survey methodology, remote field data collection devices and the WHO's Health Mapper application, which map and monitor health services, including health infrastructure, human resources, and availability of key services. Depending on the situation of districts in the country, SAM creates maps and summary statistics.

Consortium

This project is funded by WHO.

Contact Information

World Health Organization
Avenue Appia 20
1211 Geneva 27, Switzerland
Tel: + 41 22 791 21 11
Fax: + 41 22 791 31 11

8.1.46 Space-based logistical support for distribution of medicines

Name

Space-based logistical support for distribution of medicines (SpaceDream)

Objectives

The objectives of the proposed activity were to undertake a study and a pilot project, to understand how space-based technologies can support the logistical efforts to supply antiretrovirals for the treatment of HIV/AIDS within affected areas of sub-Saharan Africa.

Description

SpaceDream addresses the interest and need to apply space based tools and technologies for the location of patients and specifically the seamless transportation of Antiretrovirals to patients in a comprehensive but robust supply chain management system as support to the “3 by 5 Initiative” focusing on simplified, standardised tools for delivering ART and effective, reliable supply of medicines and diagnostics.

SpaceDream investigates where existing Earth Observation, Navigation and Telecommunication systems and their different combinations can be used to develop a system to provide better access to medicine with focus on ART to developing countries in sub-Saharan Africa. SpaceDream aims to investigate specifically:

- The use of Earth Observation for basic mapping of infrastructure (such as roads, if they exist);
- The use of GIS for integrating the different mapping information as well as AIDS related information as base level information to be permanently updated by the local users;
- The use of Navigation for the tracking of ARV transportation;
- The use of Telecommunication for relaying this information and data bases and allowing the different levels of stakeholders to communicate with each other and thus promoting information exchange in the whole ARV supply chain;
- The training of users as well as their feedback, also using space added value when possible.

In a later phase, this first system architecture and its pilot application shall be used to demonstrate the use of those dedicated space means and, if successful, to extend the case to other countries, organisations, etc.

The study and pilot project were to be defined and carried out in a manner that emphasises sustainability of the approach taken, in order to facilitate scaling up of the project.

Consortium

This project was conducted in coordination with ESA. Participants were Comunità di Sant'Egidio, LUXSpace, CISAS ‘G.Colombo’ This project has been completed.

Contact Information

Fabio Riccardi
Comunità di Sant'Egidio
Email: fariccardi@tiscali.it
Didier Schmitt
European Space Agency
didier.schmitt@esa.int
www.spacedream.it

8.1.47 Strengthening the European-South African Science and Technology Advancement Programme (ESASTAP2)

Name

Strengthening the European-South African Science and Technology Advancement Programme (ESASTAP2)

Objectives

ESASTAP2 seeks to build on the foundation laid out by the European-South African Science and Technology Advancement Programme (ESASTAP), an FP6 Specific Support Action project.

Description

ESASTAP2 (Strengthening the European-South African Science and Technology Advancement Programme) is based on the Science and Technology (S&T) agreement signed between South Africa (SA) and the European Union (EU). ESASTAP2's long-term aim, and the criteria on which the project will be judged, is to increase S&T cooperation and priorities determination between SA, the EU and its member states; enhance the awareness of S&T capability in SA and in EU member states; promote SA's knowledge and participation in the Framework Programmes and other cooperation mechanisms, and increase the participation of SA S&T organisations in FP7. ESASTAP2 will seek mechanisms to facilitate greater bilateral and multilateral cooperation and to provide a greater input into future Framework Programmes and proposals, for sustainability and growth of the SA and EU S&T communities.

Consortium

ESASTAP2 is coordinated by South Africa's Department of Science and Technology (DTS). This project was funded by EC FP7 Capacities. The starting date was 01/09/2008 with a duration of 36 months.

Contact Information

Mrs. Mamohloding Tlhagale
Tel: +27 12 843 6340
Fax: +27-866810056
Email: contact@esastap.org.za
http://www.esastap.org.za

8.1.48 Telemedicine Task Force for sub-Saharan Africa (TTF)

Name

Telemedicine Task Force for sub-Saharan Africa (TTF)

Objectives

The TTF was set up to investigate the potential for satellite telecommunication technology as a tool for supporting healthcare systems in sub-Saharan Africa. It is composed of the relevant African Organizations, the World Health Organisation (WHO), the European Commission (EC) and ESA.

Description

Satellite solutions delivering information and communication technologies can help improve health in sub-Saharan Africa; this was the main conclusion of a dedicated telemedicine task force that met in Botswana. To make these solutions a reality, some short-term, concrete actions have been suggested in a pilot projects proposal. Three activities are proposed: one focussing on the health workforce (scaling-up numbers, improving performance, increasing quality); a second on clinical services (increasing health service coverage, reaching isolated areas) and a third aimed at strengthening the intelligence gathering capacity of health systems and their ability to use information for decision making.

These demonstration projects are meant to be used to inform and to help develop a framework for extending eHealth, which should be considered as part of the European Union Strategy for Africa commitment to utilise Information and Communication Technologies (ICT) to enhance interconnectivity in Africa. The potential of this greater interconnectivity, which will be supported under the tenth European Development Fund (EDF), to extend the reach of health and health services will be tested and demonstrated through these proposed projects.

Consortium

In the Task Force there was participation from the main relevant African organisations, the World Health Organization, the European Commission and the European Space Agency.

Contact Information

Giorgio Parentela
Email: Giorgio.Parentela@esa.int

8.1.49 TIGER

Name

TIGER

Objectives

The initiative's primary objective is to help African countries overcome problems faced in the collection, analysis and dissemination of water-related geo-information by exploiting EO technology.

Description

In 2002, responding to the urgent need for action in Africa stressed by the Johannesburg World Summit on Sustainable Development (WSSD), the European Space Agency (ESA), within the context of the Committee of Earth Observation Satellites (CEOS), launched the TIGER initiative. The overall objective of the initiative is to assist African countries to overcome problems faced in the collection, analysis and use of water related geo-information by exploiting the advantages of Earth Observation (EO) technology.

Since 2005, under the guidance of the African Ministerial Council on Water (AMCOW), with contributions from CEOS (e.g., ESA, the Canadian Space Agency), UNESCO and the African Development Bank and in collaboration with the Economic Commission for Africa (UN-ECA) and several other African and international organizations (e.g. CSIR, CRTM, Ramsar-Africa, South African Department of Water Affairs), TIGER has supported African partners with access to space-borne data and products, by offering specific training on EO applications for water management, by funding North-South collaborative projects aimed at developing tailored EO-based water information systems, and by supporting take-off, operationalisation and technology transfer of those demonstrated systems to African water authorities.

TIGER has completed its first implementation period (2005-2007) involving more than 150 African institutions (water authorities, universities, technical centres) through its projects and training activities. The results and achievements of these first years of activity were recognized at the First African Water week organized in Tunis on 25-29 March 2008 with a direct recommendation "International initiatives like … TIGER which provide useful tools to the countries to strengthen their capacities for ensuring water security should be encouraged and supported". As a direct response to this African request, a new implementation period of the TIGER initiative was proposed aiming at supporting African countries to enhance their capacities to better understand, monitor and manage water resources.

A major component of this new phase is devoted to supporting African scientists to develop the scientific skills and technical capacity to make the best use of Earth Observation technology to better understand, assess and monitor the status of the water resources in Africa as well as the potential impacts of climate change, hence establishing sound scientific bases for developing effective adaptation or mitigation measures at the political level in the continent. A call for proposals was issued in 2009, which resulted in the selection of 20 projects submitted by African scientists, water authorities and technical institutions.

Consortium

The TIGER process aims at fostering partnerships between the TIGER beneficiaries and space agencies worldwide, the expert centres in earth observation applications for water management (e.g., key universities and international research organisations), the private sector (e.g., value adding companies), NGOs and finance and development agencies. All these organisations represent collectively the TIGER Stakeholder Group.

Contact Information

European Space Agency (ESA)
Earth Observation Application Department - EOP/AEP
Via Galileo Galilei - Casella Postale 64
00044 Frascati – ITALY
Email: tiger@esa.int
http://www.tiger.esa.int

8.1.50 UNITAR Operational Satellite Applications Programme (UNOSAT)

Name

UNITAR Operational Satellite Applications Programme (UNOSAT)

Objectives

UNOSAT sets a program, which is based on satellite technologies, including earth observation to telecommunication, positioning and navigation. The goal is to support emergency response operations and also states with respect to poverty, hazards, conflicts and crises.

Description

UNOSAT is a technology intensive programme active in all aspects of applied research relating to satellite solutions, from earth observation to telecommunication, positioning and navigation. UNOSAT delivers satellite solutions, geographic information to organizations within and outside the UN system to make a difference in the lives of communities exposed to poverty, hazards, and conflict, or affected by humanitarian and other crises.

UNOSAT is a program of the United Nations Institute for Training and Research (UNITAR). In 2003 UNOSAT created a new humanitarian

rapid mapping service that is today fully developed and has been activated over 100 times by UN relief and coordination agencies. This work implies very quick acquisition and processing of satellite imagery and data for the creation of map and GIS layers in support of emergency response and humanitarian relief coordination (UNDAC teams, impact assessment missions, damage estimates, etc). UNOSAT remains engaged beyond the emergency phase by supporting early response and recovery activities. A growing number of national and international development projects receive support from UNOSAT for strategic territorial planning and advanced GIS applications. The expertise accumulated by our technical team with extensive field experience allows us to provide a technological edge to local capacity development.

Furthermore, UNOSAT develops and implements integrated training modules and programmes that sometimes include the design and realization of GIS and cartographic centres that can continue to exist after the conclusion of a cooperation project, for example as self-supporting geo-information centres. The topics this project is involved in are: satellite imagery search and procurement assistance, image processing, map production including information extraction & analysis, research and methodology (design and guidance), field and remote technical assistance, including strategic consulting, training and capacity development, integrated satellite communication solutions, data storage and information & communication technology solutions.

Consortium

This project is developed by UNITAR and is currently ongoing.

Contact Information

UNOSAT / UNITAR
International Environment House
Chemin des Anémones 11-13
CH-1219 Châtelaine,
Geneva - Switzerland
Tel: +41 22 767 4020 (UNOSAT Operations)
Tel: +41 22 917 8720 - 8650 (UNOSAT Management)
Fax: +41 22 917 8993
Email: unosat@unitar.org
http://www.unitar.org/unosat/

8.1.51 United Nations International Boundary Evidence project (UNIBED)

Project name

United Nations International Boundary Evidence project (UNIBED)[48]

Objectives

The creation and maintenance of a worldwide geodatabase, based on boundaries information, including accurately depicting geometry, status, claims, date of treaty agreements is the target of the United Nations International Boundary Evidence project.

Description

The creation of a worldwide geodatabase is only possible with the informational and evidenced sources of authorities, which were set through international treaties like treaty maps, coordinates, treaty text description and interpretation of satellite imagery.

The web interface represents the essential tool to collect the information for a database in a complete and comprehensive way. This database is forwarded for the service of key UN personnel in the Department of Political Affairs, Department of Peacekeeping Operations, the UN Treaty Section and senior management

Consortium

This project is developed by the UN

Contact Information

Guillaume Le Sourd
United Nations
1 UN Plaza, DC1-512,
New York 10017, United States
Tel: +1 917 367 9534
Email: lesourd@un.org

8.2 Private Sector Initiatives

In the following subsections a list of projects in Africa using satellite applications by few private entities are listed. The list is not exhaustive but provides an overview of the types of projects developing.

[48] http://proceedings.esri.com/dvd/uc/2009/uc/abstracts/a1769.html

8.2.1 GEO-Africa (AfricaSat1)

Name

GEO-Africa (AfricaSat1)

Objectives

To provide permanent and high resolution Earth Observation data, which are fully dedicated to Africa and its specific needs, in order to encourage the continent's sustainable development.

Description

The GEO-Africa satellite (AfricaSat1), conceived by EADS Astrium, will capture images over the entire continent. It will provide the sensing of up to 100 scenes of 300 x 300 km per day, with a possible total coverage of Africa every 4 days. Receiving stations will be in Africa making access to its images easier. Additionally, fast access (in 30 minutes) to any area of the continent will be guaranteed to deal with crisis situations. GEO-Africa will bring important socio-economic benefits to the continent, in terms of natural resources management, crisis management, urban planning and infrastructure development. The AfricaSat1 system will be built based on compatibility with existing national Low Earth Observation missions in Africa and synergies with on-going African framework programs, such as the African Resources Management Constellation (ARMC).

Consortium

EADS Astrium (system designer), African Union, Algerian Space Agency (ASAL), Nigerian National Space Research & Development Agency (NASDRA), Egyptian National Authority for Space Science and Remote Sensing (NARSS), South African Department of Science and Technology (DST), Kenya, New Partnership for Africa's Development (NEPAD), United Nations Economic Commission for Africa (UNECA), African Association of Remote Sensing of the Environment (AARSE), Regional Centre for Mapping of Resources for Development (RCMRD), Lake Chad Basin Commission (LCBC), Faculty of Geo-Information Science and Earth Observation of the University of Twente (ITC), Group on Earth Observation Secretariat.

This project has completed the design phase and is currently in the development phase.

Responsible Body

GEO-Africa Core Team

Funding Body

European Development Funds (under assessment) and African Space Agencies.

Contact Information

Andre Nonguierma, GIS Officer, United Nations Economic Commission for Africa,
PO Box 3001, Addis Ababa, Ethiopia

Email: ANonguierma@uneca.org

8.2.2 Other 3 billion (o3b)

Name

o3b (Other 3 billion)

Objectives

To launch a constellation of 16 satellites into orbit to provide inexpensive Internet service to developing countries.

Description

O3b Networks plans to provide a high-capacity satellite connection to the Internet on which telephone companies and Internet service providers in emerging markets will be able to build their high-speed mobile and web-access offers.

Consortium

Google, HSBC, SES, North Bridge Venture Partners, Allen & Company, and Liberty Global

Responsible Body

o3b Networks

Funding Body

Google, HSBC, SES, and Liberty Global

Contact Information

Maher Simon VP - International Carrier Accounts
Tel: +44 7833 453018
Skype Name: simon.j.maher
Email: simon.maher@o3bnetworks.com

http://www.o3bnetworks.com/

8.2.3 mHealth for Development

Name

mHealth for Development

Objective

mHealth's objective is harnessing mobile technologies for the provision of health-related service.

Description

In many parts of the world, epidemics and a shortage of healthcare workers continue to present grave challenges for governments and health providers. Yet in these same places, the explosive growth of mobile communications over the past decade offers a new hope for the promotion of quality healthcare. Among those who had previously been left behind by the 'digital divide,' billions now have access to reliable technology.

The mHealth project case studies are organized by primary application area, from least specialized (education and awareness) to most specialized (diagnostic and treatment support). Although several projects offer multiple applications, they are categorized here by their more specialized function. The main topics are:

- Education and Awareness
- Remote Data Collection
- Remote Monitoring
- Communication and Training for Healthcare Workers
- Disease and Epidemic Outbreak Tracking
- Diagnostic and Treatment Support

In particular for Africa the project "HIV Confidant"[49] has been implemented in South Africa. In places where HIV-positive status remains a stigma, successful outreach efforts must address people's privacy and confidentiality concerns. The HIV Confidant project aims to encourage HIV/AIDS testing by ensuring secure distribution of test results through the use of handheld computers and standard encryption techniques. Dimagi, a US-based software company, implemented the HIV Confidant project in 2003 at the Africa Centre for Health and Population Studies in South Africa. In the pilot, 45,000 adults were tested for HIV, and results were shared with participants through a secure PDA-based system. People who were tested were provided with a unique ID code, and results were given only to those who provided the code. The HIV Confidant system runs on Palm m500 and Handspring Visor PDAs, but can be adapted for non-Palm devices for greater flexibility and extended reach.

There are many other projects under this mHealth initiative in Africa. Some are

- *Project Masiluleke* in South Africa with sponsoring organisations: Sponsoring Organization and Partners: Praekelt Foundation, iTeach, National Geographic, Nokia, Siemens Networks, MTN, Ghetto Ruff, Children of South African Legacies, Aricent and frog design
- *Text to Change (TTC) – HIV Prevention Through SMS Quiz* in South Africa with sponsoring organisations: Text to Change (TTC), Zain (previously Celtel), the local NGO AIDS Information Centre (AIC), the Dutch Ministry of Foreign Affairs and Merck
- *The Dokoza System* in South Africa with sponsoring organisation. Dokoza, State Information Technology Agency (SITA), Centre for Public Service Innovation (CPSI), Centre for Scientific and Industrial Research (CSIR) and the Meraka Institute, with the cooperation of South Africa's National Department of Health
- *EpiHandy* in Uganda, Zambia, Burkina Faso with the Center for International Health, Norway as a sponsoring organisation.

Consortium

Rockefeller Foundation, United Nations Foundation, and Vodafone Foundation, and U.S. President's Emergency Plan for AIDS Relief (PEPFAR), Orange Healthcare, and telecommunications service providers in the countries where they carry out the mHealth project

Responsible Body

UN Foundation/ mHealth Alliance

Funding Body

Ted Turner donated $ 1 billion in 1998, but now a public charity.

For the "HIV Confidant" project in South Africa the sponsoring partners was Dimagi, Inc. (privately held software company).

Contact Information

Media Contact
Adele Waugaman
(202) 778-1635
awaugaman@unfoundation.org

8.2.4 Réseau en Afrique Francophone pour la Télémédecine (RAFT)

Name

Réseau en Afrique Francophone pour la Télémédecine (RAFT)

[49] http://www.dimagi.com/content/hiv-confidant.html

Objective

The deployment of telemedicine and IT enabled diagnostic devices in Africa.

Description

Provision of healthcare through tele-consultations, tele-echography, and tele-education of healthcare providers.

Consortium

Geneva University Hospitals, UNESCO, Canton Genève, le Service d'Informatique Médicale des Hôpitaux Universitaires de Genève, l'Université Numérique Francophone Mondiale, la Fondation Health On the Net, Fonds de Solidarité Numérique, and le Réseau Universitaire International Genevois.

Responsible Body

Geneva University Hospitals

Funding Body

Canton Genève

Contact Information

Prof. Antoine Geissbuhler
Email: antoine.geissbuhler@hcuge.ch, ag@hcuge.ch
http://raft.hcuge.ch

8.2.5 South African Large Telescope

Name

South African Large Telescope

Objective

The development of a large teleshope for the southern hemisphere.

Description

The South Africa Large Telescope is the largest optical telescope in the southern hemisphere. South Africa contributed about a third of its total value. The rest was contributed by partners from the international community. The telescope was completed in November 2005.

Consortium

National Research Foundation South Africa, Nicolaus Copernicus Astronomical Centre Poland, Hobby-Eberly Telescope Board Texas, Rutgers University, USA, Georg-August-Universität Göttingen Germany, University of Wisconsin - Madison USA, Carnegie Mellon University, USA, University of Canterbury New Zealand, Universities of Central Lancashire, Southampton, Nottingham, Keele, the Armagh Observatory in Northern Ireland and the Open University, University of North Carolina - Chapel Hill USA, Dartmouth College USA, Inter-University Centre for Astronomy and Astrophysics India, American Museum of Natural History USA.

Funding Body

Institutions in South Africa, United States of America, Germany, Poland, India, the United Kingdom and New Zealand.

Contact Information

Tel: +27 (0)23 5711205
Fax: +27 (0)23 5712456
Email: salt@salt.ac.za
http://www.salt.ac.za/telescope/overview/

9. Conclusions and Recommendations

9.1 The setting

Africa is often looked at as one continent but the needs change from region to region. Although some African countries have reached mature levels of space technology and use of satellite applications, most countries are struggling to maintain the basic levels needed for simple use of satellite applications. Space is transversal and touches many policy areas (e.g. environment, agriculture, health, disaster management), thus, it is important that it receives more attention for growth and sustainable development. Africa has the highest number of Regional Economic Communities (RECs) which target cooperation and integration of their Member States and receive strong support from them. Eight of these RECs have become pillars of the African Union (AU) but the links between the AU and RECs are not always strong. There is often a strong debate in Africa about the effectiveness of the AU supporting integration and development whereas the RECs are often portrayed as the most effective instrument due to their closer links with the Member States. The AU appears to be lacking sufficient recourses to achieve its goals and needs to be strengthened.

There are only five years left before for the Millennium Development Goals (MDGs) target. It is essential that European and international efforts align their resources and engage in closer coordination with each other and with African actors. In particular, the United Nations is engaged in various activities in Africa and even though not all are directly engaged in space related projects and programmes, many of them are using space-based information for their work. The UN activities are not always fully coordinated between the agencies as “one UN”. Currently though, there are some pilot projects in African countries to implement the “one UN” approach, such as the one in Tanzania[50].

The First Action Plan (2008-2010) for the implementation of the Africa-EU strategic partnership is currently under evaluation, including its various components such as the GMES in Africa. The Second Action Plan (2011-2013), which is soon to be adopted, is expected to keep the same eight thematic areas as the first one. The eight thematic areas have proven to be suitable for setting the basis of cooperation, but it is necessary to capture the progress made, lessons learned and set out concrete actions to be followed. Thus, Area 8 which includes the space component, will be maintained. It is important though, to properly capture the transverse nature of space in the other areas and to ensure the involvement of the RECs for future developments. Furthermore, European Member States continue to support the efforts of the EU by engaging themselves in direct agreements with various African countries, but the coordination between the various Member States and the EU is loose which results in duplication of efforts and would need to be strengthened.

Africa is in need of maintaining these long-lasting partnerships and finding new effective instruments to support its efforts to achieve sustainable development. A comprehensive mapping of activities and actors is essential in setting the way forward. Space activities in the African continent are characterised by strong collaboration between the African countries and European and international actors. There are various initiatives at all levels but there is lack of a comprehensive overview tackling the three main questions in Africa regarding the use of satellite applications: 1) “who is who”; 2) “who is doing what”; and 3) “who is cooperating with whom”. This study provides a comprehensive overview of the main African actors and their activities in the continent as well as gathers information regarding projects run by the main European players and the United Nations organisations. This mapping was conducted in consultation with European and African stake holders and the presented material reflects mainly a European perspective. From the information gathered a social, legal and economic, political and technological (SLEPT) microenvironmental factor analysis is used as a tool to draw up policy recommendations.

50 http://www.untanzania.org/inside.asp?mainId=3&contID=8

9.2 Conclusions and Recommendations by Factor

Social factor

Space applications can assist in providing solutions to the African population regarding their basic needs such as providing food and water security, healthcare, education, early warning, disaster management and emergency response, etc. Nevertheless, the benefits of space applications are not sufficiently communicated to decision makers as well as the wider population and there is not enough basic education at various levels to perform, manage and operate space-based assets. Various space projects in different areas have been developed for Africa but very few are sustainable beyond the pilot phase. This is due to the fact that often the local community of the end users is not involved from the beginning and does not have the feeling of ownership. Often, appropriate bodies have not been identified within the government structure to take up the responsibility for running the project and its maintenance. The projects developed in Africa are often conceived by developed countries without being able to capture properly the societal need and infrastructure restrictions of the underdeveloped countries. The recommendations are:

- *Enhance communication about the benefits of space.* Appropriate information mechanisms need to be set up to communicate the benefits of various space applications and projects at local community level. Such communication mechanisms could be local information centres, radio and TV info notes.
- *Enhance education regarding space and its applications.* Education regarding space should be enhanced at school and university level. Adequate learning material on the benefits of space and its contribution to skills development should be provided. Capacity building at all levels of education (primary, secondary, tertiary and continuous learning) should be emphasised. Educational programs are a long-term measure with longer social benefits.
- *Enhance the involvement of the community in ensuring continuation and sustainability after the pilot phase is completed.* The African citizen should be involved in the ownership of each project. This would enhance continuation beyond the completion of the project. Involvement at the community level is expected to be the most effective. A coherent mechanism of knowledge transfer has to be ensured from the very beginning of pilot-projects.

Legal factor

In Africa the areas of telecommunications and satellite imaging are fragmented into small national markets and the licence requirements and regulations vary from country to country. Current regulatory aspects regarding satellite applications may hinder private initiatives by African and foreign investors and delay sustainable development. Furthermore, there is very little coordination between various policy and regulatory frameworks in various African countries regarding space and the space component of other sectors. Often different African government or different ministries within a government purchase the same asset from satellites. This duplication increases administrative and economic overheads. Current taxation systems on goods import affects the development of satellite applications equipment since equipment, devices and services need to be imported. It appears that taxation is duplicate for the same goods and services at various levels resulting in accumulative excessive burden for the importer. The recommendations are:

- *Implement appropriate regulations.* It is necessary to foster initiatives to build institutional capacity in making treaties, setting standards, formulating policy, drafting regional integration protocols and harmonising regulations at a pan-Africa level.
- *Coordinate policies and regulations.* African governments should coordinate their policies and regulations regarding commercial and non-commercial space applications (e.g. data policy). They should foster policy and regulatory harmonisation to create larger common markets based on regional economic communities while increasing private sector participation. This would facilitate exchange of assets and reduced duplication in purchasing the same asset.
- *Harmonise licensing procedures.* Establish common licensing and authorisation procedures instead of having them country by country. Creation of an internet based licence application and dissemination of regulatory information would significantly assist potential investors.
- *Facilitate cross border exchange and harmonise information infrastructure.* With regard to the purchase of information from satellites, initiatives should be

promoted for investment in access and cross-border information infrastructure through effective partnerships between public, private and not-for-profit sectors, in order to achieve universal access and full inter-country exchange of information and data.

Economic factor

Even though during the past 10 years there have been significant efforts by the REC's in integrating the various markets, many of them are still fragmented. At the same time the low income levels in Africa and limited economies of scale make it unattractive for Foreign Direct Investment (FDI). Licence fees for satellite systems are generally very high and the cost of satellites is also high, which makes it difficult for companies to survive. There is a need to have low cost systems to facilitate improved access. The African market is mostly dominated by government activities in the field of space, satellite applications and related sectors. It is important that the market is opened up. The recommendations are:

- *Enhance institutional coordination.* Common economic targets, complementarities and how they can be achieved need to be identified. The current institutional framework in Africa is appropriate but concrete links of cooperation need to be established between the main African actors like the AU, UNECA and the REC's. Common strategic partnerships between them are essential for sustainable development. Adequate resources such as funding, personnel and technical expertise are essential for the success of the partnership.
- *Align funding mechanisms.* The various funds provided from Europe, United Nations, European member states, private bodies, etc need to be aligned in terms of funding, time scale and expected targets in order to better utilise the available resources, avoid duplication and create complementarities. The funding mechanisms should also be aligned such that they cover the entire cycle of a satellite applications project, from pilot project to hand over, operation, capacity building and maintenance.
- *Refocus allocation of funding.* A cost-benefit analysis should be performed before projects are started. The criteria should not necessarily be based on direct monetary factors but should be based factors such as benefits to the citizen, safety, job creation, etc.
- *Improve economic regulatory frameworks for FDI,* making them far more open, permitting profit repatriation and providing economic measures and other incentives to attract investment.
- *Enhance and regulate the pre-competition environment.* Promote liberalization and privatization in order to open access to satellite operators and service providers through partnerships with local stakeholders in terms of public entities, private firms and not-for-profit sectors. The pre-competition framework has to facilitate the involvement of African SMEs.
- *Improve taxation.* African countries are also encouraged to accelerate the conclusion of double taxation treaties (DTTs). DTTs can make it more attractive for foreign investors to invest in a country by helping them to avoid paying taxes twice on the same transaction[51]. Taxation is a leverage to be considered also in order to support demand for satellite services in terms of tax-emption or negative taxation – financial support - in order to improve citizens' rights as members of society (freedom of self-expression, right to be informed, freedom of press, etc).

Political factor

The political commitment of African leaders to work together and with other international partners to solve problems for the African citizen and achieve societal, economic and political integration is being materialised step by step through the African Union and the various Regional Economic Communities. Nevertheless, political commitment for space and satellite applications is lacking. This is mainly due to the fact that there is insufficient understanding of the benefits and communication of the success stories. The context of space policy is different in Europe and in African and also varies from country to country in Africa. The main objectives of a space policy for Africa would be to focus on implementing those policy aspects that would allow successful use of space based information and applications for sustainable development. Space policy is uncoordinated at the continental level as well as the regional level. Furthermore, within each African country there are insufficient links between the space

[51] The majority of African countries have signed multilateral agreements dealing with the protection of FDI, such as the Convention establishing the Multilateral Investment Guarantee Agency (MIGA) and the Convention on the Settlement of Investment Disputes between States and Nationals of Other States.

component and other policy areas (e.g. agriculture, environment, health, transport, security). The link between the priorities of the various African countries and how space can assist in achieving them is missing. Even though at the continental level the appropriate institutional mechanisms are present for harvesting the benefits of space, there is insufficient coordination between the various institutions.

- *Capture adequately Member States needs and translate them into concrete areas where space can assist in meeting them.* The national needs of African countries where space competence is present need to be translated and reflected in other national policies (e.g. agriculture, environment, health, transport, security) in a coherent manner. Coordination meetings, workshops and conferences can be used as tools for the appropriate capturing of the need.
- *Identify responsible actors.* At the beginning of every project the appropriate potential governmental actors needed for the sustainability of projects need to be identified. Communication and establishment of appropriate links with the respective authorities should be part of the tasks before the completion of the pilot project.
- *Establish appropriate governance links.* The AU, ECA and REC's are important actors in Africa. The coordination and interaction between them regarding space based information and applications should be strengthened and formalised.
- *Establish harmonized international relations between African countries.* This would increase political stability and facilitate FDI's attractiveness which would further technological and industrial development.
- *Increase the involvement and commitment of political leaders.* African leaders and politicians need to be more aware of and committed to the benefits of space and use of satellite applications for the sustainable development of Africa.
- *Coordinate space policy and other policy areas.* Space policy and the space component in other policies (e.g. agriculture, environment, health, transport, and security) need to be identified in a systematic way. Coordination should be achieved at the three levels: national, regional and continental.
- *Develop the basis for a pan-African space policy.* The common aspects of the space component should be treated at regional and continental levels. After establishing national needs, African countries should work together on the regional and continental levels to tackle their common goals. As a second step, a pan-African Space Policy that can reflect common interests such as disaster management and emergency response, climate change (including desertification, deforestation, drought etc.), telecommunications and GMES would better facilitate the coordination of efforts. The area that is in vital need of such a pan-African approach is disaster management and emergency response. Thus, a common policy in disaster management and emergency response could set the first area of the pan-African space policy.
- *Formalise institutional relations for space based information and applications.* As a first step, existing institutions are sufficient enough to work together in building sustainable development for Africa but there is lack of cooperation and coordination between institutions and agencies involved in space based information and applications. The first point of coordination that is needed is in the field of access, use and sharing of space based information. In particular, maps and GIS would be the first step to formalise interactions and institutionalise use of these assets on the regional level. A particular topic where this coordination is essential is early warning and emergency response in disaster management. An effort should be made to foster horizontal agreements to establish institutional groups. A second step to be targeted in the long run would be to set the basis for institutionalisation at the continental level. Particular fields of coordination could be environmental monitoring and disaster management. Discussions about the establishment of a continental institution which would work in cooperation with the AU date back to 1991. The AU project for the establishment of an African Space Agency was finally launched last month. Such an institution would be beneficial for centralise purchase of space data like maps, satellite images etc. and would be beneficial for avoiding duplications. It could also be involved as a central point to the outsider with respect to disaster management and emergency response. Nevertheless, this should be a long term plan starting first at national and regional levels since different countries have different needs and different levels of development.

- *Enhance communication of the benefits of space.* The success stories of the use of satellite applications for the benefit of the African citizen and for sustainable development for Africa need to be communicated in way that political figures are able to use as arguments for supporting their case. Relevant information material needs to be prepared by consortiums including representatives from technical experts, end users, entrepreneurs, administrators, societal and political groups, etc.
- *Promote bilateral and/or multilateral investment treaties (BITs).* This should be conducted in conjunction with other countries who aim at protecting and promoting FDI to clarify the terms under which FDI can take place between the partner countries.

Technological factor

The different African countries focus their space activities and run projects under three main categories: a) Infrastructure development; b) use of satellite applications; and c) capacity building. From these categories the majority of the projects focus on space applications and there is a need for more projects on capacity building. A pan-African University is currently being established where space will be covered by the South Africa unit.

- *Increase projects on capacity building.* Projects with a focus on capacity building should be enhanced. These projects should cover all levels including schools, universities, government employees, politicians, and the public at large.
- *Identify relevant space technology in relation to providing solutions to needs of African population.* Developed countries often initiate projects without being able to sufficiently capture the needs of the African population. African-led initiatives should be given priority, which capture African needs. These projects should be chosen at a regional, national and continental level. Appropriate workshops should be taking place that involve the local community and the technical experts.
- *Implement technological innovation scoreboard.* This should be implemented in order to map and to monitor the degree of technology. This tool will help to understand the current degree of technology and whether this technology is suitable for the needs of the end user.
- *Increase cooperation between those that have technology know-how and users.* The end user often has very limited understanding or receives insufficient capacity building from those that have the technology know-how. After training they often move out to more senior managerial (not hand-on) levels. Increased intensity of education at the various user levels could be enhanced. Community to community training may reduce the cost of implementation.
- *Enhance transfer-of-knowledge mechanisms through initiatives of mutual learning platforms.* Academic cooperation with all other actors in order to facilitate know-how development and exchange should be enhanced. Existing academic networks could provide the initial necessary links needed. The pan-African University, currently under establishment, is expected to play an important role in the development of sustainable development of Africa.

Specific recommendations per actor

In the following table when the word space is used it is meant space based information and applications through the use of satellites, unless stated otherwise.

Actor	Proposed Actions
European Union (European Commission and other European institutions)	• strengthen its neighbour policy towards Africa focusing on Space as a strategic leverage crossing different thematic issues, such as extension of infrastructure, extra-EU transport, energy, industrial development and trade, etc • enhance thematic coordination of European activities of European Actors • enhance coordination of EU financial instruments, European Investment Bank and Member States' contributions • establish a closer dialogue with the UN agencies present and active in Africa and coordinate funding. • establish direct dialogue with Regional Economic Communities in Africa, the eight pillars of the AU and the others. • promote the use of space in other areas of the strategic partnership for the 2011-2013 Action Plan. • promote the benefits of space for Africa's sustainable development to African politicians through the High Representative of the European Union for Foreign Affairs and Security Policy and External Action Service representatives in African countries. • regularly maintain an overview mapping of funded projects, bilateral agreements, best practices, etc. • assist the African Union with obtaining adequate resources for its targets • move towards a co-funding (Africa-European) system with African states and the Africa Development Bank to increase sustainability • perform assessments of public funding activities and share best practices with European Member States. • implement better mechanisms for follow-up impact assessment of pilot projects.
EU Member States	• improve coordination between national activities in Africa and EU activities • maintain an overview of national activities regarding space based information and applications for Africa • perform assessments of public funding activities and share best practices with other European Member States and the EU.
European Space Agency	• ensure that the actors involved in ESA projects properly deal with the transition from pilot projects to operational projects and assure the commitment of African government relevant bodies to continuation and local acquisition in terms of knowledge end technology transfer • promote the coordination of space efforts through African space agencies for the space faring nations of Africa and through other governmental and technical bodies where space is relevant for those that are users of space and those that are not • participate in education regarding the use of space at different levels
EUMETSAT	• promote coordination of space efforts on a regional level through its links with African government bodies and technical agencies • increase support for capacity building efforts in Africa • enhance the role of Africa to acquire data in-situ in order to merge with satellite data
United Nations	• promote the use of space at the level of decision makers and through UNECA for the sustainable development of Africa • coordinate the efforts between the different UN agencies and programmes regarding space use as "one UN". • Establish in UNECA a specific department for space and coordinate and support the RECs in this area. • establish coordination mechanisms with the EU regarding space activities in Africa • perform cost benefit analysis of the use of space as part of awareness campaigns in order to underline shadow-prices and cross-sectional benefits • increase coordination of activities through the joint secretariat between the AU, UNECA and AfDB.

Actor	Proposed Actions
African Union	• enhance its international relationships in order to show local needs that could be satisfied through space technology • enhance its role as a catalyser of Africa's Vision for Space. The main objectives of space policy for Africa would be to focus on implementing those policy aspects that would allow successful use of space based information and applications for sustainable development. • increase the resources of the AUC in relation to space • establish a position responsible for space as a shared competence among African Member States which deals with space topics such as telecommunications, navigation, remote sensing, etc and is responsible for maintaining a database of African capabilities and projects and promoting the use of space for sustainable development • encourage Africa entrepreneurship • promote a bottom-up approach to projects • establish closer coordination with the RECs and establish policies that cover regional needs and identify how space can assist in development • develop a capacity mapping database at the continental level (experts, institutions, data, information, infrastructure etc.) • map the different ongoing activities conducted by regional and international agencies and identify end users. • collect in a systematic way and disseminate case studies and lessons learned, promoting the benefits of space with arguments to inform politicians and assist them in supporting a case for space • promote cost benefit analysis of projects that use space assets. • increase coordination of activities through the joint secretariat between the AU, UNECA and AfDB. • develop mechanisms for increasing co-funding activities with African capital • promote education on space at all levels of education: primary, secondary, tertiary and continuous learning (egg. school, university, on the job training, etc.) • perform studies on the economic benefits of space for African development • promote the implementation of space policy and space components in other policies for capacity building at various institutional levels and different actors (end-users, decision makers and trainers) • ensure sustainability by continuous training to mitigate "brain drain" • prepare and conduct workshops for key decision makers
Regional Economic Communities	• promote a fair investment climate in order to attract FDI for development and deployment of space • take space up as a topic for sustainable regional development • create a map of existing mechanisms and capacities in the region (experts, institutions, data, information, infrastructure etc.) • conduct an evaluation of the use of space from different mechanisms in the region • promote space based information and applications as a key development component for sustainable development • promote the development and adoption at the regional level of a space policy reflecting the needs of the regions • open intra-regional dialogues for continental integration • perform cost benefit analysis of the use of space for development and comparison with no use of space. A particular case would be disaster management and disaster forecasting. • establish regional centres for single point acquiring of data and maintaining a common database • promote regional development in download capabilities, information processing and capacity building • facilitate the establishment of a critical mass of providers and users at the regional level • organise regionally early warning and emergency response in disaster management and coordinate with UN efforts

Actor	Proposed Actions
	• facilitate cooperation between universities, schools, technical centres and the local community for training on space assets and sustainability • coordinate with member states to harmonise appropriate rules and regulations, licensing procedures, taxation etc. regarding space • facilitate horizontal cooperation to establish institutional groups • strengthen multi-institutional mechanisms
African Member States	• develop a map of national needs and space relevance • develop a map of national capabilities (experts, institutions, data, information, infrastructure etc.) • facilitate local activities and commitment to build up capacity and avoid reliance on foreign actors • conduct an evaluation of the use of space from different mechanisms in the country • explore ways to promote and benefit from public-private partnerships • facilitate courses in space based information and applications and recognise such professionals in civil service positions. • coordinate with other member states to harmonise appropriate rules and regulations, licensing procedures, taxation etc. regarding space • strengthen multi-institutional mechanisms • coordinate through the RECs to develop a common strategy on needs and how space can provide solutions and share costs for data acquisition, processing and capacity building. *Passive users*[52] *and Active users*[52] • identify other policy areas where space could be beneficial • promote the development and adoption at the regional level of a space policy or space policy component in other policy areas reflecting the needs of the nation • establish a centralised national approach and creation of centres for acquiring, distributing and maintaining databases of space information to avoid duplication of acquisition • promote the implementation of space policy and space components in other policies for capacity building at various institutional levels and ensure sustainability by continuous training to mitigate “brain drain” • promote on the job training • facilitate communication between information providers, and users in the country to bridge the gap between the two communities. *Active developers*[52] • promote cases of success to other countries and assist them in development • develop mechanisms to attract FDI
African Space Agencies	• coordinate their efforts on a regional level • establish links with other technical bodies in Africa that could benefit from space • establish links with academic networks • coordinate their efforts on a continental level • strategically inform and assist in making politicians and decision makers aware of the benefits of space and provide them with arguments and cases • identify a strategic plan for international relationships with European entities such as EU, ESA and EUMETSAT
African Technical Organisations	• strengthen existing networks and create new networks for cooperation and coordination of activities between the technical actors • strengthen the existing institutions to be able to take up new technolo-

[52] *Passive users* are African countries that do not have any space capabilities. They only receive passively information already processed by others.
Active users are Africa countries that have the capacity to process information offered.
Active developers: are those African countries that themselves have the capacity in space activities and typically have a space agency and more advanced space policy components either self standing or as parts of other policies.

Actor	Proposed Actions
	gies • map and monitor the current local degree of technical feasibility for space • focus on GIS and mapping activities • strengthen capacity building in radar and image processing technologies • explore opportunities to acquire information via African receiving stations • promote and lobby professionals in the fields of space based information and applications • increase the involvement of the local community • strategically inform and assist in making politicians and decision makers aware of the benefits of space and provide them with arguments and cases • facilitate data sharing among institutions

Annex 1: Space Agencies, Treaties, Laws and Policies in Africa

A1.1 Space Agencies in Africa

National Space Research and Development Agency (NASRDA), Nigeria
http://www.nasrda.org

Department of Science and Technology, South Africa
http://www.space.gov.za/

Agence Spatiale Algérienne (ASAL) Algeria
http://www.asal-dz.org/

Centre Royal de Télédétection Spatiale (CRTS), Morocco
http://www.crts.gov.ma/

National Authority for Space Science and Remote Sensing (NARSS), Egypt
http://www.narss.sci.eg/

A1.2 Overview of Space Policies, Laws and Treaties in Africa

	1967 Outer Space Treaty	1968 Rescue Agreement	1972 Liability Con-vention	1976 Registration Convention	1979 Moon Agree-ment	National Space Law	National Space Pol-icy
Nigeria	R	R	R	-	-	-	✓
Republic of South Africa	R	R	S	-	-	✓	✓
Morocco	R	R	R	R	-	-	-
Egypt	-	S	S	-	-	-	-
Algeria	R	-	R	R	-	-	-
Angola	-	-	-	-	-	-	-
Benin	R	-	R	-	-	-	-
Botswana	S	R	R	-	-	-	-
Burkina Faso	R	-	-	-	-	-	-
Burundi	S	-	S	S	-	-	-
Cameroon	S	R	-	-	-	-	-
Cape Verde	-	-	-	-	-	-	-
Central African Republic	S	-	S	-	-	-	-
Chad	-	-	-	-	-	-	-
Comoros	-	-	-	-	-	-	-
Congo	S	-	-	-	-		-
Cote d'Ivoire	-	-	-	-	-	-	-
Democratic Republic of Congo	S	S	S	-	-	-	-
Djibouti	-	-	-	-	-	-	-
Equatorial Guinea	R	-	-	-	-	-	-
Eritrea	-	-	-	-	-	-	-
Ethiopia	S	-	-	-	-	-	-
Gabon	-	R	R	-	-	-	-
Gambia	S	R	S	-	-	-	-
Ghana	S	S	S	-	-	-	-
Guinea	-	-	-	-	-	-	-
Guinea- Bissau	R	R	-	-	-	-	-

	1967 Outer Space Treaty	1968 Rescue Agreement	1972 Liability Convention	1976 Registration Convention	1979 Moon Agreement	National Space Law	National Space Policy
Kenya	R	-	R	-	-	-	-
Lesotho	S	S	-	-	-	-	-

R Ratified

S Signed

✓ Existing national space policy

✓ Existing national space law

Annex 2: Project Overview for European Actors

This Annex lists in tables some projects were the European Commission and ESA are involved. More projects and details of these projects can be found in the main part of the report.

A2.1 European Space Agency

Projects	Objectives	Responsible/ Implementers	Partners	Status
GMES Space Component	Provision of EO data for monitoring environment and supporting civil security	EC, ESA	ASI, BNSC, CDTI, CNES, CSA, DLR, EC, EUMETSAT, ESA	pre-operational; Sentinel-1 to be launched in 2011
TIGER	Help African nations overcome difficulties in collection, analysis and dissemination of water-related geo-information Through EO exploitation	ESA	expert centres in EO applications for water management, private sector, NGOs, finance & development agencies (collectively representing the *TIGER Stakeholder Group*)	first implementation period over (2005-2007); second implementation period ongoing
TTF	Investigate potential for satellite communication technology as support for healthcare systems in sub-Saharan Africa	ESA	African organisations, WHO, EC, ESA	pilot project proposed
SpaceDream	space-based technologies for support to supply antiretrovirals for treatment of HIV/AIDS in sub-Saharan Africa	ESA	Comunita di Sant'Egidio, LUXSpace, CISAS 'G.Colombo', ESA	Ongoing
GISA	study potential role of geographic and space techniques related to HIV/AIDS and ARV issues in sub-Saharan Africa. GIS for tracking epidemics	ESA	WHO, Medecins Sans Frontieres, ESA, UNAIDS, Telecoms Sans Frontieres	Ongoing
Identifying Natural Mosquito Habitats	Development and demonstration of EO technology for identifying natural mosquito habitats and predicting Malaria risk in Africa	ESA	ESA	Ongoing

IWAREMA	provide local Zambian policy makers with necessary tools for effective water resource management	ESA	Zambian Water Authorities, ESA	Ongoing

A2.2 European Commission

Projects	Objectives	Responsible/ Implementers	Partners	Status
AMMA	International project to improve understanding of the West African monsoon and its variability using a multi-scale approach	French initiative; EC FP6	Agencies from FR, UK, US and Africa; CSEI UCAD, IPD, DIREL; Medias-France	since 2002
IST-Africa (2010-2011)	Multi-stakeholder initiative focused on capacity building; open to collaboration and interested in having space projects	IIMC; EC FP7 & FP6	DST; MCST-Botswana; MCST-Lesotho; ICT PITU-Mozambique; ME-Namibia, NCB-Mauritius; COSTECH; UNCST; MoHEST; MESRS-Burundi; MPO-ICT-Rwanda; ANTIC-Cameroon; MRS; MCIT-Egypt	Ongoing
CAAST-Net	A four year joint Africa-Europe platform dedicated to advancing biregional cooperation in science and technology	MOST-KE; AU-ACU; EC FP7	MoHEST; RA; UNCST; MINEDUC; DR-MENRS; CSIR-STEPRI; DGESC; MINRESI; MRS; DST; AU-ACU; PT-DLR; IRD; RCN; JyU-UniPID; VINNOVA; CIRAD; FCT	ongoing
SCHEMA	Explores the use of EO techniques to elaborate generic method helping experts in building vulnerability and hazard impact maps associated with tsunamis and disaster scenarios	GEOSCIENCES CONSULTANTS sarl; EC FP6	ALGOSYSTEMS; HIDROMOD; University of Bologna; University of Coventry; National Observatory of Athens; CRTS; ACRI-ST; Bulgarian Academy of Science; JRC-IPSC; TUBITAK-MRC-EMSC	ongoing 01/08/07 - 01/08/10
VGT4-AFRICA	Operational and timely distribution system of vegetation data from SPOT satellites and creating high level derived products to all African countries	Flemish Institute for Technological Research; EC FP6	Flemish Institute for Technological Research; EC-JRC-IES; MEDIAS-France	completed 01/01/05 - 01/01/08
AFSAGA	The project analyses the potential applications available from Galileo and satellite communications in South Africa and the SADC region.	Thales Alenia; EC FP6	CSIR; Thales Alenia	completed 01/03/07 - 01/03/08
Net-ADDeD	Aims at developing and validating technical features for improved performance of operation of hybrid satellite wireless technologies.	Astrium SAS; EC FP6	Astrium SAS; CNES; Cemagref; Universite Blaise-Pascal; Sabanci University; Telemedicine Technologies; MICNTI Groupement des Retraites Educateurs sans	completed 01/04/07 - 01/04/09

Projects	Objectives	Responsible/ Implementers	Partners	Status
			Frontieres; Universite Medicale Virtuelle Francophone; Istanbul Medical Faculty; Faculte de Medecine et de Pharmacie Universite Hassan II; FORTHnet SA; Connected Schools	
EGNOS	Improves the open public service offered by the USA's GPS, making it suitable for safety critical applications. The possibility of extending EGNOS coverage to Africa is being evaluated	EC; ESA; Eurocontrol	EC; ESA; Eurocontrol; AENA; DFS; ENAV; NATS; NAV; Skyguide	ongoing
EAMNet	Aims to construct a network linking EO information providers, user networks and centres of excellence in Europe and Africa in the area of coastal and marine observations towards sustainable development in Africa	Plymouth Marine Laboratory; EC FP7	PML; UCT; Portuguese Institute of Marine Research; DMI; EUMETSAT; NERC/NOCS; Institute of Marine Sciences University of Dar-es-Salaam; University of Ghana; MeteoFrance; Egypt National Institute of Oceanography & Fisheries	ongoing started end 2009 duration 36 months
DevCoCast	Providing reliable, worldwide and continuous access to EO data and derived environmental information for sustainable development. A core infrastructure and early success of GEOSS.	VITO; EC FP7	VITO; ACMAD; AGRHYMET; CONAB; CREAN; CSIR; DMI; EMBRAPA; EUMETSAT; INPE; INTA; ITC; JRC-IES; UCT/MA-RE; NERC/NOCS; PML	ongoing started 01/05/08 duration 36 months
AIDA	Aims to fill ICT information gap, and collect and share ICT for Disaster and Risk Management knowledge and solutions in Africa	VITO; EC FP7	VITO; ARU; CSIR; EUMETSAT; GEOSAT Technology; IER; ITC; FUTM; NET Qi; TSF	Completed 01/06/08 - 01/06/10
AMESD	Aims to extend the operational use of EO technologies and data to environmental and climate monitoring applications in order to provide all African nations with the resources needed to manage their environment more effectively and ensure long term sustainable development in the region.	AUC; EU-EDF	AUC; Thales Alenia; IRD; BLR; AGRYMET; CICOS; BDMS; ICPAC; MOI; EUMETSAT; EC-JRC	Ongoing started October 2007 duration 48 months
EO2HEAVEN	Will monitor changes induced by human activities, with emphasis on atmospheric, river, lake and coastal marine pollution. Three different cases (in Europe and South Af-	EC FP7	ATOS; Technical University Dresden; 52 North Initiative for Geospatial Open Source Software; Umweltbundesamt; SPOT image; NEVANTROPIC; EC-DG; JRC; CSIR; Open Geospatial	Ongoing started 01/02/10 duration 36 months

Projects	Objectives	Responsible/ Implementers	Partners	Status
	rica) will be assessed and solutions evaluated.		Consortium; Fraunhofer; Bureau de recherche gro-logiques et minieres; Sticht-ing International Institute for Geo-information Science and Earth Observation; University of Kwazulu-Natal; ICT Policy Implementation Technical Unit	
AEGOS	Aims at setting up the preparatory phase for the building of an information system containing and making accessible data and knowledge on African geological resources in-cluding mineral resources, raw material, groundwater and energy (georе-sources).	EC FP7	Beak Consultants; BGR; BGS; BRGM; CIFEG; CGS; CzechGS; DNG; GD-SUM-UNZA; GSD; GSE; GSM; GTK; INETI; IRA; IRD; JRC-IES; MMI; PGI; RBINS; SEAMIC; TNO; UEMOA; DGS; DNG M; EUMETSAT; MAWF; RMCA; SGU	Ongoing started 09/02/09 duration 30 months
GEONetCab	Create conditions for im-provement and increase of GEO capacity building activities and frameworks with special emphasis on developing countries	EC FP7	Stichting International Insti-tute for Geo-information Science and Earth Observa-tion; CNRS; IRD; Karlova Universita; Centrum Badan Kosmicznych Polska Akade-mia Nauk; CSIR; CRAST; Umvoto Africa	Ongoing started October 2009 duration 36 months
MIA-VITA	Aims at developing tools and integrated cost effec-tive methodologies to mitigate risks from vari-ous hazards on active volcanoes	EC FP7	BRGM; INGV; IST; CNRS; NILU; KELL; INESC-ID; Hohenheim University; Uni-versity of Cambridge; DDSC; DPCI; INMG; MINIMIDT; CVGHM; PHIVOLCS	Ongoing started 01/10/08 duration 48 months
M2ERA	Aims to reinforce S&T cooperation and partner-ship between Morocco and Europe in order to foster the integration of Morocco into the European Re-search Area.	EC FP7	MENARA; MENESFCRS; CNRST; Aster; TETHYS; CIRPS consortium	Ongoing started 01/12/08 duration 36 months
ESASTAP2	Is based on the S&T agreement signed be-tween South Africa and the European Union and seeks to build on the foundation laid by the European-South African Science and Technology Advancement Programme (ESASTAP).	EC FP7	DTS	Ongoing started 01/09/08 duration 36 months

Appendix 3: Questionnaires

A3.1 Questions to African actors for obtaining countries profiles

Q1. Is there any agreement between your country and/or national space agency and another African country in the field of space activities and the use of satellite applications?

Q2. Is there any agreement between your country and/or national space agency and another European country in the field of space?

Q3. More generally, is there any other partnership or programme with other African countries that your country may have been engaged into, and that fully or partially relies on space technologies for its implementation? (in the field of education, development aid agriculture, or any other).

Q4. Does your country have a space agency, space policy, space programme and who is the responsible body?

A3.2 Questions to African actors for drawing policy perspectives in European African Partnership through the use of satellite applications

Part 1: Political

1. How should African groupings in relation to need, capabilities, social and political relations be split?
    - Via regions: North, East, South, West, Centre
    - Via Languages: French, English etc.
    - Via interests: desertification, deforestation, disease etc.
2. The African Union is a direct partner to the European Union. How is this viewed? Is this functional/effective? Does it reflect the African reality? Should other constructions be considered? Regional Communities? (e.g. Economic Community of West Africa, South African Development Community etc.), UN ECA, other?
3. In outreach and capacity building in Space, which are the main European and African players?
4. Which are the actions for sustainable development that Africans are undertaking themselves?
5. What is the level of horizontal cooperation between African countries themselves?
6. Are the African leaders convinced of the benefits of Space?
7. How should African leaders be addressed regarding the benefits of using satellite applications for sustainable development in Africa?
8. What should be the next steps at the policy level?
9. How high/low are space & space applications on the political agenda?
10. Why do African nations feel a need to cooperate with Europe?

Part 2: Economic

1. Are current funding mechanisms in place adequate for sustainable development?
    a. Which can be considered as success stories?
    b. Which can be considered as lessons for avoidance?
    c. What should be done next?
2. In relation to the previous question, funding Mechanisms to discuss are:
    a. European funding
    b. African funding
    c. European African co-funding

d. Public Private Partnerships

3. Is there an overview of funding levels in pan-African or national levels?
4. Which are the actions for sustainable development that Africans are undertaking themselves?

Part 3: Social

1. Are the benefits of space well communicated to the citizens?
2. Are the citizens able to use them?
3. What is considered to be the most beneficial space application for African society?
4. What happens after a project is completed? Is there continuation? How is this monitored? Is there an African authority or other body that takes up the responsibility after a project is completed?
5. What happens to projects conducted in partnership with non-African partners after external partners leave the project? Does enough local expertise/infrastructure remain in Africa for sustainability?
6. Has there been measurable evidence of the results of space applications to the local communities in achieving the millennium goals? Please give examples.
7. Lessons learned: good and bad practices? What is needed and what can be done next?

Part 4: Technological

1. Technology levels can be seen as active or passive user, technology developer. User needs mapping currently is being performed for GMES. What else has been done? What other mapping of needs are missing and what should be the next steps?
2. What thematic topics are missing?
3. What can be done to facilitate technology development? How should this be looked at the different technology levels?
4. How is sustainability viewed in this context?
5. Which mechanisms exist for successfully training/educating the next generation of Africans in space activities and applications using satellites?

A3.3 Questions to European actors for obtaining countries profiles

Q1. Is there any agreement between your country and/or national space agency and an African counterpart in the field of space ?

Q2. More generally, is there any other partnership or programme with Africa that your country may be engaged into, and that fully or partly relies on space technologies for its implementation ? (in the field of education, development aid, agriculture or any other domain)?

A3.4 Questions aux acteurs africains pour l'élaboration des perspectives politiques du partenariat européen-africain à travers l'utilisation des applications de satellites

Première Partie: Volet politique.

1. De quelle manière faut-il effectuer des regroupements de pays par rapport aux besoins, aux capacités, aux relations sociales et politiques?
   - Par régions: le nord, l'est, le sud, l'ouest, le centre
   - Par langues: français, anglais etc.
   - Par intérêts: la désertification, la déforestation, les maladies etc.
2. L'Union Africaine est un partenaire direct de l'Union Européenne. Comment cela est-il perçu? Est-ce que c'est fonctionnel/efficace ? Est- ce que cela reflète la réalité africaine? D'autres constructions devraient-elles être considérées, telles que des communautés régionales? (par exemple La Communauté Économique de l'Afrique Ouest, la Communauté de Développement sud-africaine, la Commission Economique pour l'Afrique etc.) ou autres ?
3. Quels sont les principaux acteurs européens et africains en termes de travail de proximité et de renforcement des capacités spatiales?

4. Quelles sont les actions pour le développement durable que les Africains entreprennent eux-mêmes?
5. Quel niveau de coopération horizontale existe-t-il entre les pays africains ?
6. Les leaders africains sont-ils convaincus des avantages de l'Espace?
7. Comment les aborder pour promouvoir auprès d'eux les avantages de l'utilisation des applications des satellites pour le développement durable en Afrique?
8. Quelles devraient être les étapes suivantes au niveau de la politique menée?
9. Quelle place l´espace et les applications spatiales occupent-ils dans l'ordre du jour politique?
10. Pourquoi les nations africaines ressentent-elles le besoin de coopérer avec l'Europe?

Deuxième partie: Volet économique

1. Les mécanismes actuels de financement sont-t-ils adéquats pour le développement durable?
    a. Lesquels peut-on considérer comme des succès?
    b. Lesquels doit-on éviter à l'avenir ?
    c. Quelle est la prochaine action à entreprendre en ce domaine?
2. En relation avec la question précédente, voici les mécanismes de financement à envisager :
    a. Financement européen
    b. Financement africain
    c. Cofinancement africain et européen
    d. Partenariats public- privé
3. Y-a-t-il une vue d'ensemble des niveaux de financement au niveau panafricain ou national?
4. Quelles sont les actions pour le développement durable que les Africains entreprennent eux-mêmes?

Troisième Partie : Volet social

1. Les avantages de l'espace sont-ils bien expliqués aux citoyens?
2. Les citoyens sont-ils capables de les utiliser?
3. Quelle est l'application spatiale qui est perçue comme la plus avantageuse pour la société africaine?
4. Qu'arrive-t-il lorsqu'un projet est achevé? Y a-t- il une continuité? Comment est-ce géré? Y a-t-il une autorité africaine ou un organe quelconque qui prend le relais?
5. Qu'arrive-t-il aux projets menés en partenariat avec des partenaires non africains après le départ de ces derniers? Reste-t-il assez d'expertise et d'infrastructures locales pour la durabilité?
6. Y a-t-il eu des preuves quantifiables des résultats des applications des satellites pour les communautés locales dans l'accomplissement des objectifs du millénaire pour le développement? Pouvez-vous donner des exemples?
7. Les leçons apprises: bonnes et mauvaises pratiques? Quels sont les besoins actuels et qu'est-ce qui peut être entrepris par la suite?

Quatrième partie: Volet technologique

1. On peut considérer plusieurs niveaux de technologie : utilisateur passif, utilisateur actif, développeur de technologie. Les besoins de l'utilisateur sont actuellement en cours de classification pour le GMES. Quelles autres tâches ont été accomplies? Qu'est-ce qui manque encore à l'identification des besoins? Quelles devraient être les prochaines étapes?
2. Quels sont les sujets thématiques manquants?
3. Que peut-on faire pour faciliter le développement technologique à différents niveaux?
4. Comment le développement durable est-il perçu dans ce contexte?
5. Quels sont les mécanismes existants pour former et éduquer la prochaine génération d'Africains à l'espace et aux applications utilisant les satellites?

A3.3 Questions to European actors for drawing policy perspectives in European African Partnership through the use of satel-lite applica-tionsPart 1: Po-litical

1. Why is Europe interested in Euro-pean-African cooperation? What are the underlying motives?
2. How strong is the European interest in Africa?
3. How is the increasing involvement of non-European players in Africa (like China) perceived by Europeans?
4. How should African groupings in rela-tion to need, capabilities, social and political relations be split?
    - Via regions: North, East, South, West, Centre
    - Via Languages: French, English etc.
    - Via interests: desertification, de-forestation, disease etc.
5. The African Union is the direct part-ner to the European Union. How is this viewed? Is this func-tional/effective? Does it reflect the African reality? Should other con-structions be considered ? Regional Communities? (eg. Economic Com-munity of West Africa, South African Development Community etc.) other? Should there be a more 'peer to peer' involvement?
6. In outreach and capacity building in Space, which are the main European and African players?
7. Which are the actions for sustainable development that Africans are under-taking themselves? And how are these perceived by Europeans?
8. Are the African leaders convinced of the benefits of Space?
9. How to address them?
10. What should be the next steps at the policy level?
11. How high/low are space & space ap-plications on the political agenda?
12. Why do African nations feel such a strong need to cooperate with Europe?

Part 2: Economic

1. Are current funding mechanisms in place adequate for sustainable devel-opment?
    a. Which can be considered as suc-cess stories?
    b. Which can be considered as les-sons for avoidance?
    c. What should be done next?
2. Funding Mechanisms to discuss
    a. European funding
    b. African funding
    c. European African co-funding
    d. Public Private Partnerships
3. How is the progress on the actions for sustainable development that Afri-cans are undertaking themselves viewed by the Europeans?

Part 3: Social

1. Are the benefits of space well com-municated to the citizens?
2. Are the citizens able to use them? How do they differ from our European understanding?
3. What is the most beneficial space ap-plication for African society from a European perspective?
4. What happens after a project fin-ishes? Is there continuation? How is this monitored? Are the European Stakeholders involved?
5. What happens to projects in partner-ship after external partners leave the scenes? Does enough local exper-tise/infrastructure remain? What is the European involvement?
6. Has there been measurable evidence of the results of space to the commu-nities in achieving the millennium goals?
7. Lessons learned: good and bad prac-tices? What is needed and what can be done next?

Part 4: Technological

1. Technology levels can be seen as passive user, active user, technology developer. User needs mapping are currently being performed for GMES. What else has been done? What other mapping of needs are missing and what should be the next steps?
2. What thematic topics are missing?
3. What can be done to facilitate technology development? How should we look at the different levels?
4. How is a 'system of systems' build up approach perceived?
5. How is sustainability viewed by Europeans in this context?
6. Which mechanisms exist for successfully training/educating the next generation in space and related technologies?

List of Acronyms

AARSE	African Association of Remote Sensing for Environment
AAU	Association of African Universities
ACMAD	African Centre for Meteorological Applications
ACP	Africa/Caribbean/Pacific countries
ACSRT	African Centre for Studies and Research on Terrorism
ACST	African Cluster on Science and Technology
ADOS	African Doppler Survey (ended in 1986)
AEG	African Expert Group
AfDB	African Development Bank
AFREF	African Geodetic Reference Frame
AICC	African Institute on Climate Change
ALC	African Leaders Conference
AMCEN	African Ministerial Conference on the Environment
AMCOST	African Ministerial Conference on Science and Technology
AMESD	African Monitoring for Sustained Environment and Development
AOCRS	African Organization of Cartography and Remote Sensing
AOSTI	African Observatory of Science, Technology and Innovation
APF	African Peace Facility
APM	Anti-Personnel Landmines
APO	African Press Organisation
APRM	African Peer Review Mechanism
APSA	African Peace and Security Architecture
ARAPKE	African Regional Action Plan for Knowledge Economy (AU initiative)
ARC	AGRHYMET Regional Centre (specialized institute of CILSS)
ARMC	African Resource Management Constellation
ASAL	Algerian Space Agency (Agence Spatiale Algerienne)
AU PSC	AU Peace and Security Council
AUC	African Union Commission
AUC-HRST	AUC's Department on Human Resource Science and Technology
AXIS	African Internet Exchange System
BASIC	Building Africa's Scientific and Institutional Capacity
CAAST-Net	Network for the Coordination and Advancement of Africa=EU Science and Technology Cooperation
CEMAC	Economic and Monetary Community of Central Africa
CILSS	Permanent Interstate Committee for Drought Control in the Sahel
ClimDev Africa	African Climate Information for Development Programme

CODIST	ECA's Committee on Development Information, Science and Technology
CPA	Africa's Science and Technology Consolidated Plan of Action
CRM	Climate Risk Management
CRTS	Royal Centre for Remote Sensing
CSIR	Council for Scientific and Industrial Research
CSOs	Civil Society Organisations
DCI	Development Cooperation Instrument
DGESC	Direccao Geral de Ensino Superior a Ciencia, Cape Verde
DR-MENRS	Ministry of National Education and Scientific Research
DRSRS	Department of Resources Surveys and Remote Sensing, Kenya
DST	Department of Science and Technology, South Africa
EAC	East African Community
EAMNet	Europe-Africa Marine EO Network
ECA	Economic Commission for Africa
ECOSOCC	AU Economic, Social and Cultural Council
ECOWAS	Economic Community of East African States
EDCTP	European and Developing Countries Clinical Trials Partnership
EDF	European Development Fund
EEG	European Expert Group
EESC	European Economic and Social Committee
EGNOS	European Geostationary Navigation Overlay Service
ENPI	European Neighbourhood Policy Instrument
EP	European Parliament
EPAs	Economic Partnership Agreements
ERA	European Research Area
ERW	Explosive Remnants of War
EUROGI	European Umbrella Organization for Geographic Information
FARA	Forum for Agricultural Research in Africa
FP6	Framework Programme 6
FP7	Framework Programme 7
GCCA	Global Climate Change Alliance
HESS	High Energy Stereoscopic System
IAA	International Academy of Astronautics
IAG	International Association of Geodesy
ICA	Infrastructure Consortium for Africa
ICPAC	IGAD Climate Prediction and Application Centre
ICSU	International Council for Science
IGAD	Intergovernmental Authority on Development
INCO	Instruments for Cooperation
IOC	Indian Ocean Commission
ITRF	International Terrestrial Reference Frame
KMD	Kenya Meteorological Department

MDGs	Millennium Development Goals
MEAs	Multilateral Environmental Agreements
MinESS	Ministry of Education, Sport and Science, Ghana
MINISTR	Ministry of Science and Technology, Rwanda
MINRESI	Ministry of Scientific Research and Innovation, Cameroon
MOST-KE	Ministry of Science and Technology
MPC	Mediterranean Partner Countries
MRS	Ministere de la Recherche Scientifique, Senegal
NARSS	National Authority for Remote Sensing and Space Sciences, Egypt
NASRDA	National Space Research and Development Agency, Nigeria
NCST	National Council for Science and Technology, Kenya
NEPAD	New Partnership for Africa's Development
NRF	National Research Foundation, South Africa
OOSA	United Nations Office for Outer Space Affairs
PAIPO	Pan African Intellectual Property Organisation
PAP	Pan-African Parliament
PAU	Pan African University
PIDA	Programme for Infrastructure Development in Africa
PPPs	Public-Private Partnerships
PRSAO	West African Regional Health Programme (Programme Regional Sante en Afrique de l'Ouest)
PRSPs	Poverty Reduction Strategy Papers
PSC	Partnership Steering Committee
PUMA	Preparation for the Use of Meteosat second generation in Africa
RA	Research (Africa) (Pty) Ltd., South Africa
RASCOM	Regional African Satellite Communications Organisation
RCMRD	Regional Centre for Mapping of Resources for Development, Kenya
RECs	Regional Economic Communities
RECTAS	Regional Centre for Training in Aerospace Surveys
SAAO	South African Astronomical Observatory
SADC	South African Development Community
SALT	South African Large Telescope
SALW	Small Arms and Light Weapons
SBAS	Satellite-based Augmentation Services
SICAs	Specific International Cooperation Actions
SSLT	Surrey Satellite Technology
UNCST	Ugandan National Council for Science and Technology
UN-ECA	United Nations Economic Commission for Africa
YD	Yamoussoukro Decision

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Participants in the stakeholder workshop organised by ESPI (from left): Michel Bosco (EC), Christine Leurquin (SES), Cecile Desnos (ESPI), Christina Giannopapa (ESPI), Lionel Poncelet (Belgian EU Council Presidency 2/2010), Sebastian Rieder (Eurisy), Vincent Gabaglio (EUMETSAT)



Round table from left: Colin Hicks (President Eurisy), Adigun Ade Abiodun (Former Chairman UNCOPUOS, Founder of the African Space Foundation, Nigeria), Driss el Hadani (Director Royal Centre for Remote Sensing, Morocco), Peter Martinez (Division Head of Space Science and Technology, South African Astronomical Observatory), Jean-Francois Mayence (Belgian EU Council Presidency 2/2010), Marie-Elisabeth de Vel (International Relations ESA) and Kai-Uwe Schrogl (Director ESPI)



UN SPIDER workshop organisers and ESPI Resident Fellow. From right: Michael Leitgab (UNOOSA), Andre Nonguerma (UNECA), Christina Giannopapa (ESPI), Juan-Carlos Villagràn de León (UNOOSA), Agnieszka Lukaszczyk (SWF).

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